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Kidney Transplantation

Number: 0493

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Scope of Policy

This Clinical Policy Bulletin addresses kidney transplantation.

I. **Medically Necessary**

A. **Kidney Transplantation**

Aetna considers kidney transplantation medically necessary for members who meet the transplanting institution's selection criteria. In the absence of an institution's selection criteria, Aetna considers kidney transplantation medically necessary when *all* of the following criteria below are met:

1. Member has completed an evaluation and been accepted by the kidney transplant committee at the kidney transplantation center. **Note**: Frequently requests for evaluation for transplantation are confused with requests for the transplantation itself. While the transplant evaluation of persons with kidney disease may be indicated, the medical necessity for transplantation itself depends on the results of the evaluation; *and*
2. Member meets transplanting institution's protocol eligibility criteria regarding age; *and*
3. Absence of malignancy (except for non-melanomatous skin cancers or low-grade prostate cancer) or the malignancy has had curative therapy (e.g., surgical resection of non-invasive squamous cell or basal cell skin cancer) or the estimated risk of recurrence of the malignancy is less than 10% within the next 2 years. For example, renal cell carcinoma treated by nephrectomy with no evidence of metastatic disease 2 years after the nephrectomy, prostate cancer with negative prostate-specific antigen levels after treatment, surgically treated colon cancer, thyroid cancer with normal thyroglobulin levels after therapy, and others. Women should have a negative Pap smear within the past 3 years and mammography, where indicated, within the past 2 years; *and*
4. Absence of systemic infection; *and*
5. Absence of symptomatic HIV infection, as defined by *all* of the following:

    a. CD4 count greater than 200 cells/mm^3 for more than 6 months; *and*

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b. HIV-1 RNA (viral load) undetectable; *and*
c. On stable anti-viral therapy for more than 3 months; *and*
d. No other complications from AIDS, such as opportunistic infection (e.g., aspergillus, coccidiomycosis, resistant fungal infections, tuberculosis), Kaposi's sarcoma or other neoplasms; *and*

6. Attending physician determines that there is no prohibitive cardiovascular risk; *and*
7. Attending physician determines that there is no prohibitive pulmonary risk; *and*
8. Attending physician determines that there is no prohibitive hepatic risk; *and*
9. Severity of disease:

    a. Member is already on hemodialysis or continuous ambulatory peritoneal dialysis (CAPD); *or*
    b. Member has chronic renal failure with anticipated deterioration to end stage renal disease, where member is seeking precertification for cadaveric kidney transplantationFootnote1* *or*
    c. Member has end stage renal disease, evidenced by a creatinine clearance below 20 ml/min or development of symptoms of uremia, and member is seeking precertification for a living donor kidney transplantation; *and*

    Footnote1* **Note**: Given waiting periods for cadaveric donors averaging 1 to 4 years, kidney transplantation is considered medically necessary for persons with severe chronic renal failure with anticipated progression to end stage renal disease. Severe chronic renal failure is defined as a creatinine clearance of less than 30 ml/min;

Kidney transplant is not considered medically necessary for persons who do not meet the transplanting institution's protocol selection criteria, or in the absence of a protocol, for persons who have *any* of the following (not an all-inclusive list):

1. Active vasculitis; *or*
2. Age over 70 years with severe co-morbidities; *or*
3. Life threatening extra-renal congenital abnormalities; *or*
4. Ongoing alcohol or drug abuse; *or*
5. Severe neurological or mental impairment, in persons without adequate social support, such that the person is unable to adhere to the regimen necessary to preserve the transplant; *or*
6. Untreated coagulation disorder.

B. **Combined Kidney/Pancreas Transplantation**

1. For persons undergoing kidney transplantation due to diabetic nephropathy, a combined kidney/pancreas transplantation may be considered medically necessary under some circumstances (see CPB 0587 - Pancreas Kidney Transplantation).
2. Other multi-organ transplants (e.g., kidney/heart, liver/kidney) should be referred to Aetna's National Medical Excellence Program for review.

C. **Renal Auto-Transplantation and Ex-Vivo Bench Surgery**

Aetna considers auto-transplantation and ex-vivo repair medically necessary where repair of the kidney, ureter, renal artery or its branches are not amenable to in-situ reconstruction.

D. **Belatacept (Nulojix)**

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Aetna considers the use of belatacept (Nulojix) medically necessary for the prevention of acute rejection in kidney transplant recipients who are sero-positive for the Epstein Barr virus (EBV).

Aetna considers belatacept experimental, investigational, or unproven for the prophylaxis of organ rejection in other transplanted organs because its effectiveness for the prevention of acute rejection in organ transplant other than kidney has not been established.

E. **Equine Anti-Thymocyte Immune Globulin**

Aetna considers equine anti-thymocyte immune globulin (Atgam) medically necessary for the following indications:

1. Prophylaxis or treatment of allograft rejection episodes in renal transplantation in combination with conventional therapy; *or*
2. Moderate to severe aplastic anemia in persons who are not suitable candidates for bone marrow transplantation.

Aetna considers equine anti-thymocyte immune globulin experimental, investigational, or unproven for all other indications.

F. **Renal Auto-Transplantation**

Aetna considers renal auto-transplantation medically necessary for the treatment of individuals with loin pain hematuria syndrome who have failed non-surgical therapies including analgesics.

II. Experimental, Investigational, or Unproven

Aetna considers the following experimental, investigational, or unproven:

A. **Gene Microarrays for Diagnosis of Rejection**

Aetna considers the use of gene microarrays (e.g., the Kidney Microscope Diagnostic System (MMDx-Kidney)) in diagnosis of rejection of kidney transplantation experimental, investigational, or unproven because of insufficient evidence of their effectiveness.

B. **Experimental, Investigational, or Unproven Markers of Acute Rejection**

Aetna considers measurement of cytokines (e.g., cytokine-14, interleukin-1 beta (IL-1β), IL-2, IL-4, IL-6, granulocyte-macrophage colony-stimulating factor (GM-CSF), monocyte chemoattractant protein-1 (MCP-1), and tumor necrosis factor-alpha (TNF-α); not an all-inclusive list) for the diagnosis of acute renal allograft rejection experimental, investigational, or unproven because the effectiveness of this approach has not been established.

C. **Experimental, Investigational, or Unproven Markers of Rejection Risk**

Aetna considers the following experimental, investigational, or unproven because the effectiveness of this approach has not been established:

1. Clarava as a pre-transplant prognosis test for the risk of early acute rejection in kidney transplant candidates;
2. Human leukocyte antigen-G-14-base-pair-insertion/deletion polymorphism;
3. Interleukin-2-330 T/G promoter;
4. Interleukin-10-1082 (G/A) promoter polymorphisms testing for evaluating the risk of developing kidney graft rejection;

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5. Perfusate biomarkers produced during hypothermic machine perfusion for prediction of graft outcomes in kidney transplantation;
6. Pleximark (measurement of donor and third party-induced CD154+T-cytotoxic memory cells) for evaluation of acute cellular rejection following kidney transplantation; and
7. Tuteva as a post-transplant test for acute cellular rejection, including sub-clinical rejection, in kidney transplant recipients.

D. **Bisphosphonates**

Aetna considers bisphosphonates experimental, investigational, or unproven for the treatment of low bone mineral density after kidney transplantation because their effectiveness of this indication has not been established.

E. **Pre-Conditioning Therapy**

Aetna considers pre-conditioning therapy (e.g., immune-adsorption or rituximab) in ABO-incompatible kidney transplantation experimental, investigational, or unproven because the effectiveness of this approach has not been established.

F. **Biomarkers of Acute Kidney Injury**

Aetna considers urinary neutrophil gelatinase-associated lipocalin (NGAL) and liver-type fatty acid-binding protein (L-FABP) experimental, investigational, or unproven as biomarkers of acute kidney injury following kidney transplantation because the effectiveness of this approach has not been established.

G. **FASL mRNA**

Aetna considers Fas ligand (FASL) mRNA detection experimental, investigational, or unproven as a diagnostic marker for acute renal rejection because the effectiveness of this approach has not been established.

H. **Urinary Biomarkers (e.g., Chemokines, Extracellular Vesicles, and Monocyte Chemoattractant Protein-1)**

Aetna considers urinary biomarkers (e.g., chemokines including CXCL9 and CXCL10 (e.g., One Lambda CXCL10 urine test), alone or in combination; extracellular vesicles; monocyte chemoattractant protein-1 (MCP-1/CCL2) experimental, investigational, or unproven for detection and monitoring of renal graft rejection because the effectiveness of this approach has not been established.

I. **Donor-Derived Cell-Free DNA Testing**

Aetna considers donor-derived cell-free DNA testing (e.g., Allosure, Prospera Kidney, VitaGraft Kidney Baseline + 1st Plasma Test, VitaGraft Kidney Subsequent, and VitaGraft Kidney 2.0) experimental, investigational, or unproven for consideration of tapering of immunosuppression therapy in kidney transplant recipients, and monitoring acute rejection following renal transplantation because the effectiveness of these approaches has not been established.

J. **Measurement of Angiotensin II Type 1 (AT1) Receptors or AT1 Antibodies for Evaluation of Renal Transplantation Candidates / Recipients**

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Aetna considers measurement of angiotensin II type 1 (AT1) receptors or AT1 antibodies experimental, investigational, or unproven for evaluation of renal transplantation candidates / recipients because the effectiveness of this approach has not been established.

K. **Complement Inhibitors (e.g., Eculizumab) for the Treatment of Antibody-Mediated Rejection in Renal Transplantation Recipients**

Aetna considers complement inhibitors (e.g., eculizumab) experimental, investigational, or unproven for the treatment of antibody-mediated rejection in renal transplantation recipients because their effectiveness for this indication has not been established.

L. **Belimumab for the Treatment of Antibody-Mediated Rejection in Renal Transplantation Recipients**

Aetna considers belimumab experimental, investigational, or unproven for the treatment of antibody-mediated rejection in renal transplantation recipients because its effectiveness for this indication has not been established. See CPB 0818 - Belimumab (Benlysta).

M. **Genotyping Donors and Recipients Before Renal Transplantation**

Aetna considers genotyping donors and recipients before renal transplantation experimental, investigational, or unproven because its effectiveness for this indication has not been established.

N. **Recombinant Human Erythropoietin**

Aetna considers recombinant human erythropoietin (e.g., Epogen, Procrit, and Retacrit) experimental, investigational, or unproven for nephron-protection in persons undergoing kidney transplantation because its effectiveness for this indication has not been established. See also CPB 0195 - Erythropoiesis Stimulating Agents.

O. **TruGraf Blood Gene Expression Test**

Aetna considers the TruGraf blood gene expression test (TruGraf Kidney (Eurofins Transplant Genomics, Inc.)) experimental, investigational, or unproven for the management of kidney transplant recipients because its effectiveness has not been established.

P. **Plasminogen Activator Evaluation**

Aetna considers plasminogen activator evaluation experimental, investigational, or unproven as part of a hypercoagulable workup prior to kidney transplantation because the effectiveness of this approach has not been established.

Q. **Evaluation of Urine Immunocytology**

Aetna considers evaluation of urine immunocytology for T cells experimental, investigational, or unproven for the diagnosis of acute kidney rejection because its role has not been established.

R. **Soluble CD30 Level**

Aetna considers measurement of pre-transplantation soluble CD30 level experimental, investigational, or unproven as a predictor of acute rejection in kidney transplantation because its clinical value has not been established.

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S. **DNA Methylation**

Aetna considers the use of DNA methylation experimental, investigational, or unproven as a biomarker of post-transplant complications in kidney transplantation because its clinical value has not been established.

T. **Artificial Intelligence/Machine Learning Method for Predicting Graft Survival**

Aetna considers artificial intelligence/machine learning method experimental, investigational, or unproven for predicting graft survival in kidney transplantation because the effectiveness of this approach has not been established.

U. **Panniculectomy in Kidney Transplant Candidates**

Aetna considers panniculectomy experimental, investigational, or unproven if used to improve kidney transplant outcomes or improve kidney transplant candidacy because the effectiveness of this approach has not been established.

V. **myOLARIS-KTdx**

Aetna considers myOLARIS-KTdx experimental, investigational, or unproven for diagnosis and prognosis of kidney allograft rejection because the effectiveness of this approach has not been established.

III. Related Policies

- CPB 0195 - Erythropoiesis Stimulating Agents
- CPB 0587 - Pancreas Kidney Transplantation
- CPB 0818 - Belimumab (Benlysta)

Table:

CPT Codes / HCPCS Codes / ICD-10 Codes

Code	Code Description
CPT codes covered if selection criteria are met:	
50300	Donor nephrectomy, (including cold preservation); from cadaver donor, unilateral or bilateral
50320	Donor nephrectomy, (including cold preservation); open from living donor
50323	Backbench standard preparation of cadaver donor renal allograft prior to transplantation, including dissection of allograft and removal or perinephric fat, diaphragmatic and retroperitoneal attachments, excision of adrenal gland, and preparation of ureter(s), renal vein(s), and renal artery(s), ligating branches, as necessary
50325	Backbench standard preparation of living donor renal allograft (open or laparoscopic) prior to transplantation, including dissection and removal of perinephric fat and

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Code	Code Description
	preparation of ureter(s), renal vein(s), and renal artery(s), ligating branches, as necessary
50327	Backbench reconstruction of cadaver or living donor renal allograft prior to transplantation; venous anastamosis, each
50328	arterial anastamosis, each
50329	ureteral anastamosis, each
50340	Recipient nephrectomy (separate procedure)
50360	Renal allotransplantation, implantation of graft; without recipient nephrectomy
50365	Renal allotransplantation, implantation of graft; with recipient nephrectomy
50370	Removal of transplanted renal allograft
50380	Renal autotransplantation, reimplantation of kidney
50547	Laparoscopy, surgical; donor nephrectomy (including cold preservation); from living donor

CPT codes not covered for indications listed in the CPB:

***DNA methylation as a biomarker of post transplantation complications in kidney transplantation, Urinary extracellular vesicles for detection and monitoring of renal graft rejection, Artificial intelligence/machine learning method* - no specific code**

0018M	Transplantation medicine (allograft rejection, renal), measurement of donor and third-party- induced CD154+T-cytotoxic memory cells, utilizing whole peripheral blood, algorithm reported as a rejection risk score
0319U	Nephrology (renal transplant), RNA expression by select transcriptome sequencing, using pretransplant peripheral blood, algorithm reported as a risk score for early acute rejection
0320U	Nephrology (renal transplant), RNA expression by select transcriptome sequencing, using posttransplant peripheral blood, algorithm reported as a risk score for acute cellular rejection

Other CPT codes related to the CPB:

77051 - 77057	Breast Mammography[female candidates should have a negative result within the past two years]
88141 - 88175	Cytopathology [female candidates should have a negative result within the past three years]
90918 - 90940	End stage renal disease services and hemodialysis
96365	Intravenous infusion, for therapy, prophylaxis, or diagnosis (specify substance or drug); initial, up to 1 hour
96366	Intravenous infusion, for therapy, prophylaxis, or diagnosis (specify substance or drug); each additional hour (List separately in addition to code for primary procedure)
96367	Intravenous infusion, for therapy, prophylaxis, or diagnosis (specify substance or drug); additional sequential infusion of a new drug/substance, up to 1 hour (List separately in addition to code for primary procedure)
96368	Intravenous infusion, for therapy, prophylaxis, or diagnosis (specify substance or drug); concurrent infusion (List separately in addition to code for primary procedure)
99512	Home visit for hemodialysis

Other HCPCS codes related to the CPB:

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Code	Code Description
G0101	Cervical or vaginal cancer screening; pelvic and clinical breast examination[female candidates should have a negative result within the past three years]
G0123	Screening cytopathology, cervical or vaginal (any reporting system), collected in preservative fluid, automated thin layer preparation; screening by cytotechnologist under physician supervision[female candidates should have a negative result within the past three years]
G0124	requiring interpretation by physician [female candidates should have a negative result within the past three years]
G0141 - G0148	Screening, cytopathology, other
G0202 - G0206	Mammography
G0308 - G0327	End stage renal disease services
S9335	Home therapy, hemodialysis; administrative services, professional pharmacy services, care coordination, and all necessary supplies and equipment (drugs and nursing services coded separately), per diem
S9339	Home therapy; peritoneal dialysis, administrative services, professional pharmacy services, care coordination and all necessary supplies and equipment (drugs and nursing visits coded separately

ICD-10 codes covered if selection criteria are met:

N18.5	Chronic kidney disease, Stage V
N18.6	End stage renal disease
N39.8	Other specified disorders of urinary system [loin pain hematuria syndrome]

ICD-10 codes contraindicated for this CPB:

D65 - D68.9	Coagulation defects [untreated]
C00.0 - C96.9, D00.00 - D09.9	Malignant neoplasms and carcinoma in situ [other than low grade prostate cancer and non-melatomatous skin cancers]
E75.00 - E75.19 E75.23 E75.25 - E75.29 E75.4	Disorders of sphingolipid metabolism and other lipid storage disorders [severe neurological or mental impairment]
F10.10 - F19.99	Mental and behavioral disorders due to psychoactive substance [ongoing alcohol or drug abuse]
F84.2	Rett's syndrome [severe neurological or mental impairment]
G11.0 - G12.9 G13.8 G20.A1 - G26 G30.0 - G32.89 G80.3	Hereditary and degenerative diseases of the central nervous system [severe neurological or mental impairment]

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Code	Code Description
G90.01 - G91.9 G93.7 G93.89 - G93.9 G94 G95.0 - G95.9 G99.0, G99.2	
I77.6	Arteritis, unspecified [active vasculitis]
Q00.0 - Q56.4 Q65.00 - Q99.9	Congenital anomalies [extrarenal congenital abnormalities]

Kidney Microscope Diagnostic System (MMDx-Kidney):

CPT codes not covered for indications listed in the CPB:

0088U	Transplantation medicine (kidney allograft rejection), microarray gene expression profiling of 1494 genes, utilizing transplant biopsy tissue, algorithm reported as a probability score for rejection

Belatacept (Nulojix):

HCPCS codes covered if selection criteria are met:

J0485	Injection, belatacept, 1 mg

ICD-10 codes covered if selection criteria are met:

T86.10 - T86.19	Complications of kidney transplant [for the prevention of acute rejection in kidney transplant recipients who are sero-positive for the Epstein Barr virus (EBV)] [not covered for the prophylaxis of organ rejection in other transplanted organs]
Z48.22	Encounter for aftercare following kidney transplant
Z94.0	Kidney transplant status [for the prevention of acute rejection in kidney transplant recipients who are sero-positive for the Epstein Barr virus (EBV)] [not covered for the prophylaxis of organ rejection in other transplanted organs]

ICD-10 codes not covered for indications listed in the CPB:

T86.00 - T86.09, T86.20 - T86.99	Complications of transplanted organs and tissue [excludes kidney]

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Code	Code Description

Soluble CD30:

83520	Immunoassay for analyte other than infectious agent antibody or infectious agent antigen; quantitative, not otherwise specified [not covered for measurement of pre-transplantation soluble CD30 level as a predictor of acute rejection in kidney transplantation]

Panniculectomy:

CPT code not covered for indications listed in the CPB:

15830	Excision, excessive skin and subcutaneous tissue (includes lipectomy); abdomen, infraumbilical panniculectomy
15847	Excision, excessive skin and subcutaneous tissue (includes lipectomy), abdomen (eg, abdominoplasty) (includes umbilical transposition and fascial plication) (List separately in addition to code for primary procedure

ICD-10 codes not covered for indications listed in the CPB:

N18.6	End stage renal disease

Anti-thymocyte globulin:

HCPCS code covered if selection criteria are met:

J7504	Lymphocyte immune globulin, antithymocyte globulin, equine, parenteral, 250 mg
J7511	Lymphocyte immune globulin, antithymocyte globulin, rabbit, parenteral, 25mg

ICD-10 codes covered if selection criteria are met::

D61.01 - D61.9	Other aplastic anemias and other bone marrow failure syndromes [moderate to severe]
T86.10 - T86.19	Complications of kidney transplant [prophylaxis of allograft rejection episodes]
Z94.0	Kidney transplant status

Measurement of cytokines:

83520	Immunoassay for analyte other than infectious agent antibody or infectious agent antigen; quantitative, not otherwise specified [(e.g., cytokine-14, interleukin-1 beta [IL-1β], IL-2, IL-4, IL-6, granulocyte-macrophage colony-stimulating factor [GM-CSF], monocyte chemoattractant protein-1 [MCP-1], and tumor necrosis factor-alpha [TNF-α]; not an all-inclusive list) for the diagnosis of acute renal allograft rejection]

ICD-10 codes not covered for indications listed in the CPB:

T86.10 - T86.19	Complications of transplanted kidney [not covered for the diagnosis of acute renal allograft rejection]

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Code	Code Description
Z13.89	Encounter for screening for other disorder [Not covered for evaluating the risk of developing kidney graft rejection]
Z94.0	Kidney transplant status [not covered for the diagnosis of acute renal allograft rejection]

***Bisphosphonate*:**

HCPCS code covered if selection criteria are met:

J1436	Injection, Etidronate disodium, per 300 Mg
J1740	Injection, Ibandronate sodium, 1 mg
J2340	Injection, Pamidronate disodium, per 30 mg
J3489	Injection, Zoledronic acid, 1 mg

ICD-10 codes not covered for indications listed in the CPB:

Z94.0	Kidney transplant status[low bone mineral density]

***Interleukin-2-330 T/G and interleukin-10-1082 (G/A) promoter polymorphisms testing*:**

***There is no specific code for interleukin-2-330 T/G and interleukin-10-1082 (G/A) promoter polymorphisms testing*:**

***There is no specific code for interleukin-2-330 T/G and interleukin-10-1082 (G/A) promoter polymorphisms testing*:**

Pre-conditioning
Therapy

HCPCS code covered if selection criteria are met:

J9312	Injection, rituximab, 10 mg

ICD-10 codes not covered for indications listed in the CPB:

T86.10 - T86.19	Complications of transplanted kidney [ABO-incompatible kidney transplantation]

***Perfusate biomarkers produced during hypothermic machine perfusion for prediction of graft outcomes in kidney transplantation*:**

CPT codes not covered for indications listed in the CPB:

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Code	Code Description

***Perfusate biomarkers produced during hypothermic machine perfusion for prediction of graft outcomes in kidney transplantation* - no specific code:**

ICD-10 codes not covered for indications listed in the CPB:

Z94.0	Kidney transplant status [not covered for prediction of graft outcomes in kidney transplantation]

***Urinary neutrophil gelatinase-associated lipocalin (NGAL) and liver-type fatty acid-binding protein (L-FABP)*:**

No specific code

ICD-10 codes not covered for indications listed in the CPB:

T86.19	Other complication of kidney transplant [acute kidney injury following kidney transplantation]

***Fas ligand (FASL) mRNA detection*:**

CPT codes not covered for indications listed in the CPB:

83520	Immunoassay for analyte other than infectious agent antibody or infectious agent antigen; quantitative, not otherwise specified [fas-ligand (FASL) mRNA detection as a diagnostic marker for acute renal rejection]

ICD-10 codes not covered for indications listed in the CPB:

T86.10 - T86.19	Complications of kidney transplant [not covered as a diagnostic marker for acute renal rejection]
Z94.0	Kidney transplant status[not covered as a diagnostic marker for acute renal rejection]

***Urinary monocyte chemoattractant protein-1 (MCP-1/CCL2)*:**

CPT codes not covered for indications listed in the CPB:

83520	Immunoassay for analyte other than infectious agent antibody or infectious agent antigen; quantitative, not otherwise specified [urinary monocyte chemoattractant protein-1 (MCP-1/CCL2) for the detection and monitoring of renal graft rejection]

ICD-10 codes not covered for indications listed in the CPB:

T86.10 - T86.19	Complications of kidney transplant [not covered for the detection and monitoring of renal graft rejection]

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Code	Code Description
Z94.0	Kidney transplant status [not covered for the detection and monitoring of renal graft rejection]

***Donor-Derived Cell-Free DNA Testing (e.g., Allosure, Prospera)*:**

CPT codes not covered for indications listed in the CPB:

***Donor-Derived Cell-Free DNA Testing (e.g., Allosure)* - no specific code :**

0493U	Transplantation medicine, quantification of donor-derived cell-free DNA (cfDNA) using next-generation sequencing, plasma, reported as percentage of donor-derived cell-free DNA
0508U	Transplantation medicine, quantification of donor-derived cell-free DNA using 40 single- nucleotide polymorphisms (SNPs), plasma, and urine, initial evaluation reported as percentage of donor-derived cell- free DNA with risk for active rejection
0509U	Transplantation medicine, quantification of donor-derived cell-free DNA using up to 12 single-nucleotide polymorphisms (SNPs) previously identified, plasma, reported as percentage of donor-derived cell-free DNA with risk for active rejection
0540U	Transplantation medicine, quantification of donor-derived cell-free DNA using next-generation sequencing analysis of plasma, reported as percentage of donor-derived cell-free DNA to determine probability of rejection
0544U	Nephrology (transplant monitoring), 48 variants by digital PCR, using cell-free DNA from plasma, donor-derived cell-free DNA, percentage reported as risk for rejection

ICD-10 codes not covered for indications listed in the CPB:

T86.10 - T86.19	Complications of kidney transplant
Z79.620	Long term (current) use of immunosuppressive biologic [monoclonal antibodies]
Z79.621	Long term (current) use of calcineurin inhibitor
Z79.623	Long term (current) use of mammalian target of rapamycin (mTOR) inhibitor
Z79.624	Long term (current) use of inhibitors of nucleotide synthesis
Z94.0	Kidney transplant status

***Measurement of angiotensin II type 1 (AT1) receptors or AT1 antibodies*:**

CPT codes not covered for indications listed in the CPB:

***Measurement of angiotensin II type 1 (AT1) receptors or AT1 antibodies* - no specific code:**

ICD-10 codes not covered for indications listed in the CPB:

Z94.0	Kidney transplant status

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Code	Code Description

***Complement inhibitors*:**

HCPCS codes not covered for indications listed in the CPB:

***Pexelizumab* - no specific code :**

J0596	Injection, c1 esterase inhibitor (recombinant), ruconest, 10 units
J0597	Injection, c-1 esterase inhibitor (human), berinert, 10 units
J0598	Injection, c-1 esterase inhibitor (human), cinryze, 10 units
J0599	Injection, c-1 esterase inhibitor (human), (haegarda), 10 units
J1299	Injection, eculizumab, 2 mg

***Belimumab*:**

HCPCS codes not covered for indications listed in the CPB:

J0490	Injection, belimumab, 10 mg

ICD-10 codes not covered for indications listed in the CPB:

T86.10 - T86.19	Complications of kidney transplant
Z94.0	Kidney transplant status

***Recombinant Human Erythropoietin for nephron-protection in persons undergoing kidney transplant*:**

HCPCS codes not covered for indications listed in the CPB:

J0885	Injection, epoetin alfa, (for non-ESRD use), 1000 units
J0888	Injection, epoetin beta, 1 mcg, (for non-ESRD use)
Q5106	Injection, epoetin alfa-epbx, biosimilar, (Retacrit) (for non-ESRD use), 1000 units
S9537	Home therapy; hematopoietic hormone injection therapy (e.g., erythropoietin, G-CSF, GM-CSF); administrative services, professional pharmacy services, care coordination, and all necessary supplies and equipment (drugs and nursing visits coded separately), per diem

ICD-10 codes not covered for indications listed in the CPB:

Z94.0	Kidney transplant status [not covered for nephron-protection in persons undergoing kidney transplant]

***TruGraf blood gene expression test*:**

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Code	Code Description

CPT codes not covered for indications listed in the CPB:

81558	Transplantation medicine (allograft rejection, kidney), mRNA, gene expression profiling by quantitative polymerase chain reaction (qPCR) of 139 genes, utilizing whole blood, algorithm reported as a binary categorization as transplant excellence, which indicates immune quiescence, or not transplant excellence, indicating subclinical rejection

ICD-10 codes not covered for indications listed in the CPB:

Z94.0	Kidney transplant status

Plasminogen Activator Evaluation:

CPT codes not covered for indications listed in the CPB:

85415	Fibrinolytic factors and inhibitors; plasminogen activator

ICD-10 codes not covered for indications listed in the CPB:

Z76.82	Awaiting organ transplant status [Kidney transplant]

Urinary Biomarkers :

CPT codes not covered for indications listed in the CPB:

0526U	Nephrology (renal transplant), quantification of CXCL10 chemokines, flow cytometry, urine, reported as pg/mL creatinine baseline and monitoring over time

ICD-10 codes not covered for indications listed in the CPB:

T86.10 - T86.19	Complications of transplanted kidney [not covered to the biomarkers for renal allograft rejection]

MyOLARIS-KTdx test:

CPT codes not covered for indications listed in the CPB:

0542U	Nephrology (renal transplant), urine, nuclear magnetic resonance (NMR) spectroscopy measurement of 84 urinary metabolites, combined with patient data, quantification of BK virus (human polyomavirus 1) using real-time PCR and serum creatinine, algorithm reported as a probability score for allograft injury status

ICD-10 codes not covered for indications listed in the CPB:

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Code	Code Description
T86.10 – T86.19	Complications of kidney transplant

Background

Chronic renal failure (CRF) occurs in approximately 2 out of 10,000 people. It results in the accumulation of fluid and waste products in the body, causing azotemia and uremia. Azotemia is the build-up of nitrogen waste products in the blood. It may occur without symptoms. Uremia is the state of ill health resulting from renal failure since most body systems are affected by CRF. Treatment of the underlying disorders may help prevent or delay development of CRF.

Chronic renal failure is slowly progressive over a number of years and most often results from any disease that causes gradual destruction of the internal structures of the kidneys. It can range from mild dysfunction to severe kidney failure, termed end stage renal disease (ESRD). In the early stages, there may be no symptoms. In fact, progression may be so gradual that symptoms do not occur until kidney function is less than 1/10 of normal. Because of the reversible nature of acute renal failure, all patients with this diagnosis should be supported with dialysis, at least for some period of time, to allow return of renal function.

The 3 diseases most commonly leading to CRF and treated by kidney transplantation are

I. type 1 diabetes mellitus,
II. glomerulonephritis, and
III. hypertensive nephrosclerosis, accounting for about 75 % of the total candidate population.

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Numerous subsets of patients in several study populations have shown that patients have a better survival if they receive a renal transplant than if they remain on dialysis therapy.

Patients with ESRD have 3 options for renal replacement therapy:

I. hemodialysis;
II. chronic ambulatory peritoneal dialysis; or
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The choice should be based on the relative risks and benefits. With the increasing appreciation that transplantation results are superior to those of chronic dialysis, the indications for transplantation have been broadened. Improvements in peri-operative care and immunosuppression have allowed many patients who would previously have been denied transplantation consideration as acceptable candidates. The best recipients for transplantation are young individuals whose renal failure is not due to a systemic disease that will damage the transplanted kidney or cause death from extra-renal causes.

The time a patient has spent on dialysis is an independent predictor of a poorer outcome from renal transplantation. Pre-emptive renal transplantation generally leads to better outcomes than transplantation after dialysis is initiated, and should be pursued in most cases for live donor transplants. The current

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shortage of cadaveric kidneys makes it unlikely that pre-emptive transplants will be a practical option for recipients of cadaveric kidney transplants.

No specific cause of intrinsic and irreversible renal failure is considered a contraindication to kidney transplantation. Nonetheless, all patients still should have reversible causes of renal dysfunction excluded before considering renal replacement therapy e.g., obstructive nephropathy has to be removed, chronic pyelonephritis secondary to recurrent infection has to be adequately treated, and reflux has to be fixed.

The evaluation of all transplant candidates, in addition to a standard medical work-up, should include cytomegalovirus (CMV) antibody titer; creatinine clearance; serology for syphilis, and hepatitis B (HBV) and C (HCV) viruses; evaluation of parathyroid status; coagulation profile; Pap smear; ABO and histocompatibility typing; urologic evaluation (including a voiding cystourethrogram in selected patients to assess outlet obstruction and reflux); gastro-intestinal evaluation (as warranted by history of ulcer, diverticulitis, or other symptoms); and psychosocial evaluation.

Patients with renal failure induced by diabetes (Kimmelstiel-Wilson disease) make up the greatest population of patients currently referred for transplantation. Actually, this has become the treatment of choice because persons with diabetes clearly do better with transplantation than with dialysis. In fact, both graft and patient survival for 1 to 2 years are reported to be as good in persons with diabetes as in other patients, whereas on chronic dialysis, less than 20 % of persons with diabetes survive 5 years. If diabetic patients can undergo transplantation before extensive damage occurs in other organs, such as the eye and heart, rehabilitation will be more satisfactory. Even patients with diseases in which the transplanted kidney may eventually be damaged by recurrent disease (e.g., lupus erythematosus, cystinosis, and amyloidosis) are often better palliated by transplantation than by dialysis. Indeed, the current results of transplantation mandate serious consideration of this therapy in virtually any patient with terminal renal disease. Not only is the quality of life far better with transplantation than with dialysis, but because the mortality of patients in the first year after transplantation is now less than 5 %, survival is also superior.

Careful attention must be given to eradication of all infections including those of the urinary tract, lungs, teeth, and skin. Since cardiovascular complications are as common as infection as a cause of post-transplantation mortality, the patient's cardiovascular status should be carefully evaluated and optimized. In older patients and diabetic patients, this might require stress testing, cardiac catheterization, or even pre-transplant coronary artery bypass. Age is never an absolute contraindication for kidney transplantation. Although infants have had successful transplantations, most centers maintain infants on dialysis until body size is increased to 10 to 20 kg. Older patients are becoming more numerous in transplant clinics. Older age (greater than 65 years) never precludes transplantation, but it increases the risk of complications. Transplant centers usually encourage older patients who have multiple medical problems (rather than isolated kidney failure) to remain on dialysis. On both ends of the age spectrum, however, transplantation is becoming more common. Malignancy is considered a contraindication for kidney transplantation, as is severe atherosclerotic or pulmonary disease. Patients with active liver disease are also usually excluded. Both hepatitis B and C can result in eventual liver failure in some patients after transplantation.

The proper timing of transplantation is a delicate decision because the progression of renal dysfunction is variable and premature imposition of the risks of transplantation is not justified. However, dialysis or transplantation should not be withheld until advanced uremic symptoms, such as pericarditis, cardiac failure, severe anemia, osteodystrophy and neuropathy, ensure because these complications may become irreversible.

There are 3 sources of donor kidneys for kidney transplantation:

I. living related donors;
II. cadaver donors; and
III. living unrelated donors. A donor left kidney is usually transplanted to the right iliac fossa, with the renal artery anastomosed end-to-end to the hypogastric artery, and the renal vein end-to-end to the common iliac vein.

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The ureter is implanted into the bladder and under special conditions a uretero-ureteral anastomosis or uretero-pyelostomy may be performed. Auto-transplantation has developed as an outgrowth of the technique used in renal transplantation. The simultaneous development of an apparatus that could preserve kidneys extracorporeally for long periods of time and of preservation solutions led to extracorporeal renal repair (work-bench surgery) and subsequent auto-transplantation for conditions mentioned above.

On rare occasions, kidneys with lesions of the renal artery or its branches are not amenable to in-situ reconstruction. In these circumstances, temporary removal of the kidney, ex-vivo preservation, microvascular repair (work-bench surgery), and auto-transplantation may permit salvage.

Some examples of clinical conditions where the renal artery or its branches are not amenable to in-situ reconstruction such that a person might benefit from auto-transplantation and/or ex-vivo repair include but are not limited to:

- Abdominal aortic aneurysms that involve the origin of the renal arteries; or
- Disease of the major vessels extends beyond the bifurcation of the main renal artery into the segmental branches; or
- Extensive atheromatous aortic disease when an operation on the aorta itself may prove hazardous; or
- Multiple vessels supplying the affected kidney are involved; or
- Persons who have large aneurysms, arteriovenous fistulas, or malformations of the kidney; or
- Traumatic arterial injuries.

Patients with chronic kidney disease have significant abnormalities of bone remodeling and mineral homeostasis and are at increased risk of fracture. The fracture risk for kidney transplant recipients is 4 times that of the general population and higher than for patients on dialysis. Ebeling (2007) noted that organ transplant candidates should be assessed and pre-transplantation bone disease should be treated. Preventive therapy initiated in the immediate post-transplantation period is indicated in patients with osteopenia or osteoporosis, as further bone loss will occur in the first several months following transplantation. Long-term organ transplant recipients should also have bone mass measurement and treatment of osteoporosis. Bisphosphonates are the most promising approach for the management of transplantation osteoporosis. Active vitamin D metabolites may have additional benefits in reducing hyper-parathyroidism, particularly after kidney transplantation. The author stated that large, multi-center treatment trials with oral or parenteral bisphosphonates and calcitriol are recommended.

In a Cochrane review, Palmer et al (2007) assessed the use of interventions for treating bone disease following kidney transplantation. Randomized controlled trials (RCTs) and quasi-RCTs comparing different treatments for kidney transplant recipients of any age were selected. All other transplant recipients, including kidney-pancreas transplant recipients were excluded. Two authors independently evaluated trial quality and extracted data. Statistical analyses were performed using the random effects model and the results expressed as relative risk (RR) with 95 % confidence intervals (CI) for dichotomous variables and mean difference (MD) for continuous outcomes. A total of 24 trials (n = 1,299) were included. No individual intervention (bisphosphonates, vitamin D sterol or calcitonin) was associated with a reduction in fracture risk compared with placebo. Combining results for all active interventions against placebo demonstrated any treatment of bone disease was associated with a reduction in the RR of fracture (RR 0.51, 95 % CI: 0.27 to 0.99). Bisphosphonates (any route), vitamin D sterol, and calcitonin all had a beneficial effect on the bone mineral density (BMD) at the lumbar spine. Bisphosphonates and vitamin D sterol also had a beneficial effect on the BMD at the femoral neck. Bisphosphonates were more effective in preventing BMD loss when compared head-to-head with vitamin D sterols. Few or no data were available for combined hormone replacement, testosterone, selective estrogen receptor modulators, fluoride or anabolic steroids. Other outcomes including all-cause mortality and drug-related toxicity were reported infrequently. The authors concluded that treatment with bisphosphonates, vitamin D sterol or calcitonin after kidney transplantation may protect

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against immunosuppression-induced reductions in BMD and prevent fracture. However, they state that adequately powered clinical studies are needed to ascertain if bisphosphonates are better than vitamin D sterols for fracture prevention in this population. Moreover, the optimal route, timing, and duration of administration of these interventions remains unknown.

Acute rejection is an immune process that begins with the recognition of the allograft as non-self and ends in graft destruction. Histological features of the allograft biopsy are currently used for the differential diagnosis of allograft dysfunction. In view of the safety and the opportunity for repetitive sampling, development of non-invasive biomarkers of allograft status is an important objective in transplantation. Khatri and Sarwal (2009) stated that in the past 10 years, microarray technology has revolutionized biological research by allowing the screening of tens of thousands of genes simultaneously. These investigators reviewed recent studies in organ transplantation using microarrays and highlighted the issues that should be addressed in order to use microarrays in the diagnosis of rejection. Microarrays have been useful in identifying potential biomarkers for chronic rejection in peripheral blood mononuclear cells, novel pathways for induction of tolerance, and genes involved in protecting the graft from the host immune system. Microarray analysis of peripheral blood mononuclear cells from chronic antibody-mediated rejection has identified potential non-invasive biomarkers. In a recent study, correlation of pathogenesis-based transcripts with histopathological lesions is a promising step towards inclusion of microarrays in clinics for organ transplants. The authors concluded that despite promising results in diagnosis of histopathological lesions using microarrays, the low dynamic range of microarrays and large measured expression changes within the probes for the same gene continue to cast doubts on their readiness for diagnosis of rejection. They stated that more studies are needed to resolve these issues. Dominating expression of globin genes in whole blood poses another challenge for identification of non-invasive biomarkers. In addition, studies are also needed to demonstrate effects of different immunosuppression therapies and their outcomes.

Hartono et al (2010) noted that urinary cell and peripheral blood cell mRNA profiles have been associated with acute rejection of human renal allografts. Emerging data support the idea that development of non-invasive biomarkers predictive of antibody-mediated rejection is feasible. The demonstration that intra-graft microRNA expression predicts renal allograft status suggests that non-invasively ascertained microRNA profiles may be of value. These researchers stated that they are pleased with the progress to date, and anticipate clinical trials investigating the hypotheses that non-invasively ascertained mRNA profiles will minimize the need for invasive biopsy procedures, predict the development of acute rejection and chronic allograft nephropathy, facilitate preemptive therapy capable of preserving graft function, and facilitate personalization of immunosuppressive therapy for the allograft recipient.

Mihovilovic and colleagues (2010) evaluated urine immuno-cytology for T cells as a method for non-invasive identification of patients with acute renal allograft rejection in comparison to renal biopsy. In this prospective study, a cohort of 56 kidney, or kidney-pancreas transplant recipients was included. Patients either received their transplant at the University Hospital "Merkur", or have been followed at the "Merkur" Hospital. Patients were subject to either protocol or indication kidney biopsy (a total of 70 biopsies), with simultaneous urine immunocytology (determination of CD3-positive cells in the urine sediment). Acute rejection was diagnosed in 24 biopsies; 23 episodes were T-cell mediated (6 grade IA, 5 grade IB, 1 grade IIA, 1 grade III and 10 borderline), while in 1 case acute humoral rejection was diagnosed. A total of 46 biopsies did not demonstrate acute rejection. CD3-positive cells were found in 21 % of cases with acute rejection and in 13 % of cases without rejection (non-significant). A finding of CD3-positive cells in urine had a sensitivity of 21 % and specificity of 87 % for acute rejection (including borderline), with positive predictive value of 45 % and negative predictive value of 68 %. The authors concluded that although tubulitis is a hallmark of acute T cell-mediated rejection, detection of T cells in urine sediment was insufficiently sensitive and insufficiently specific for diagnosing acute rejection in this cohort of kidney transplant recipients.

Belatacept, a selective T-cell co-stimulation blocker, is a cytotoxic T-lymphocyte-associated antigen 4-immunoglobulin. It is designed to block CD28, a critical activating receptor on T cells, by binding and saturating its ligands B7-1 and B7-2. In phase II and III clinical trials, belatacept was compared with cyclosporine (in combination with basiliximab, mycophenolate mofetil, and steroids). Advantages observed with belatacept include superior graft function, preservation of renal structure and improved cardiovascular

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risk profile. Concerns associated with belatacept are a higher frequency of cellular rejection episodes and more post-transplant lymphoproliferative disorder (PTLD) cases especially in Epstein-Barr virus (EBV) sero-negative patients, who should be excluded from belatacept-based regimens (Wekerle and Grinyo, 2012).

On June 15, 2011, the Food and Drug Administration approved belatacept (Nulojix) for the prevention of acute rejection in adult kidney transplant recipient. Nulojix is approved for use with other immunosuppressants, specifically basiliximab, corticosteroids, and mycophenolate mofetil. The approval of Nulojix was base on 2 open-label, randomized, multi-center, controlled phase III clinical trials that enrolled more than 1,200 patients and compared 2 dose regimens of Nulojix with another immunosuppressant, cyclosporine. These trials demonstrated that the recommended Nulojix regimen is safe and effective for the prevention of acute organ rejection.

Nulojix carries a Boxed Warning for an increased risk of developing PTLD. The risk of PTLD is higher for transplant patients who have never been exposed to EBV. Transplant patients who have not been exposed to EBV have more difficulty mounting an effective immune response to the virus if they get infected after transplant; typically they get exposed to the virus at time of transplant, as it is carried in around 80 % of donated organs. Patients should be tested for EBV and should only receive Nulojix if the test shows they have already been exposed to EBV. Another Boxed Warning on the Nulojix label, as well as labels of other immunosuppressants, warns of an increased risk of serious infections and other cancers. Common adverse reactions observed in transplant patients in the trials included anemia, constipation, kidney or bladder infection, and swollen legs, ankles, or feet. Any transplant patients, including those receiving Nulojix, should limit the amount of time spent in sunlight because of the risk of skin cancer and should not get live vaccines because of the risk of infection.

Chen and colleagues (2012) stated that the question of whether high pre-transplantation soluble CD30 (sCD30) level can be a predictor of kidney transplant acute rejection (AR) is under debate. These investigators performed a meta-analysis on the predictive efficacy of sCD30 for AR in renal transplantation. PubMed (1966 to 2012), EMBASE (1988 to 2012), and Web of Science (1986 to 2012) databases were searched for studies concerning the predictive efficacy of sCD30 for AR after kidney transplantation. After a careful review of eligible studies, sensitivity, specificity, and other measures of the accuracy of sCD30 were pooled. A summary receiver operating characteristic curve was used to represent the overall test performance. A total of 12 studies enrolling 2,507 patients met the inclusion criteria. The pooled estimates for pre-transplantation sCD30 in prediction of allograft rejection risk were poor, with a sensitivity of 0.70 (95 % CI: 0.66 to 0.74), a specificity of 0.48 (95 % CI: 0.46 to 0.50), a positive likelihood ratio of 1.35 (95 % CI: 1.20 to 1.53), a negative likelihood ratio of 0.68 (95 % CI: 0.55 to 0.84), and a diagnostic odds ratio of 2.07 (95 % CI: 1.54 to 2.80). The area under curve of the summary receiver operating characteristic curve was 0.60, indicating poor overall accuracy of the serum sCD30 level in the prediction of patients at risk for AR. The authors concluded that the results of the meta-analysis showed that the accuracy of pre-transplantation sCD30 for predicting post-transplantation AR was poor. They stated that prospective studies are needed to clarify the usefulness of this test for identifying risks of AR in transplant recipients.

Lv and colleagues (2012) noted that results from published studies on the association of donor or recipient interleukin (IL)-6 -174G/C (rs1800795) polymorphism with AR of renal allograft are conflicting. These investigators performed a meta-analysis to estimate the possible association. Studies were identified by searching PubMed and Embase until July 1, 2011. Meta-analysis was performed in a fixed/random effects model using Revman 5.0.25 and STATA10.0. A total of 7 studies addressing the association between donor high producer genotype (G/G and G/C) of IL-6 -174G/C polymorphism and AR of renal allograft were identified. Pooled odds ratio (OR) based on 341 cases (whose recipient developed AR) and 702 controls (whose recipient did not develop AR) was 0.59 (95 % CI: 0.26 to 1.33; p = 0.20), with a strong between-study heterogeneity. No association was observed in the subgroup analysis based on ethnicity. A total of 13 studies evaluating the association between recipient IL-6 -174G/C polymorphism and AR were identified. Pooled OR based on 451 cases (patients did not develop AR) and 848 controls was 1.00 (95 % CI: 0.72 to 1.37; p = 0.98), with a weak between-study heterogeneity. The authors concluded that donor high producer genotype (G/G and G/C) of IL-6 -174G/C polymorphism had a tendency of decreased risk for AR, although it was not statistically significant. Recipient high producer genotype was not associated with AR of renal allograft. Moreover, they stated that additional well-designed studies with larger sample size are needed to

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support these findings, especially for the association between donor high producer genotype (G/G and G/C) of IL-6 -174G/C polymorphism and acute renal allograft rejection.

In an observational cross-sectional study, De Serres et al (2012) determined the utility of a non-invasive cytokine assay in screening of AR. A total of 64 patients from 2 centers were recruited upon admission for allograft biopsy to investigate acute graft dysfunction. Blood was collected before biopsy and assayed for a panel of 21 cytokines secreted by peripheral blood mononucleated cells (PBMCs). Patients were classified as acute rejectors or non-rejectors according to a classification rule derived from an initial set of 32 patients (training cohort) and subsequently validated in the remaining patients (validation cohort). Although 6 cytokines (interleukin-1 beta [IL-1β], IL-6, tumor necrosis factor-alpha [TNF-α], IL-4, granulocyte-macrophage colony-stimulating factor [GM-CSF], and monocyte chemoattractant protein-1 [MCP-1]) distinguished acute rejectors in the training cohort, logistic regression modeling identified a single cytokine, IL-6, as the best predictor. In the validation cohort, IL-6 was consistently the most accurate cytokine (area under the receiver-operating characteristic curve, 0.85; $p = 0.006$), whereas the application of a pre-specified cut-off level, as determined from the training cohort, resulted in a sensitivity and specificity of 92 % and 63 %, respectively. Secondary analyses revealed a strong association between IL-6 levels and AR after multi-variate adjustment for clinical characteristics ($p < 0.001$). The authors concluded that in this pilot study, the measurement of a single cytokine can exclude AR with a sensitivity of 92 % in renal transplant recipients presenting with acute graft dysfunction. Moreover, they stated that prospective studies are needed to determine the utility of this simple assay, particularly for low-risk or remote patients.

Wu and associates (2013) stated that cytokines have been implicated in the AR of solid organ transplantation. Many studies have investigated the association between recipient or donor IL-4 polymorphism and AR, with different studies reporting inconclusive results. These investigators searched PubMed and Embase until June 2012 to identify eligible studies investigating the association between IL-4 polymorphism with AR after solid organ transplantation. Statistical analysis was performed using STATA10.0. A total of 12 studies were included. Pooled ORs suggested

I. no significant association was detected between recipient or donor IL-4 -590C/T polymorphism and acute rejection of solid allograft;
II. no significant association was detected between recipient IL-4 -33C/T polymorphism and AR of solid allograft;
III. when stratified by transplantation type, IL-4 -590C/T polymorphism was associated with AR of liver transplantation (T/T+C/T versus C/C: OR = 0.36, 95 % CI: 0.14 to 0.90); and
IV. significantly decreased risk of AR was detected in recipient IL-4 -590multiply*T- negative/donor T-positive genotype pairs than all other recipient-donor IL-4 -590T/C pairs (OR = 0.14, 95 % CI: 0.03 to 0.66).

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The authors concluded that the findings of this meta-analysis suggested that recipient IL-4 -590C/T polymorphism was associated with AR of liver transplantation, but nor renal or heart transplantation. It was also suggested that combined recipient IL-4 -590multiply*T- negative/donor T-positive genotype may suffer decreased risk of AR of solid allograft. Moreover, they stated that further well-designed studies with larger sample size were needed to verify these findings, with focus on the association of IL-4 polymorphism with AR in patients with liver transplantation and studies investigating combined recipient-donor genotype.

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An UpToDate review on “Clinical manifestations and diagnosis of acute renal allograft rejection” (Chon and Brenna, 2014a) does not mention measurement of cytokines as a management tool.

Furthermore, an UpToDate review on “Investigational methods in the diagnosis of acute renal allograft rejection” (Chon and Brenna, 2014b) states that “The introduction of potent immunosuppressive drugs in the past three decades has led to a dramatic reduction in the incidence of acute rejection in kidney transplant recipients. At the present time, renal allograft biopsy with conventional histologic evaluation remains the gold standard for diagnosing acute rejection among patients with a deterioration in kidney function as detected by measuring serum creatinine levels. However, the lack of additional markers of rejection makes it difficult to optimize anti-rejection therapy for transplant recipients. The evaluation of methods other than conventional renal biopsy and/or measurement of the serum creatinine to help diagnosis acute kidney rejection has been the focus of a large number of investigators. This topic review will discuss some of the methods undergoing investigation for the diagnosis of acute rejection …. Measuring the levels of urinary or circulating proteins and cytokines, circulating soluble interleukin-2 (IL-2) receptor, the urinary concentration of soluble adhesion molecules, or cellular activation with urinary flow cytometry may be helpful in diagnosing acute allograft rejection …. Interleukin-6 (IL-6) may be a potential biomarker for acute rejection. In an observational study of 32 patients who presented with acute graft dysfunction, of six tested cytokines (including IL-1beta, IL-6, TNF-alpha, IL-4, GM-CSF, and MCP-1), IL-6 best predicted acute rejection. Among a validation cohort of 32 additional patients, using a prespecified IL-6 cutoff level of 85 pg/ml, IL-6 assay had a sensitivity of 92 and specificity of 63 percent for the diagnosis of acute rejection. These results need to be confirmed in a larger prospective trial”.

Kim et al (2014) stated that antibody-mediated rejection (AMR), also known as B-cell-mediated or humoral rejection, is a significant complication after kidney transplantation that carries a poor prognosis. Although fewer than 10 % of kidney transplant patients experience AMR, as many as 30 % of these patients experience graft loss as a consequence. Although AMR is mediated by antibodies against an allograft and results in histologic changes in allograft vasculature that differ from cellular rejection, it has not been recognized as a separate disease process until recently. With an improved understanding about the importance of the development of antibodies against allografts as well as complement activation, significant advances have occurred in the treatment of AMR. The standard of care for AMR includes plasmapheresis and intravenous immunoglobulin that remove and neutralize antibodies, respectively. Agents targeting B cells (rituximab and alemtuzumab), plasma cells (bortezomib), and the complement system (eculizumab) have also been used successfully to treat AMR in kidney transplant recipients. However, the high cost of these medications, their use for unlabeled indications, and a lack of prospective studies evaluating their safety and effectiveness limit the routine use of these agents in the treatment of AMR in kidney transplant recipients.

Gupta et al (2014) stated that although several strategies for treating early AMR in kidney transplants have been investigated, evidence on treatment of late AMR manifesting after 6 months is sparse. In this single-center series, these researchers presented data on 23 consecutive patients treated for late AMR. Late AMR was diagnosed using Banff 2007 criteria along with presence of donor-specific antibodies (DSA) and acute rise in serum creatinine (SCr). Response to therapy was assessed by improvement in SCr, histologic improvement, and decline in DSA strength. Overall, 17 % (4/23) had documented non-adherence while 69 % (16/23) had physician-recommended reduction in immunosuppression before AMR. Eighteen patients (78 %) were treated with plasmapheresis or low-dose IVIG + rituximab; 11 (49 %) with refractory AMR also received 1 to 3 cycles of bortezomib. While there was an improvement ($p = 0.02$) in mean SCr (2.4 mg/dL) at the end of therapy compared with SCr at the time of diagnosis (2.9 mg/dL), this improvement was not sustained at most recent follow-up. Eleven (48 %) patients had no histologic resolution on follow-up biopsy. Lack of histologic response was associated with older patients (OR = 3.17; $p = 0.04$), presence of cytotoxic DSA at time of diagnosis (OR = 200; $p = 0.04$), and severe chronic vasculopathy (cv greater than or equal to 2) on index biopsy (OR = 50; $p = 0.06$). The authors concluded that a major setting in which late AMR occurred in this cohort was reduction or change in immunosuppression. They stated that these data demonstrated an inadequate response of late AMR to current and novel (bortezomib) therapies. They stated that the benefits of therapy need to be counter-weighed with potential adverse effects especially in older patients, large antibody loads, and chronic allograft vasculopathy.

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Eskandary et al (2014) noted that despite major advances in transplant medicine, improvements in long-term kidney allograft survival have not been commensurate with those observed shortly after transplantation. The formation of DSA and ongoing AMR processes may critically contribute to late graft loss. However, appropriate treatment for late AMR has not yet been defined. There is accumulating evidence that bortezomib may substantially affect the function and integrity of alloantibody-secreting plasma cells. The impact of this agent on the course of late AMR has not so far been systematically investigated. The BORTEJECT Study is a RCT designed to clarify the impact of intravenous bortezomib on the course of late AMR. In this single-center study (nephrological outpatient service, Medical University Vienna) these researchers plan an initial cross-sectional DSA screening of 1,000 kidney transplant recipients (functioning graft at greater than or equal to 180 days; estimated glomerular filtration rate (eGFR) greater than 20 ml/min/1.73 m2). DSA-positive recipients will be subjected to kidney allograft biopsy to detect morphological features consistent with AMR. Forty-four patients with biopsy-proven AMR will then be included in a double-blind placebo-controlled intervention trial (1:1 randomization stratified for eGFR and the presence of T-cell-mediated rejection). Patients in the active group will receive 2 cycles of bortezomib (4 × 1.3 mg/m2 over 2 weeks; 3-month interval between cycles). The primary end-point will be the course of eGFR over 24 months (intention-to-treat analysis). The sample size was calculated according to the assumption of a 5 ml/min/1.73 m2 difference in eGFR slope (per year) between the 2 groups (alpha: 0.05; power: 0.8). Secondary end-points will be DSA levels, protein excretion, measured glomerular filtration rate, transplant and patient survival, and the development of acute and chronic morphological lesions in 24-month protocol biopsies. The authors concluded that the impact of anti-humoral treatment on the course of late AMR has not yet been systematically investigated. Based on the hypothesis that proteasome inhibition improves the outcome of DSA-positive late AMR, these investigators suggested that their trial has the potential to provide solid evidence towards the treatment of this type of rejection.

Hou et al (2014) noted that the human leukocyte antigen-G may have a positive role in graft acceptance in human organ transplant. Several studies have reported an association between the human leukocyte antigen-G-14-base-pair-insertion/deletion polymorphism and risk of developing kidney graft rejection, but the results are inconclusive. These researchers performed a meta-analysis to evaluate this association. They included 5 case-control studies that evaluated the association between human leukocyte antigen-G-14-base-pair-insertion/deletion polymorphism and risk of developing kidney transplant rejection, including a total 907 patients (rejection, 271 patients; no rejection, 636 patients). There was no significant association between the human leukocyte antigen-G-14-basepair-insertion/deletion polymorphism and risk of developing kidney transplant rejection in the allele contrast, homozygous, heterozygous, recessive, or dominant genetic models for all rejection or acute rejection. In 2 studies, there was a significant association between human leukocyte antigen-G-14-base-pair-insertion/deletion polymorphism and chronic graft rejection in the allele contrast model (+14 versus -14: OR, 0.68; 95 % CI: 0.48 to 0.96; p = 0.618), heterozygous model (+14/-14 versus -14/-14: OR, 0.44; 95 % CI: 0.23 to 0.83; p = 0.248), and dominant genetic model ([+14/+14 and +14/-14] versus -14/-14: OR, 0.48; 95 % CI: 0.30 to 0.78; p = 0.355). The authors concluded that there may be no association between 14-base-pair polymorphisms and risk of developing kidney allograft rejection. They stated that additional studies with larger sample size and better study design are justified.

Atgam

Atgam (anti-thymocyte globulin equine) is lymphocyte-selective immunosuppressant. It is prepared from purified concentrated sterile gamma globulin, primarily monomeric IgG, from the serum of hyper-immune horses immunized with human thymus lymphocytes.

Anti-thymocyte globulin equine acts to reduce the number of circulating, thymus dependent lymphocytes. Subsequently, this is believed to alter the function of Tlymphocytes involved in humoral immunity that are responsible for cell-mediated immunity.

Atgam (anti-thymocyte immune globulin, equine) has been FDA-approved in adults for moderate to severe aplastic anemia unsuitable for bone marrow transplant, renal allograft transplant rejection, and renal allograft transplant rejection prophylaxis.

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Atgam (anti-thymocyte immune globulin, equine) has shown efficacy in the treatment of aplastic anemia and allograft renal transplant rejection/prophylaxis. These two disease states represent conditions in which a hyperactive immune state is detrimental to patient outcomes. In renal transplant rejection, 3-year overall graft survival rates have been shown to be significantly higher in the anti-thymocyte immune globulin treated populations when compared to standard therapy. For the prophylaxis of renal transplant rejection, those patients who received anti-thymocyte immune globulin in addition to standard therapy experienced had longer delays in 14 and 28 day first rejection episodes at rates of 25% vs. 65% and 45% vs. 85%, respectively. In aplastic anemia, a review of several trials found that though patients respond in 54% of the cases, the response is evident in 98% of those patients as early as 6 months with the majority within 3 months. Only 3% of responders experienced a hematologic relapse. Response was defined as a rise in neutrophil and reticulocyte counts.

Each lot of anti-thymocyte globulin equine is tested by the manufacturer to assure its ability to inhibit rosette formation between human peripheral lymphocytes and sheep red blood cells in vitro before release. Antibody activity is measured against human red blood cells and platelets and determined to be within acceptable limits. Only lots that test negative for antihuman serum protein antibody, anti-glomerular basement membrane antibody and pyrogens are released.

A Medicare National Coverage Determination states that the FDA has approved lymphocyte immune globulin, anti-thymocyte globulin (equine), for the management of allograft rejection episodes in renal transplantation. The Centers for Medicare and Medicaid Services has stated that these biologics are viewed as adjunctive to traditional immunosuppressive products such as steroids and anti metabolic drugs. At present, lymphocyte immune globulin preparations are not recommended to replace conventional immunosuppressive drugs, but to supplement them and to be used as alternatives to elevated or accelerated dosing with conventional immunosuppressive agents.

Anti-thymocyte immune globulin, equine is supplied as Atgam 50 mg/ml 5 ml ampules in packages of five.

The recommended dose for aplastic anemia is 10 to 20 mg/kg once daily for 8-14 days, then every other day up to 21 total doses if needed.

Recommended dosing for treatment of renal transplant rejection is 10 to 15 mg/kg IV once daily for 14 days, then every other day for 14 more days if needed up to 21 total doses.

Recommended dosing for prophylaxis of renal transplant rejection is 15 mg/kg IV once daily for 14 days, then every other day for 14 days for a total of 21 doses in 28 days; begin first dose within 24 hours before or after transplantation.

Atgam (anti-thymocyte immune globulin, equine) should not be used in persons with aplastic anemia secondary to neoplastic disease, storage disease, myelofibrosis, or Fanconi's syndrome. Atgam should not be used in persons known to have been exposed to myelotoxic agents or radiation.

Black Box warnings:

- Only physicians experienced in immunosuppressive therapy in the treatment of renal transplant or aplastic anemia patients should use lymphocyte immune globulin, anti-thymocyte globulin equine.
- Patients receiving lymphocyte immune globulin, anti-thymocyte globulin equine should be treated in facilities equipped and staffed with adequate laboratory and supportive medical resources.

Discontinue treatment with Atgam (anti-thymocyte immune globulin, equine) if any of the following occurs:

- Symptoms of anaphylaxis
- Severe and continuous thrombocytopenia in renal transplant members
- Severe and continuous leukopenia in renal transplant members.

To identify those at greatest risk of systemic anaphylaxis, skin testing potential recipients is strongly recommended before commencing treatment. A conservative, conventional approach would first employ epi-

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cutaneous (prick) testing with undiluted ATGAM. If the subject does not show a wheal ten minutes after pricking, proceed to intradermal testing with 0.02 mL of a 1:1000 v/v (volume/volume) saline dilution of ATGAM with a separate saline control injection of similar volume. Read the result at 10 minutes: a wheal at the Atgam site 3 or more mm larger in diameter than that at the saline control site (or a positive prick test) suggests clinical sensitivity and an increased possibility of a systemic allergic reaction should the drug be dosed intravenously.

Interleukin-2 / Interleukin-10 Gene Polymorphisms and Graft Rejection Risk

Hu and colleagues (2015) stated that IL-2-330 T/G promoter polymorphism is involved in the AR risk of kidney transplantation. However, results from published studies on the association between recipient IL-2-330 T/G polymorphism and AR risk are conflicting and inconclusive. These investigators searched Medline, Embase, Web of Science, and Cochrane Central Register from their inceptions through January 2015 for relevant studies. Data concerning publication information, population characteristics, and transplant information were extracted. Odds ratios were calculated for the association between IL-2-330 T/G polymorphism and AR risk. This meta-analysis included 8 case-control studies with 1,405 cases of renal transplant recipients. The pooled estimate showed that IL-2-330 T/G polymorphism was not associated with AR risk: TT versus TG+GG: OR (fixed), 0.93; 95 % CI: 0.72 to 1.21; p = 0.60; GG versus TG+TT: OR (fixed), 1.15; 95 % CI: 0.76 to 1.72; p = 0.51; TG versus TT+GG: OR (fixed), 1.01; 95 % CI: 0.78 to 1.31; p = 0.91; T versus G: OR (fixed), 0.93; 95 % CI: 0.77 to 1.13; p = 0.48. None of subgroup analyses yielded significant results in the association between IL-2-330 T/G polymorphism and AR risk. Meta-regression confirmed that there was no significant correlation between the pre-selected trial characteristics and the study results. The authors concluded that the findings of this meta-analysis suggested that IL-2-330 T/G polymorphism may not be associated with AR risk in renal transplant recipients.

Xiong and co-workers (2015) noted that IL-10 is an important immune-modulatory cytokine. Several studies focused the association between IL-10 promoter gene polymorphisms and graft rejection risk in kidney transplantation recipients. However, the results of these studies remain inconclusive. The se researchers performed a meta-analysis to further examine the associations. PubMed, Embase, and Ovid Medline databases were searched. Two independent authors extracted data, and the effects were estimated from an OR with 95 % CIs. Subgroup and sensitivity analyses identified sources of heterogeneity. A total of 16 studies including 595 rejection patients and 1,239 stable graft patients were included in order to study the IL-10 -1082 (rs1800896 G/A), -819 (rs1800871 C/T), -592 (rs1800872 C/A) and IL-10 (-1082,-819,-592) polymorphisms. The -1082 G/A polymorphism was not associated with an increased graft rejection risk (OR = 1.03; 95 % CI: 0.85 to 1.25, p = 0.74 for GA+AA versus GG model). Moreover, all of the -819 C/T (OR = 1.06, 95 % CI: 0.79 to 1.42, p = 0.70 for TA+TT versus CC model), -592 C/A (OR = 1.10, 95 % CI: 0.85 to 1.42, p = 0.47 for AC+AA versus CC model) and IL-10 (-1082,-819,-592) polymorphisms (OR = 1.00, 95 % CI: 0.79 to 1.27, p = 0.98 for I+L versus H model) did not increase the graft rejection risk. In addition, these investigators also performed subgroup analysis by ethnic group (mainly in Europeans or Asians) and rejection type (acute or chronic). There was also lack of evidence of a significant association between the IL-10 gene polymorphism and graft rejection risk. The present meta-analysis indicated that the IL-10 gene polymorphism was not associated with graft rejection risk in kidney transplantation recipients. The authors concluded that this meta-analysis found evidence that the IL-10 polymorphism does not increase the risk of graft rejection in kidney transplantation recipients. They stated that further chronic rejection and other ethnic population studies are needed to confirm these results.

Hu and associates (2016) examined the association between IL-10-1082 (G/A) promoter polymorphism and AR in renal transplant recipients. These investigators searched MEDLINE, EMBASE, Web of Science, and Cochrane Central Register from the inception to March 2015 for relevant studies. Data concerning publication information, population characteristics, and transplant information were extracted. Odds ratios was calculated for the association between IL-10-1082 GG genotype (or IL-10-1082 G allele) and AR risk. This meta-analysis included 22 case-control studies including 2,779 cases of renal transplant recipients. The pooled estimate showed that the IL-10-1082 GG genotype was not significantly associated with AR risk (OR random = 1.07, 95 % CI: 0.80 to 1.43, p = 0.64). Similarly, the pooled estimate showed that the IL-10-1082 G allele was not significantly associated with AR risk (OR fixed = 1.02, 95 % CI: 0.90 to 1.16, p = 0.74).

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None of subgroup analyses yielded significant results in the association between IL-10-1082 GG genotype (or IL-10-1082 G allele) and AR risk. Meta-regression confirmed that there was no significant correlation between the pre-selected trial characteristics and the study results. The authors concluded that he findings of this meta-analysis suggested that IL-10-1082 G/A polymorphism is not significantly associated with AR risk in renal transplant recipients.

Preconditioning Therapy in ABO-Incompatible Living Kidney Transplantation

Lo and associates (2016) stated that ABO-incompatible (ABOi) kidney transplantation is now an established form of renal replacement therapy, but the safety and effectiveness of the different types of pre-conditioning therapies are unclear. These investigators synthesized the totality of the published evidence about the effects of any form of pre-conditioning therapies in living donor ABOi kidney transplantation on graft and patient outcomes. They searched MEDLINE, Embase, and Clinicaltrial.gov databases (inception through June 2015) to identify all studies that described the outcomes of adult living donor ABOi kidney transplantations using any form of pre-conditioning therapies. Two independent reviewers identified studies, extracted data, and assessed the risk of bias. Data were summarized using the random effects model, and heterogeneity was explored using subgroup analyses. These researchers assessed confidence in the evidence using the Grading of Recommendations Assessment, Development, and Evaluation framework. A total of 83 studies (54 case reports and case series, 25 cohort, 2 case-control, and 2 registry studies) involving 4,810 ABOi transplant recipients were identified. Overall, confidence in the available evidence was low. During a mean follow-up time of 28 (standard deviation [SD], 26.6) months, the overall graft survival for recipients who received immune-adsorption or apheresis was 94.1 % (95 % CI: 88.2 % to 97.1 %) and 88.0 % (95 % CI: 82.6 % to 91.8 %), respectively. For those who received rituximab or underwent splenectomy, the overall graft survival was 94.5 % (95 % CI: 91.6 % to 96.5 %) and 79.7 % (95 % CI: 72.9 % to 85.1 %), respectively. Data on other longer-term outcomes, including malignancy, were sparse. The authors concluded that rituximab or immune-adsorption appeared to be promising pre-conditioning strategies before ABOi kidney transplantation. However, the overall quality of evidence and the confidence in the observed treatment effects were low. The increased use of ABOi kidney transplantation needs to be matched with randomized trials of different types, dosing, and frequency of pre-conditioning therapies so that this scarce resource can be used most effectively and efficiently.

Treatment of Low Bone Mineral Density after Kidney Transplantation

Kan and colleagues (2016) noted that in patients with low BMD after kidney transplantation, the role of bisphosphonates remains unclear. These researchers performed a systematic review and meta-analysis to examine the safety and effectiveness of bisphosphonates. They retrieved trials from PubMed, EMBASE, and the Cochrane Central Register of Controlled Trials (CENTRAL) from inception through May 2015. Only RCTs that compared bisphosphonate-treated and control groups of patients with low BMD after kidney transplantation were included. The primary outcomes were the percent change in BMD, the absolute change in BMD, and the BMD at the end of study at the lumbar spine. The results were expressed as the MD or RR with the 95 % CI. These investigators used a random-effects model to pool the outcomes. They included a total of 17 RCTs with 1,067 patients. Only 1 included trial was found to be at low risk of bias. The rest of the included studies were found to have high to uncertain risk of bias. Compared with the control group, those who received bisphosphonates had a significant increase in percent change in BMD (MD =5.51, 95 % CI: 3.22 to 7.79, $p<0.00001$) and absolute change in BMD (MD=0.05, 95 % CI: 0.04 to 0.05, $p<0.00001$), but a non-significant increase in BMD at the end of the study (MD=0.02, 95 % CI: -0.01 to 0.05, $p=0.25$) at the lumbar spine. Bisphosphonates resulted in a significant improvement in percent change in BMD (MD= 4.95, 95 % CI: 2.57 to 7.33, $p<0.0001$), but a non-significant improvement in absolute change in BMD (MD =0.03, 95 % CI: -0.00 to 0.06, $p=0.07$) and BMD at the end of the study (MD=-0.01, 95 % CI: -0.04 to 0.02, $p=0.40$) at the femoral neck. No significant differences were found in vertebral fractures, non-vertebral fractures, adverse events, and gastro-intestinal adverse events. The authors concluded that bisphosphonates appeared to have a beneficial effect on BMD at the lumbar spine; and did not significantly decrease fracture events in recipients. Moreover, they stated that the results should be interpreted cautiously due to the lack of robustness and the heterogeneity among studies.

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Furthermore, an UpToDate review on “Bone disease after renal transplantation" (Yarlagadda et al, 2016) stated that "Bisphosphonates -- We do not give bisphosphonates to prevent bone loss among renal transplant recipients, because there is a risk that they may worsen low-turnover (i.e., adynamic) bone disease. However, some studies have suggested that bisphosphonates may provide a benefit …. However, despite these reports that suggest a benefit, we do not use bisphosphonates to prevent bone loss. This is because many transplant recipients have low-turnover bone disease, which could be made worse by bisphosphonates. Low-turnover bone disease is a mineral and bone disease of chronic kidney disease (CKD-MBD) that is associated with over-suppression of PTH. An adverse effect of bisphosphonates on low-turnover bone disease was suggested by a study of 72 new renal transplant recipients who were randomly assigned to either pamidronate with calcitriol plus calcium or only calcitriol plus calcium. At 6 months, adynamic bone disease was observed in all patients receiving pamidronate compared with 50 % in the control group. If antiresorptive therapies worsen low-turnover bone disease, they could increase the risk of atypical fractures related to low-turnover bone disease. Thus, the overall risk of fracture could increase even if BMD is improved by bisphosphate therapy. The prevalence of bisphosphonate-associated atypical fractures is not known, but a potential case report has been documented. Among transplant recipients, it is not clear whether antiresorptive therapy has an effect on the overall number of fractures, although trials that have evaluated bisphosphonates were not powered for fracture outcomes”.

Palmer and colleagues (2019) noted that individuals who have CKD have important changes to bone structure, strength, and metabolism. Children experience bone deformity, pain, and delayed or impaired growth. Adults experience limb and vertebral fractures, avascular necrosis (AVN), and pain. The fracture risk after kidney transplantation (KTx) is 4 times that of the general population and is related to CKD-MBD occurring with end-stage kidney failure, steroid-induced bone loss, and persistent hyperparathyroidism following transplantation. Fractures may reduce quality of life (QOL) and lead to being unable to work or contribute to community roles and responsibilities. Earlier versions of this review have found low certainty evidence for effects of treatment. This was an update of a review first published in 2005 and updated in 2007. This review update examined the benefits and harms of interventions for preventing bone disease following KTx. These investigators searched the Cochrane Kidney and Transplant Register of Studies up to May 16, 2019 through contact with the Information Specialist using search terms relevant to this review. Studies in the Register were identified through searches of CENTRAL, Medline, and Embase, conference proceedings, the International Clinical Trials Register (ICTRP) Search Portal and ClinicalTrials.gov; RCTs and quasi-RCTs evaluating treatments for bone disease among KTx recipients of any age were eligible. Two authors independently assessed trial risks of bias and extracted dats. Statistical analyses were performed using random effects meta-analysis. The risk estimates were expressed as a RR for dichotomous variables and MD for continuous outcomes together with the corresponding 95 % CI. The primary efficacy outcome was bone fracture. The primary safety outcome was acute graft rejection. Secondary outcomes included death (all cause and cardiovascular), MI, stroke, musculoskeletal disorders (e.g., skeletal deformity, bone pain), graft loss, nausea, hypercalcemia or hypocalcemia, kidney function, serum parathyroid hormone (PTH), and BMD.

In this 2019 update, a total of 65 studies (involving 3,598 subjects) were eligible; 45 studies contributed data to the meta-analyses (2,698 subjects). Treatments included bisphosphonates, vitamin D compounds, teriparatide, denosumab, cinacalcet, parathyroidectomy, and calcitonin. Median duration of follow-up was 12 months; 43 studies evaluated bone density or bone-related biomarkers, with more recent studies evaluating proteinuria and hyperparathyroidism. Bisphosphonate therapy was usually commenced in the peri-operative transplantation period (within 3 weeks) and regardless of BMD. Risks of bias were generally high or unclear leading to lower certainty in the results. A single study reported outcomes among 60 children and adolescents. Studies were not designed to measure treatment effects on fracture, death or cardiovascular outcomes, or graft loss. Compared to placebo, bisphosphonate therapy administered over 12 months in KTx recipients may prevent fracture (RR 0.62, 95 % CI: 0.38 to 1.01; low certainty evidence) although the 95 % CI included the possibility that bisphosphonate therapy might make little or no difference. Fracture events were principally vertebral fractures identified during routine radiographic surveillance. It was uncertain whether any other drug class decreased fracture (low or very low certainty evidence). It was uncertain whether interventions for bone disease in KTx reduce all-cause or cardiovascular death, MI or stroke, or graft loss in very low certainty evidence. Bisphosphonate therapy may decrease acute graft rejection (RR 0.70, 95 % CI: 0.55 to 0.89; low certainty evidence), while it was uncertain whether any other treatment impacts graft

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rejection (very low certainty evidence). Bisphosphonate therapy may reduce bone pain (RR 0.20, 95 % CI: 0.04 to 0.93; very low certainty evidence), while it was very uncertain whether bisphosphonates prevent spinal deformity or AVN (very low certainty evidence). Bisphosphonates may increase to risk of hypocalcemia (RR 5.59, 95 % CI: 1.00 to 31.06; low certainty evidence). It was uncertain whether vitamin D compounds had any effect on skeletal, cardiovascular, death, or transplant function outcomes (very low certainty or absence of evidence). Evidence for the benefits and harms of all other treatments was of very low certainty. Evidence for children and young adolescents was sparse. The authors concluded that bisphosphonate therapy may reduce fracture and bone pain after KTx, however low certainty in the evidence indicated it is possible that treatment may make little or no difference. It was uncertain whether bisphosphonate therapy or other bone treatments prevented other skeletal complications following KTx, including spinal deformity or AVN. The effects of bone treatment for children and adolescents following KTx were very uncertain.

Urinary Biomarkers

Urinary Neutrophil Gelatinase-Associated Lipocalin for Renal Rejection

Iguchi and associates (2015) stated that neutrophil gelatinase-associated lipocalin (NGAL) and liver-type fatty acid-binding protein (L-FABP) are promising early biomarkers for acute kidney injury (AKI). In organ transplant recipients, AKI predictability based on NGAL and L-FABP remains to be elucidated. Furthermore, the association between serial NGAL and L-FABP measurements and AKI outcome is unknown. These researchers evaluated the ability of NGAL and L-FABP to predict AKI after organ transplantation and examined the association between NGAL, L-FABP and AKI outcome. A total of 25 organ transplant recipients admitted to the intensive care unit (ICU) immediately after transplant surgery were studied prospectively. Plasma NGAL (P-NGAL), urinary NGAL (U-NGAL) and L-FABP were measured from ICU admission to ICU discharge; U-NGAL and L-FABP were corrected for dilution/concentration by calculating U-NGAL/urine creatinine ratios (U-NGAL/Cr) and L-FABP/urine creatinine ratios (L-FABP/Cr). Acute kidney injury was defined according to the Kidney Disease: Improving Global Outcomes criteria. Acute kidney injury occurred in 11 patients; P-NGAL, U-NGAL/Cr and L-FABP/Cr upon ICU admission were unrelated to AKI development ($p = 0.24$, 0.22, and 0.53, respectively). There were no differences in P-NGAL, U-NGAL/Cr, and L-FABP/Cr levels from day 1 to day 6 between patients who did not recover from AKI and patients who recovered from AKI ($p = 0.82$, 0.26, and 0.61, respectively). The authors concluded that the findings of this study suggested that NGAL and L-FABP upon ICU admission were not predictive of AKI and serial NGAL and L-FABP measurements may be ineffective for monitoring the status and treatment of post-transplantation AKI.

In a prospective, single-center study, Seeman and co-workers (2017) examined the ability of urinary NGAL to distinguish AR from other causes of AKI in children after renal transplantation. A total of 15 children fulfilled the inclusion criteria (AKI) with allograft biopsy, at least 21 days after renal transplantation, no sepsis) during 2013 to 2014 in the authors' pediatric transplantation center. The mean age was 14.8 +/- 2.8 years, median time after renal transplantation was 0.4 years (range of 0.1 to 3.8). Urinary NGAL was measured in spot urine by chemiluminescent microparticle immunoassay technology. A total of 4 patients had biopsy proven AR (rejection group), 11 children had AKI of other cause (non-rejection group). The median urinary NGAL concentration in the rejection group was not significantly different from NGAL in the non-rejection group (7.3 ng/ml, range of 3.0 to 42.3 versus 8.6 ng/ml, range of 3.4 to 54.7, $p = 0.48$). There was a significant negative correlation between eGFR and urinary NGAL concentrations ($r = -0.77$, $p < 0.001$). The authors concluded that the findings of this small study suggested that in children after renal transplantation, urinary NGAL cannot be used as a specific marker for distinguishing AR from other non-rejection causes of AKI; urinary NGAL was mainly associated with graft function but not with the etiology of AKI.

Urinary Monocyte Chemoattractant Protein-1 (MCP-1/CCL2) for Renal Rejection

Wang and colleagues (2016) stated that there is a high risk for the survival of patients with an end-stage renal disease for kidney transplantation. To avoid rejection by strict medication adherence is of utmost importance

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to avoid the failure of a kidney transplant. It is imperative to develop non-invasive biomarkers to assess immunity risk, and to ultimately provide guidance for therapeutic decision-making following kidney transplantation. Urine biomarkers may represent the promising non-invasive tools that will help in predicting risk or success rates of kidney transplantations. Furthermore, composite urinary biomarkers or urinary biomarker panel array might be critical in improving the sensitivity and specificity in reflecting various risks of kidney failure during transplantation. These investigators focused on the role of such biomarkers in predicting chronic kidney disease (CKD) progression and/or cardiovascular disease (CVD) risk in renal allograft. Chemokine ligand 2 (CCL2) is a small cytokine that belongs to the CC chemokine family; CCL2 recruits monocytes, memory T cells, and dendritic cells to the sites of inflammation produced by either tissue injury or infection. The advancing age of renal transplant recipients (RTRs) correlates with the increasing CCL2 concentrations, which is reflected in the smoldering inflammation and alterations in matrix metallo-proteinases (MMPs)/tissue inhibitor metallo-proteinases (TIMPs) profiles, especially with increased plasma MMP-2 and urine TIMP-1 concentrations. The advanced age of RTRs has a negative impact on kidney allograft survival through impaired extracellular matrix degradation by the MMPs/TIMPs system. The authors concluded that "The development of novel biomarkers and detection technologies may lead to the creation of multiplex assays, which allows for the measurement of multiple specific biomarkers simultaneously in the same analyte. Such assays may prove useful for determining the nature of renal injury or the stage of disease in patients. Novel multiplex technologies such as Biomarker Panel Array (BPA) may provide a new perspective for diagnosis and prognosis of renal rejection after kidney transplantation; more importantly, these technologies could significantly improve the sensitivity and specificity in reflecting the clinical manifestations. It is important to note that the advancement of proteomics technologies may become a critical drive for the discovery of novel urinary biomarkers which could specifically and accurately reflect underlying molecular events of allograft rejection. The validation and assessment of the performance of urinary biomarkers for allograft rejection remains a significant, costly, and high-risk undertaking process".

Raza and associates (2017) noted that CCL2 is a chemoattractant for monocytes/macrophages, T cells, and natural killer cells. It is shown to be involved in the immunological responses against renal allograft. These researchers evaluated the role of urinary CCL2 expression in predicting the rejection episodes in renal transplant patients. A total of 409 urine samples were included in this study. The samples consisted of (a) biopsy-proven graft rejection (n = 165); (b) non-rejection (n = 93); (c) non-biopsy stable-graft (n = 42), and (d) healthy renal donors (n = 109). Samples were quantified for the CCL2 using the MCP-1/CCL2 ELISA kit. Data were analyzed using the Statistical Package for Social Sciences (SPSS) and MedCalc statistical software. Results showed that the CCL2 levels were significantly increased in rejection group when compared with the non-rejection, stable-graft, and control ($p < 0.05$). The receiver operating curve's characteristics illustrated that the urinary CCL2 level is a good predictor for graft rejection, with an area under the curve of 0.81 ± 0.03 with optimum sensitivity and specificity of 87 % and 62 %, respectively, at a cut-off value of 198 pg/ml. Kaplan-Meier curve also showed better cumulative rejection-free graft survival time in group with less than 198 pg/ml of CCL2 as compared to those with expression levels of more than 198 pg/ml (30 weeks versus 3 weeks; log-rank test, $p < 0.001$). The authors concluded that non-invasive measurement of CCL2 levels in urine has showed potential to predict rejection episodes. It is suggested that the CCL2, with others markers, may help in early detection and monitoring of graft rejection episodes.

Urinary Chemokines

Guzzi and colleagues (2020) stated that non-invasive tools for diagnosis or prediction of acute kidney allograft rejection have been extensively studied in recent years. Biochemical and molecular analyses of blood and urine provide a liquid biopsy that could offer new possibilities for rejection prevention, monitoring; and thus, treatment. Nevertheless, these tools are not yet available for routine use in clinical practice. In a systematic review, Medline was searched for studies examining urinary biomarkers for diagnosis or prediction of kidney allograft acute rejection published in the past 5 years (from January 1, 2015 to May 31, 2020). This review followed the Preferred Reporting Items for Systematic Reviews and Meta-analysis (PRISMA) guidelines. Studies providing targeted or un-biased urine sample analysis for the diagnosis or prediction of both acute cellular and antibody-mediated kidney allograft rejection were included, analyzed, and graded for methodological quality with a particular focus on study design and diagnostic test accuracy measures. Urinary C-X-C motif chemokine ligands were the most promising and frequently

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studied biomarkers. The combination of precise diagnostic reference in training sets with accurate validation in real-life cohorts provided the most relevant results and exciting groundwork for future studies. These researchers stated that urinary chemokines CXCL9 and CXCL10, alone or in combination with others, are the most frequently used and the most promising biomarkers; however, multi-parametric clinical and laboratory models could represent the best strategy for future studies.

The authors stated that this systematic review had several drawbacks. The heterogeneity of the included studies did not allow these investigators to detail the many facets of individual study findings, especially the more complex ones, to stick with the systematic review question. For space restraints, tables only reported the major findings of each study, limited to urinary biomarkers. A narrative synthesis of the most promising results was applied to improve readability and a meta-analysis could not be carried out. From these researchers' work, overall good quality studies emerged, many with diagnostic test accuracy analysis and some comprising a thorough validation process yielding a very good-to-excellent diagnostic performance. Although specific forms of bias were examined using QUADAS-2 publication bias could not be formally evaluated and the authors acknowledged this could over-estimate the weight of positive results. Weaknesses of the included studies were often the use of small cohorts obtained by case-control selection yielding inflated predictive values, the exclusion of confounding, unclear or out-of-date Banff classification application, the absence of validation cohorts, and lack of hypothesis-driven approach. In fact, the biomarker discovery process should not only consist of a training phase (i.e., a case-control study), but also comprise independent validation in a prospective study and confrontation with real-life clinical setting.

Urinary Extracellular Vesicles

Wu and colleagues (2021) stated that extracellular vesicles (EVs) are nanoparticles that transmit molecules from releasing cells to target cells. Recent studies link urinary EVs (uEV) to diverse processes such as infection and rejection following KTx. This, and the unmet need for biomarkers diagnosing kidney transplant dysfunction, has led to the current high level of interest in uEV. uEV provide non-intrusive access to local protein, DNA, and RNA analytics without invasive biopsy. To determine the added value of uEV measurements for detecting allograft dysfunction following KTx, these investigators systematically included all related literature containing directly relevant information, with the addition of indirect evidence regarding urine or kidney injury without transplantation. According to their varying characteristics, uEV markers after transplantation could be categorized into kidney-specific, donor-specific, and immune response-related (IR-) markers. A few convincing studies have shown that kidney-specific markers (PODXL, ion cotransporters, SYT17, NGAL, and CD133) and IR-markers (CD3, multi-mRNA signatures, and viral miRNA) could diagnose rejection, BK virus-associated nephropathy, and calcineurin inhibitor nephrotoxicity following KTx. Furthermore, some indirect proof regarding donor-specific markers (donor-derived cell-free DNA) in urine has been demonstrated. The authors concluded that this review highlighted potential uEV-marker candidates and indicated directions for further study, opening up an attractive perspective that would universally utilize uEV in clinical practice and avoid unnecessary biopsy following KTx. These researchers stated that future studies should take a multi-center approach that enables large patient numbers. This should take into account clinical properties (e.g., gender, race, age, body size, HLA-mismatch, and common chronic diseases such as hypertension, diabetes mellitus, and hepatitis virus infection) of donor and recipient; the type and proportion of donors (e.g., living donor or deceased donor); immunosuppressor-treated or -free, and the type of medicine administered. This information is of critical value for multi-variate analysis, which will provide results defining the diagnostic accuracy of uEV measurement in KTx.

The authors stated that uEV have potential as non-invasive biomarkers to monitor kidney allograft dysfunction and to establish its cause. However, there are still 2 major blanks to be filled, with potential for new and interesting directions of study. First, HLA and DNA have not yet been shown to be "donor-specific" in uEV, based on any evidence of donor-recipient mismatch. Second, urinary miRNA is of great diagnostic value for kidney allograft dysfunction, whereas the characteristics of uEV-miRNA in KTx are unknown.

Fas Ligand (FASL) mRNA Detection for Acute Renal Rejection

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Heng and colleagues (2016) stated that the value of Fas ligand (FASL) as a diagnostic immune marker for acute renal rejection is controversial; this meta-analysis aimed to clarify the role of FASL in acute renal rejection. The relevant literature was included by systematic searching the Medline, Embase, and Cochrane Library databases. Accuracy data for AR and potential confounding variables (the year of publication, area, sample source, quantitative techniques, housekeeping genes, fluorescence staining, sample collection time post-renal transplantation, and clinical classification of AR) were extracted after carefully reviewing the studies. Data were analyzed by Meta-DiSc 1.4, RevMan 5.0, and the Midas module in Stata 11.0 software. A total of 12 relevant studies involving 496 subjects were included. The overall pooled sensitivity, specificity, positive likelihood ratio (LR), negative LR, and diagnostic OR, together with the 95 % CI were 0.64 (0.57 to 0.70), 0.90 (0.85 to 0.93), 5.66 (3.51 to 9.11), 0.30 (0.16 to 0.54), and 30.63 (14.67 to 63.92), respectively. The area under the summary receiver operating characteristic curve (AUC) was 0.9389. Fagan's nomogram showed that the probability of AR episodes in the kidney transplant recipient increased from 15 % to 69 % when FASL was positive, and was reduced to 4 % when FASL was negative. No threshold effect, sensitivity analyses, meta-regression, and subgroup analyses based on the potential variables had a significant statistical change for heterogeneity. The authors concluded that current evidence suggested the diagnostic potential for FASL mRNA detection as a reliable immune marker for AR in renal allograft recipients. Moreover, they stated that further large, multi-center, prospective studies are needed to validate the power of this test marker in the non-invasive diagnosis of AR after renal transplantation.

Donor-Derived Cell-Free DNA Testing (e.g., AlloSure, Prospera Kidney, VitaGraft Kidney Baseline + 1st Plasma Test, VitaGraft Kidney Subsequent, VitaGraft Kidney 2.0)

Bloom and colleagues (2017) stated that histologic analysis of the allograft biopsy specimen is the standard method used to differentiate rejection from other injury in kidney transplants. Donor-derived cell-free DNA (dd-cfDNA) is a non-invasive test of allograft injury that may enable more frequent, quantitative, and safer assessment of allograft rejection and injury status. These investigators examined this possibility by prospectively collected blood specimens at scheduled intervals and at the time of clinically indicated biopsies. In 102 kidney recipients, these researchers measured plasma levels of dd-cfDNA and correlated the levels with allograft rejection status ascertained by histology in 107 biopsy specimens. The dd-cfDNA level discriminated between biopsy specimens showing any rejection (T cell-mediated rejection or antibody-mediated rejection [ABMR]) and controls (no rejection histologically), $p < 0.001$ (receiver operating characteristic area under the curve [AUC], 0.74; 95 % confidence interval [95% CI]: 0.61 to 0.86). Positive and negative predictive values (PPV and NPV) for active rejection at a cut-off of 1.0 % dd-cfDNA were 61 % and 84 %, respectively. The AUC for discriminating ABMR from samples without ABMR was 0.87 (95 % CI: 0.75 to 0.97). PPV and NPV for ABMR at a cut-off of 1.0 % dd-cfDNA were 44 % and 96 %, respectively. Median dd-cfDNA was 2.9 % (ABMR), 1.2 % (T cell-mediated types ≥ IB), 0.2 % (T cell-mediated type IA), and 0.3 % in controls ($p = 0.05$ for T cell-mediated rejection types ≥ IB versus controls). The authors concluded that dd-cfDNA may be used to assess allograft rejection and injury; dd-cfDNA levels of less than 1 % reflected the absence of active rejection (T cell-mediated type ≥ IB or ABMR) and levels greater than 1 % indicated a probability of active rejection. They stated that the next steps of development include studies to validate these findings and to demonstrate the clinical utility of this new type of immune monitoring of the graft.

This study had several drawbacks. First, these researchers were unable to estimate the performance of dd-cfDNA to discriminate active rejection or ABMR in patients who may have had sub-clinical rejection because there were only 34 surveillance biopsies and only 1 finding of active rejection. However, this low rejection frequency was consistent with reports by others in an era of tacrolimus-mycophenolic acid-prednisone-based maintenance immunosuppression that question the utility of protocol biopsy for this purpose. Second, the number of active rejections ($n = 27$) and sub-classes of rejection observed among these biopsy specimens was limited. However, these met the target total number of rejections prospectively stated in the statistical analysis, and indeed, the results proved this number to be sufficient to demonstrate statistically significant performance characteristics. Third, biopsy-matched blood samples were not collected for all biopsy specimens, and some of the matched blood samples were excluded due to issues such as inadequate amount of total DNA or timing of the blood draw relative to the biopsy. Of all collected blood

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samples, 4.5 % did not render results due to some aspect of sample collection or testing. Most patients completed surveillance visits in compliance (77 %) with the center schedule.

Lee and associates (2017) noted that early detection and proper management of kidney rejection are crucial for the long-term health of a transplant recipient. Recipients are normally monitored by serum creatinine measurement and sometimes with graft biopsies. Donor-derived cell-free deoxyribonucleic acid (cfDNA) in the recipient's plasma and/or urine may be a better indicator of acute rejection. These investigators evaluated digital PCR (dPCR) as a system for monitoring graft status using single nucleotide polymorphism (SNP)-based detection of donor DNA in plasma or urine. They compared the detection abilities of the QX200, RainDrop, and QuantStudio 3D dPCR systems. The QX200 was the most accurate and sensitive. Plasma and/or urine samples were isolated from 34 kidney recipients at multiple time-points after transplantation, and analyzed by dPCR using the QX200. They found that donor DNA was almost undetectable in plasma DNA samples, whereas a high percentage of donor DNA was measured in urine DNA samples, indicating that urine is a good source of cfDNA for patient monitoring. These researchers found that at least 24 % of the highly polymorphic SNPs used to identify individuals could also identify donor cfDNA in transplant patient samples. The authors concluded that these findings showed that autosomal, sex-specific, and mitochondrial SNPs were suitable markers for identifying donor cfDNA. They found that donor-derived cfDNA measurement by dPCR was not sufficient to predict a patient's clinical condition; these results indicated that donor-derived cfDNA is not an accurate predictor of kidney status in kidney transplant patients.

Agbor-Enoh et al (2018) stated that antibody-mediated rejection (AMR) often progresses to poor health outcomes in lung transplant recipients (LTRs). This, combined with the relatively insensitive clinical tools used for its diagnosis (spirometry, histopathology), led these researchers to examine if clinical AMR is diagnosed significantly later than its pathologic onset. These researchers leveraged the high sensitivity of donor-derived cell-free DNA (ddcfDNA), a novel genomic tool, to detect early graft injury after lung transplantation. They adjudicated AMR and acute cellular rejection (ACR) in 157 LTRs using the consensus criteria of the International Society for Heart and Lung Transplantation (ISHLT); and assessed the kinetics of allograft injury in relation to ACR or AMR using both clinical criteria (decline in spirometry from baseline) and molecular criteria (ddcfDNA); percent ddcfDNA was quantitated via shotgun sequencing. These investigators used a mixed-linear model to examine the relationship between and ddcfDNA levels and donor-specific antibodies (DSA) in AMR+ LTRs. Compared with ACR, AMR episodes ($n = 42$) were associated with significantly greater allograft injury when assessed by both spirometric (0.1 L versus -0.6 L, $p < 0.01$) and molecular (ddcfDNA) analysis (1.1 % versus 5.4 %, $p < 0.001$). Allograft injury detected by ddcfDNA preceded clinical AMR diagnosis by a median of 2.8 months. Within the same interval, spirometry or histopathology did not reveal findings of allograft injury or dysfunction. Elevated levels of ddcfDNA before clinical diagnosis of AMR were associated with a concurrent rise in DSA levels. The authors concluded that diagnosis of clinical AMR in LTRs lags behind DSA-associated molecular allograft injury as assessed by ddcfDNA. Moreover, these researchers stated that areas for further investigation include determination of specificity of elevated ddcfDNA levels for AMR; how this test may be best used in concert with DSAs and other biomarkers; and how pre-emptive AMR therapy may impact clinical outcomes, such as graft function and survival. These investigators anticipated that their findings would offer a foundation for future research focused on the use of molecular diagnostic tools to improve patient outcomes after transplantation.

The authors stated that in this study, they observed the following potential limitations using the 2016 ISHLT criteria for diagnosis of clinical AMR. First, the low number of positive C4d tests leading to no “definite” AMR events, similar to earlier reports. Second, the high occurrence of alternative causes of allograft dysfunction at time of AMR diagnosis, especially clinical infections. The 2010 ISHLT criteria may over-diagnose clinical infections as the criteria may not reliably distinguish microbial colonization from true clinical infection. Despite this limitation, these data suggested a positive correlation between detectable infections and DSA. A better understanding of the interactions between infection, DSA levels, and AMR awaits further investigation. The small number of AMR events was also a potential limitation of this trial.

Sigdel et al (2018) noted that standard non-invasive methods for detecting renal allograft rejection and injury have poor sensitivity and specificity. Plasma dd-cfDNA has been reported to accurately detect allograft rejection and injury in transplant recipients and shown to discriminate rejection from stable organ function in

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KTx recipients. In a retrospective, single-center study, these researchers used a novel SNP-based massively multiplexed PCR (mmPCR) methodology to measure dd-cfDNA in various types of KTx recipients for the detection of allograft rejection/injury without prior knowledge of donor genotypes. A total of 300 plasma samples (217 biopsy-matched: 38 with active rejection (AR), 72 borderline rejection (BL), 82 with stable allografts (STA), and 25 with other injury (OI)) were collected from 193 unique KTx patients; dd-cfDNA was processed by mmPCR targeting 13,392 SNPs. Median dd-cfDNA was significantly higher in samples with biopsy-proven AR (2.3 %) versus BL (0.6 %), OI (0.7 %), and STA (0.4 %) ($p < 0.0001$ all comparisons). The SNP-based dd-cfDNA assay discriminated active from non-rejection status with an AUC of 0.87, 88.7 % sensitivity (95 % CI: 77.7 % to 99.8 %) and 72.6 % specificity (95 % CI: 65.4 % to 79.8 %) at a pre-specified cut-off (greater than 1 % dd-cfDNA). Of 13 patients with AR findings at a routine protocol biopsy 6-month post-transplantation, 12 (92 %) were detected positive by dd-cfDNA. The authors concluded that this rapid, accurate, and non-invasive technology allowed for detection of significant renal injury in patients better than the current standard of care, with the potential for better patient management, more targeted biopsies, and improved renal allograft function and survival.

The authors stated that a drawback of this study was that it was a retrospective analysis of archived samples from a single center; however, the central geographical area enabled all biopsies to be carried out by a single pathologist, which may have helped minimize variability in biopsy classification. In addition, all experimenters were kept blinded during the process of data generation. The retrospective study design may have led to differences in patient characteristics across the rejection groups (e.g., the stable allografts group was enriched with younger patients who may be better suited immunologically to tolerate transplanted organs compared to older-aged patients. However, these age differences likely did not affect the validity of the findings of this study.

Furthermore, an UpToDate review on “Investigational methods in the diagnosis of acute renal allograft rejection” (Anglicheau et al, 2018) lists donor-derived cell-free DNA as an investigational method. It states that “During allograft rejection, large amounts of dd-cfDNA are released from the injured allograft into the bloodstream. Quantification of plasma levels of dd-cfDNA has been proposed as a noninvasive test for the early diagnosis of acute renal allograft rejection. Methods for measuring dd-cfDNA include quantitative reverse transcription PCR (RT-PCR), droplet digital PCR (ddPCR), and targeted next-generation sequencing. In one multicenter study, plasma levels of dd-cfDNA were measured by targeted next-generation sequencing in 102 renal transplant recipients and correlated with allograft rejection status as determined by renal allograft biopsy. The dd-cfDNA level was able to differentiate between biopsy specimens showing any form of rejection (TCMR or ABMR) and those without rejection. Using a cutoff of 1.0 %, dd-cfDNA had a PPV and NPV for active rejection of 61 and 84 %, respectively. Studies in heart, lung, and liver transplant recipients have also shown an association between increased plasma dd-cfDNA levels and acute rejection”.

Knight et al (2019) noted that there is increasing interest in the use of non-invasive biomarkers to reduce the risks posed by invasive biopsy for monitoring of solid organ transplants (SOTs). One such promising marker is the presence of dd-cfDNA in the urine or blood of transplant recipients. These researchers systematically reviewed the published literature examining the use of cfDNA in monitoring of graft health after SOT. Electronic databases were searched for studies relating cfDNA fraction or levels to clinical outcomes, and data including measures of diagnostic test accuracy were extracted. Narrative analysis was performed. A total of 95 articles from 47 studies met the inclusion criteria (18 kidneys, 7 livers, 11 hearts, 1 kidney-pancreas, 5 lungs, and 5 multi-organs). The majority were retrospective and prospective cohort studies, with 19 reporting diagnostic test accuracy data. Multiple techniques for measuring dd-cfDNA were reported, including many not requiring a donor sample. dd-cfDNA fell rapidly within 2 weeks, with baseline levels varying by organ type. Levels were elevated in the presence of allograft injury, including AR and infection, and return to baseline after successful treatment. Elevation of cfDNA levels was observed in advance of clinically apparent organ injury. Discriminatory power was greatest for higher grades of T cell-mediated and antibody-mediated AR, with high negative predictive values (NPVs). The authors concluded that cell-free DNA is a promising biomarker for monitoring the health of SOTs. Moreover, these researchers stated that future studies are needed to define how it can be used in routine clinical practice and determine clinical benefit with routine prospective monitoring.

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Hinojosa et al (2019) noted that confirmatory biopsy after treatment for AR is uncommon. Serum creatinine (SCr) and random urine albumin to urine creatinine (Ur alb:Ur Cr) ratios are used to monitor resolution of rejection. dd-cfDNA (AlloSure; CareDX, Brisbane, CA) levels greater than 1 % detect active renal allograft injury. Given the short half-life of dd-cfDNA (approximately 30 mins), these investigators believed that if patients with AR were treated appropriately, then dd-cfDNA levels could verify resolution of rejection as early as the final days of treatment. These investigators treated 3 patients for biopsy-proven renal allograft rejection, and dd-cfDNA declined less than 1 % upon completion of therapy. The authors concluded that this was the 1st report of monitoring dd-cfDNA trends before and upon completion of AR treatment; they stated that large-scale studies are needed to formally examine these findings and investigate the accuracy of dd-cfDNA as a potential biomarker for real-time monitoring of response to therapy.

Huang et al (2019) stated that donor-derived cell-free DNA (dd-cfDNA) became Medicare reimbursable in the U.S. in October 2017 for the detection of rejection in kidney transplant recipients based on results from its pivotal validation trial; however, it has not yet been externally validated. These researchers evaluated 63 adult kidney transplant recipients with suspicion of rejection with dd-cfDNA and allograft biopsy. Of these, 27 (43 %) patients had donor-specific antibodies (DSA) and 34 (54 %) were found to have rejection by biopsy. The percentage of dd-cfDNA was higher among patients with antibody-mediated rejection (ABMR; median of 1.35 %; inter-quartile range [IQR]: 1.10 % to 1.90 %) compared to those with no rejection (median of 0.38 %, IQR: 0.26 % to 1.10 %; $p < 0.001$) and cell-mediated rejection (CMR; median of 0.27 %, IQR: 0.19 % to 1.30 %; $p = 0.01$). The dd-cfDNA test did not discriminate patients with CMR from those without rejection. The area under the ROC curve (AUC) for CMR was 0.42 (95 % CI: 0.17 to 0.66). For ABMR, the AUC was 0.82 (95 % CI: 0.71 to 0.93) and a dd-cfDNA greater than or equal to 0.74 % yielded a sensitivity of 100 %, specificity 71.8 %, positive predictive value [PPV] of 68.6 %, and negative predictive value [NPV] of 100 %. The dd-cfDNA test did not discriminate CMR from no rejection among kidney transplant recipients, although performance characteristics were stronger for the discrimination of ABMR. Moreover, these researchers stated that larger studies, especially among patients with DSA, are needed to further validate these findings along with those made in the DART trial. Additionally, further studies are ongoing that will examine the utility of routine surveillance of dd-cfDNA to predict the onset of acute rejection in kidney transplant recipients.

The authors stated that this study had several drawbacks. This trial was a single center study with a relatively small number (n = 63) of patients. Nevertheless, the number and types of rejections confirmed by biopsy in this study were comparable to the number analyzed in the DART trial. Furthermore, these findings were similar to those made in the DART trial and indicated that the results from DART were reproducible. Although the large number of sensitized patients was unique to the authors' transplant center, findings from this study underscored the need to validate dd-cfDNA in varying populations. Second, the optimal dd-cfDNA threshold level for detection of rejection in this study differed slightly from the DART trial. It should be noted that the test is better interpreted as a continuous result, where the probability of rejection became higher with increasing dd-cfDNA levels. Levels exceeding the threshold should not be interpreted to mean that a patient has rejection; rather, a level above the threshold indicated only a higher probability of rejection. These investigators cautioned against relying on thresholds for interpretation of rejection, as the specific threshold would likely vary from center to center. Third, these researchers included all patients who had assessment of dd-cfDNA within 30 days of an allograft biopsy. It could be argued that the level of dd-cfDNA may have changed between the date of sample collection and performance of the biopsy. However, if this were true, a significant degree of within-individual variability would diminish the practical value of dd-cfDNA as a screening test. These investigators attempted to simulate clinical practice in this study, where scheduling constraints and the process of obtaining insurance approval can often delay performance of an allograft biopsy . Fourth, dd-cfDNA was primarily assessed on a "for-cause" basis, where dd-cfDNA was assessed in the setting of suspected allograft rejection. It was not yet known whether the prediction characteristics for dd-cfDNA will be similar when used to predict sub-clinical rejection when graft function was stable. This is currently being assessed in a large, multi-center observational study using dd-cfDNA as part of routine surveillance for allograft rejection in kidney transplant recipients. Last, 11 of the 63 patients (17 %) in this study received some form of augmented immunosuppression within 3 months of dd-cfDNA assessment. In theory, recent augmentation of immunosuppression could dampen dd-cfDNA levels, which might affect the interpretation of the assay. However, the authors did not observe a clinically meaningful

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impact on the interpretation of dd-cfDNA levels in these patients. Levels of dd-cfDNA tended to be low in recently treated patients without rejection on biopsy and were elevated among those with histologic evidence of rejection.

Huang et al (2020) noted that higher Banff inflammation and chronicity scores on kidney transplant biopsies are associated with poorer graft survival, although histology alone has limitations in predicting outcomes. These investigators examined if integrating dd-cfDNA (AlloSure) with Banff biopsy scores into a predictive model for estimated glomerular filtration rate (eGFR) over time can improve prognostic assessment versus histology alone. These researchers identified 180 kidney transplant patients with dd-cfDNA assessed within 1 month of biopsy. Using linear mixed-effects models, a prediction model of Banff histology scores and dd-cfDNA on eGFR over time was derived. Nested models were compared using the likelihood-ratio test, Akaike Information Criterion, and Bayesian Information Criterion to assess if inclusion of dd-cfDNA into a model consisting of Banff biopsy scores would improve model fit. Uni-variate models identified significant covariate-by-time interactions for cg = 3 versus < 3 (coefficient: −1.3 ml/min/1.73 m2/month; 95 % confidence interval [CI]: −2.4 to −0.2; p = 0.02) and ci + ct ≥ 3 versus < 3 (coefficient: −0.7 ml/min/1.73 m2/month; 95 % CI: −1.3 to −0.1; p = 0.03) and a trend toward significant covariate-by-time interaction for dd-cfDNA (coefficient: −0.5 ml/min/1.73 m2/month; 95 % CI: −1.0 to 0.1; p = 0.08). Addition of acute inflammation (i, t, and v), microvascular inflammation (g and ptc), and inflammation in area of interstitial fibrosis and tubular atrophy scores to chronicity scores (cg ≥ 3 and ci + ct ≥ 3) did not improve model fit. However, a model including dd-cfDNA with cg and ci + ct with covariate-by-time interactions had a better model fit compared with cg and ci + ct alone (likelihood-ratio test statistic = 21.1; df = 2; p < 0.001). The authors concluded that addition of dd-cfDNA to Banff biopsy scores provided better prognostic assessment over biopsy characteristics alone. Moreover, these researchers stated that these preliminary findings should be validated in larger, multi-center datasets. If validated, assessment of biopsies alongside dd-cfDNA could be considered as a newer paradigm for classifying rejection in kidney transplantation.

The authors stated that this study had several drawbacks. Baseline characteristics, immunologic risk, and treatment protocols may differ between transplant centers, which may potentially have an impact on eGFR response. Additionally, this was a retrospective study that relied on clinical decisions and not pre-specified protocols on dd-cfDNA and biopsy assessment. As such, it was unclear whether exclusion of dd-cfDNA tests assessed without a biopsy may have biased these findings. It should also be noted that this study used cross-sectional data and did not account for changes in dd-cfDNA or biopsy findings over time.

In a systematic review and meta-analysis, Wijtvliet et al (2020) examined the value of dd-cfDNA as a non-invasive biomarker in diagnosing kidney allograft rejection. These investigators searched PubMed, Web of Science and the Cochrane Library for original research papers published between January 1994 and May 2020 on dd-cfDNA fractions in blood of kidney allograft recipients. A single-group meta-analysis was carried out by computing pooled estimates for dd-cfDNA fractions using the weighted median of medians or quantile estimation (QE) approach. Weighted median differences in medians (WMDMs) and median differences based on the QE method were used for pair-wise comparisons. Despite heterogeneity among the selected studies, the meta-analysis revealed significantly higher median dd-cfDNA fractions in patients with antibody-mediated rejection (ABMR) than patients without rejection or patients with stable graft function. When comparing patients with T cell-mediated rejection (TCMR) and patients with ABMR, the 2 statistical approaches revealed conflicting results. Patients with TCMR did not have different median dd-cfDNA fractions than patients without rejection or patients with stable graft function. The authors concluded that dd-cfDNA may be a useful marker for ABMR, but probably not for TCMR.

Stites et al (2020) stated that the clinical importance of sub-clinical, early T cell-mediated rejection (Banff TCMR 1A and borderline lesions) remains unclear, due, in part to the fact that histologic lesions used to characterize early TCMR can be non-specific. dd-cfDNA is an important molecular marker of active graft injury. Over a study period from June 2017 to May 2019, these researchers examined clinical outcomes in 79 patients diagnosed with TCMR 1A/borderline rejection across 11 U.S. centers with a simultaneous measurement of dd-cfDNA; 42 patients had elevated dd-cfDNA (greater than or equal to 0.5 %) and 37 patients had low levels (less than 0.5 %). Elevated levels of dd-cfDNA predicted adverse clinical outcomes: among patients with elevated cfDNA, estimated glomerular filtration rate (eGFR) declined by 8.5 % (IQR -16.22 % to -1.39 %) (-3.50 ml/min/1.73 m2 IQR -8.00 to -1.00) versus 0 % (-4.92 % to 4.76 %) in low dd-

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cfDNA patients (p = 0.004), de-novo DSA formation was observed in 40 % (17/42) versus 2.7 % (p < 0.0001), and future or persistent rejection occurred in 9 of 42 patients (21.4 %) versus 0 % (p = 0.003). The use of dd-cfDNA may complement the Banff classification and to risk stratify patients with borderline/TCMR 1A identified on biopsy.

Kant et al (2020) stated that approximately 15 % of kidney transplant recipients (KTRs) developed BK viremia (BKV), with 1 % to 10 % developing BK virus-associated nephropathy (BKVAN), which histologically resembles rejection. The Diagnosing Acute Rejection in Kidney Transplant Recipients (DART) study showed that dd-cfDNA levels of less than 1 % have a NPV of 85 % for active allograft rejection. Using data from this study, these researchers examined the association of dd-cfDNA with plasma BK viral loads and biopsy findings to determine if dd-cfDNA can distinguish asymptomatic BKV from BKVAN. Data on dd-cfDNA, plasma BK viral loads, and biopsy findings from patients from the DART study were retrospectively examined. BKV was defined as 500 to 10,000 copies/ml. Presumptive BKVAN was defined as BK greater than 10,000 copies/ml. Of 102 participants with biopsies, 10 patients with BKV and BKVAN had paired dd-cfDNA, and viral loads available for analysis. Patients diagnosed with BKV and BKVAN had a median dd-cfDNA of 0.58 % (IQR 0.43 to 1.15) and 3.38 % (IQR 2.3 to 4.56, p = 0.001), respectively. dd-cfDNA titers correlated with BK PCR viral loads (R = 0.874, p = 0.01) and the presence of histologic evidence of BKVAN (100 % sensitivity, 50 % specificity); 5 of 7 patients with BKVAN, but only 2 of 7 with BKV, had biopsies meeting Banff criteria for T-cell-mediated rejection. Median dd-cfDNA in nonrejection patients was 0.43 % versus 2.84 % in rejection patients (p = 0.001). The authors concluded that higher dd-cfDNA titers were associated with higher BK viral loads, biopsy-diagnosed BVAN, as well histologic changes meeting Banff criteria for as T-cell–mediated rejection. dd-cfDNA may be a useful non-invasive test to assess for progression of BKV to BKVAN.

The authors concluded that this study had several drawbacks. First, this was a retrospective study, and had a small sample size. Three of the biopsies did not have a formal pathology report available, and in 4 cases of BKVAN, the diagnosis was made solely by histology (characteristic intranuclear inclusions) without confirmation by immuno-histochemical staining for SV40. However, limitations of biopsy sampling and associated SV40 staining were being increasingly recognized. The DART study was not able to distinguish borderline and 1A rejections using a dd-cfDNA threshold of 1 %. Recent studies using a dd-cfDNA threshold of 0.7 % were able to identify Banff Borderline and 1A rejections. Finally, there were no long-term follow-up data available on the patient cohort in this study.

Kataria et al (2021) stated that the last few years have seen an explosion in clinical research focusing on the use of dd-cfDNA in solid-organ transplants (SOT). While most of the literature published so far focuses on kidney transplants, there are several recent as well as ongoing research studies on heart, lung, pancreas and liver transplants. Though initially studied as a non-invasive means of identifying sub-clinical or acute rejection in SOT, it is rapidly becoming clear that instead of being a specific marker for allograft rejection, dd-cfDNA is more appropriately described as a marker of severe injury, although the most common etiology of this injury is allograft rejection. Multiple studies in kidney transplants have shown that while sensitivity for the diagnosis of antibody-mediated rejection is excellent, it is less so for T-cell mediated rejection. It is possible that combining dd-cfDNA with other novel urine or blood-based biomarkers may increase the sensitivity for the diagnosis of rejection. Irrespective of the etiology though, elevated dd-cfDNA appeared to portend adverse allograft prognosis and formation of de-novo donor-specific antibody. The authors concluded that although current data does not lend itself to a clear conclusion, ongoing studies may reveal the utility of serial surveillance for the management of SOT as following levels of dd-cfDNA over time may provide windows of opportunity to intervene early and prior to irreversible allograft injury. Finally, cost-effectiveness studies will be needed to guide the ideal incorporation of dd-cfDNA into routine clinical practice.

Ehlayel et al (2021) stated that non-invasive technologies to monitor kidney allograft health using high-throughput assays of blood and urine specimens are emerging out of the research realm and slowly becoming part of everyday clinical practice. HLA epitope analysis and eplet mismatch score determination promise a more refined approach to the pre-transplant recipient-donor HLA matching that may lead to reduced rejection risk. High-resolution HLA typing and multiplex single antigen bead assays are identifying potential new offending HLA antibody subtypes. There is increasing recognition of the deleterious role non-HLA

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antibodies play in post-transplant outcomes. Donor-derived cell-free DNA detected by next-generation sequencing (NGS) is a promising biomarker for kidney transplant rejection. Multi-omics techniques are shedding light on discrete genomic, transcriptomic, proteomic, and metabolomic signatures that correlate with and predict allograft outcomes. Over the next 10 years, a comprehensive approach to optimize kidney matching and monitor transplant recipients for acute and chronic graft dysfunction will likely involve a combination of those emerging technologies summarized in this review.

Martuszewski et al (2021) noted that KTx is the best treatment method for end-stage kidney disease. KTx improves the patient's quality of life and prolongs their survival time; however, not all patients benefit fully from the transplantation procedure. For some patients, a problem is the premature loss of graft function due to immunological or non-immunological factors. Circulating cell-free DNA (cfDNA) is degraded deoxyribonucleic acid fragments that are released into the blood and other body fluids; dd-cfDNA is cfDNA that is exogenous to the patient and comes from a transplanted organ. As opposed to an invasive biopsy, dd-cfDNA can be detected by a non-invasive analysis of a sample. The increase in dd-cfDNA concentration occurs even before the creatinine level starts rising, which may enable early diagnosis of transplant injury and adequate treatment to avoid premature graft loss. The authors summarized the latest promising results related to cfDNA in renal transplant patients. Moreover, these researchers stated that future multi-center studies and clinical trials on detecting dd-cfDNA (in urine and serum) in KTx patients with suspected graft rejection are still needed.

Wolf-Doty et al (2021) stated that the quantification of rejection treatment efficacy has been insufficient using traditional markers due, in part, to the lagging response of serum creatinine and histologic alterations on biopsy specimens; dd-cfDNA is a molecular marker of injury that may assess allograft injury after rejection. These investigators carried out a retrospective review of the DART study and identified 70 patients who had a clinically indicated biopsy, simultaneous dd-cfDNA measurement, and at least 1 follow-up dd-cfDNA within 3 months post-treatment; 35 patients had no biopsy-proven rejection and no rejection treatment (NR), 16 patients had no biopsy-proven rejection but did receive rejection treatment (CR), 9 patients had diagnosis of ABMR/mixed rejection on biopsy and received rejection treatment (ABMR), and 10 patients had diagnosis of TCMR and received rejection treatment (TCMR). The CR, ABMR, and TCMR groups combined to form a rejection (R) group. In the R group, median dd-cfDNA values at baseline and 1 month were 0.62 % and 0.35 % ($n = 21$ pairs, $p = 0.34$), and at baseline and 2 to 3 months were 0.77 % and 0.21 % ($n = 23$ pairs, $p = 0.002$). In TCMR, median dd-cfDNA values at baseline and 1 month were 1.13 % and 0.37 % ($n = 5$ pairs, $p = 0.63$), and at baseline and 2 to 3 months were 0.25 % and 0.12 % ($n = 9$ pairs, $p = 0.004$). In ABMR, median dd-cfDNA values at baseline and 1 month were 1.61 % and 1.2 % ($n = 6$ pairs, $p > 0.99$), and at baseline and 2 to 3 months were 3.85 % and 1.32 % ($n = 6$ pairs, $p = 0.09$). In CR, median dd-cfDNA values at baseline and 1 month were 0.31 % and 0.29 % ($n = 10$ pairs, $p = 0.38$), and at baseline and 2 to 3 months were 0.38 % and 0.1 7% ($n = 8$ pairs, $p = 0.31$). Lastly, in NR, median dd-cfDNA values at baseline and 1 month were 0.23 % and 0.18 % ($n = 21$ pairs, $p = 0.10$), and at baseline and 2 to 3 months were 0.33 % and 0.17 % ($n = 26$ pairs, $p = 0.003$). Changes in serum creatinine across 1 month and 2 to 3 months following rejection were similar. The authors concluded that longitudinal monitoring of dd-cfDNA may be useful as a dynamic biomarker to examine the health of the kidney allograft after rejection treatment. Future studies may allow dd-cfDNA to differentiate adequate versus inadequate response to current treatment regimens and suggest the need to provide alternative therapeutic options or augment treatment practices. These researchers speculated that serial surveillance of dd-cfDNA after rejection events may support strategies for precision medicine and immunosuppression, thereby improving post-transplant health.

The authors stated that this study had several drawbacks. First, the sample size in this study was small, affecting statistical significance; however, the protocol required that patients had assessment with a dd-cfDNA measurement concurrent with the clinically indicated biopsy, and a follow-up level obtained within 3 months. Although the number of patients who received rejection treatment was modest, the necessary subsets of rejection were required due to inherent differences in the natural history of TCMR and ABMR with response to treatment. Sample-size limitations were also due to lack of available dd-cfDNA levels for individual patients for both the 1 month and 2- to 3-month post-treatment analyses. Nevertheless, improvement in dd-cfDNA was evident for specific categories of rejection. Furthermore, most patients were enrolled at the time of a clinically indicated biopsy, without preceding baseline levels for comparison; thus, baseline levels were established from a separate control group. There was also a lack of histologic data, or

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uniformity in the use of biopsies, after rejection treatment was administered. Some centers performed biopsy and others relied on laboratory values (i.e., serum creatinine) to follow patients post–rejection treatment. Although repeat biopsies allowed for comparison of histologic architecture to examine if there was true resolution of pathology, they were invasive with increased hemorrhagic risk and findings could be patchy in distribution. In this study, only 6 of the 70 patients in this study had a follow-up biopsy at 1 month, and 6 patients had a follow-up biopsy in the 2- to 3-month range, making the analysis limited for histologic response to treatment. However, only 7 of 35 patients received further rejection treatment at 1 month and 4 patients at 2 to 3 months; therefore, these researchers suspected most patients had clinically recovered from the antecedent rejection events. Lastly, dd-cfDNA is a molecular marker of injury and there are other causes of allograft injury aside from rejection. Other non-rejection sources of injury, such as BK virus-associated nephropathy or de-novo DSA, were not assessed in this study and may have contributed to the elevations in dd-cfDNA. These researchers also stated that inter-observer variability among histopathologic assessment of rejection has also been shown and tissue molecular markers aligned closely with dd-cfDNA compared with traditional histology, lending to increased precision as an adjunct to traditional histopathology. From these data, decreased dd-cfDNA could confirm recovery from a rejection event, whereas persistently elevated dd-cfDNA could signal incomplete recovery and the need for closer monitoring and/or additional treatment. However, future studies should be carried out to validate these findings in a larger, prospective cohort that includes consensus on markers of rejection resolution, standardized dd-cfDNA and serum creatinine assessments, and longer follow-up to determine the effect on long-term kidney allograft health.

Bu et al (2021) noted that the use of routine monitoring of dd-cfDNA after kidney transplant may allow clinicians to identify subclinical allograft injury and intervene before development of clinically evident graft injury. To evaluate this, data from 1,092 kidney transplant recipients monitored for dd-cfDNA over a 3-year period was analyzed to examine the association of dd-cfDNA with histologic evidence of allograft rejection. Elevation of dd-cfDNA (0.5 % or more) was significantly correlated with clinical and sub-clinical allograft rejection. dd-cfDNA values of 0.5 % or more were associated with a nearly 3-fold increase in risk development of de-novo DSAs (HR 2.71) and were determined to be elevated a median of 91 days (IQR of 30 to 125 days) ahead of DSA identification. Persistently elevated dd-cfDNA (more than 1 result above the 0.5 % threshold) predicted over a 25 % decline in the eGFR over 3 years (HR 1.97); thus, routine monitoring of dd-cfDNA allowed early identification of clinically important graft injury. Biomarker monitoring complemented histology and traditional laboratory surveillance strategies as a prognostic marker and risk-stratification tool post-transplant; therefore, persistently low dd-cfDNA levels may accurately identify allograft quiescence or absence of injury, paving the way for personalization of immunosuppression trials. Moreover, these researchers states that additional interventional studies are underway to help better define how the information provided by dd-cfDNA can be used to guide clinical practice and decisions regarding immunomodulation, management of infection, treatment of all types of rejection, and control or even prevent the formation of de-novo DSAs.

The authors stated that the drawbacks of this study primarily reflected its observational, real-world design. Comparison with UNOS data suggested that clinical determination did not bias inclusion of patients across these 7 centers and that this cohort truly represented the wider transplant population. As clinicians were unblinded with regard to dd-cfDNA measurements and other clinical data, clinical treatment may have altered the natural history of disease and affected the correlations reported. Furthermore, logistical constraints led to dd-cfDNA levels and biopsies not always being concurrently obtained. To account for these barriers, these investigators allowed biopsies carried out less than or equal to 30 days after dd-cfDNA levels to be considered as paired results. While it was possible that sub-clinical rejection may have resolved before biopsy, this effect would most likely have biased the study toward the null finding and thus should not invalidate the findings reported. Verification bias was a consideration as biopsies were carried out locally and not all read or acted on centrally. However, with data showing consistent patterns despite this heterogeneity, the results identified clear direction for future work. Missing values causing ascertainment bias in the absence of a control group in the prediction model analysis was also a consideration; however, these researchers felt the large sample size limited this, where longitudinal serial samples allowed patients to be their own control. Another potential drawback was that testing was more frequently done in the 1st year of transplant; thus, there was a natural ascertainment and selection bias as allo-immune injury and infection were more common during this period; however, this followed the routine clinic schedule so again it reflected real-life practice. The authors stated that further investigation is needed to establish the optimal

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interval of monitoring as there is clear multi-factorial value considering dd-cfDNA as part of the clinical assessment of the patient. Finally, heterogeneity of dd-cfDNA levels between patients, underlying pathology, effect of interventions impacting the degree of association between dd-cfDNA measurements, and clinical evidence need to be considered. In the future, Bayesian probability evaluation incorporating knowledge of the patient's past clinical course as well as current presentation need to be considered in modeling algorithms to reduce the impact of this heterogeneity.

Agbor-Enoh et al (2021) stated that after heart transplantation, endomyocardial biopsy (EMBx) is used to monitor for acute rejection (AR). Unfortunately, EMBx is invasive, and its conventional histological interpretation has limitations. This is a validation study to assess the performance of a sensitive blood biomarker -- %ddcfDNA -- for detection of AR in cardiac transplant recipients. This prospective cohort, multi-center study recruited heart transplant subjects and collected plasma samples contemporaneously with EMBx for %ddcfDNA measurement by shotgun sequencing. Histopathology data were collected to define AR, its 2 phenotypes (ACR and AMR), and controls without rejection. The primary analysis was to compare %ddcfDNA levels (median and interquartile range [IQR]) for AR, AMR, and ACR with controls and to determine %ddcfDNA test characteristics using receiver-operator characteristics (ROC) analysis. The study included 171 subjects with median post-transplant follow-up of 17.7 months (IQR, 12.1 to 23.6), with 1,392 EMBx, and 1,834 %ddcfDNA measures available for analysis. Median %ddcfDNA levels decayed after surgery to 0.13 % (IQR, 0.03 % to 0.21 %) by 28 days. Furthermore, %ddcfDNA increased again with AR compared with control values (0.38 % [IQR, 0.31 % to 0.83 %], versus 0.03 % [IQR, 0.01 % to 0.14 %]; $p < 0.001$). The rise was detected 0.5 and 3.2 months before histopathologic diagnosis of ACR and AMR. The area under the receiver operator characteristic curve for AR was 0.92. A 0.25 %ddcfDNA threshold had a negative predictive value for AR of 99 % and would have safely eliminated 81 % of EMBx. In addition, %ddcfDNA showed distinctive characteristics comparing AMR with ACR, including 5-fold higher levels (AMR greater than or equal to 2, 1.68 % [IQR, 0.49 % to 2.79 %] versus ACR grade greater than or equal to 2R, 0.34 % [IQR, 0.28 5 to 0.72 %]), higher area under the receiver operator characteristic curve (0.95 versus 0.85), higher guanosine-cytosine content, and higher percentage of short ddcfDNA fragments. The authors found that %ddcfDNA detected AR with a high area under the receiver operator characteristic curve and negative predictive value (NPV). Monitoring with ddcfDNA demonstrated excellent performance characteristics for both ACR and AMR and led to earlier detection than the EMBx-based monitoring. This study supported the use of %ddcfDNA to monitor for AR in patients with heart transplant and paved the way for a clinical utility study.

The authors stated that this study had several drawbacks. First, variabilities in the management of post-operative immunosuppression among the 5 GRAfT centers could have affected the incidence of allograft rejection, its detection, and the levels of ddcfDNA. Furthermore, the prospective cohort study design used was beneficial because it provided a real-world analysis of the performance of ddcfDNA; however, this might have contributed to variabilities in patient care. Second, although these findings indicated significantly higher levels of ddcfDNA levels in patients with systolic dysfunction on echocardiography and rejection on biopsy, concomitant measurements of allograft function and histology were only available for 1/3 of the study subjects. Third, the use of a sub-optimal "gold standard," the EMBx, to assess a new paradigm-shifting modality, ddcfDNA. The explosion of next generation sequencing technologies coupled with computational pipelines for data processing has tremendous impact in basic science research. These advancements are being translated to highly effective diagnostic tools for application to clinical care as evidence by prenatal diagnosis. However, the advent of a paradigm-shifting scientific advancements poses challenges to conventional statistical methods that employ sensitivity and specificity measures to assess the effectiveness of a new diagnostic against an established standard, in this case the EMBx. The concept of a paradigm shift in science was first described by American physicist and philosopher Thomas Kuhn, as a "fundamental change in the basic concepts and experimental practices of a scientific discipline". The ddcfDNA offers a fundamental change in the basic concept of allograft rejection by shifting away from histologic evidence of inflammatory infiltrate and myocyte damage, to a direct quantitative measure of donor-derived DNA in the recipient's plasma. These findings indicated that ddcfDNA was elevated several weeks before histologic features of rejection were apparent, created an incorrect assignment as "false positive" and an inaccurate assessment of the test characteristics that led to a low positive predictive value (PPV). Thus, clinical researchers should question the validity of the biopsy as the "gold standard" for rejection in the current era of highly precise genomic-based assays and consider alternative approaches to

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assessing the ability of a new biomarker such as ddcfDNA to identify patients at the earliest stages of acute rejection.

Puliyanda et al (2021) examined the use of dd-cfDNA for identifying allograft rejection in pediatric patients. Between October 2017 and October 2019, a total of 67 patients, who underwent initial testing with dd-cfDNA as part of routine monitoring or in response to clinical suspicion for rejection, were included. Biopsies were carried out when dd-cfDNA of greater than 1.0 % or where clinical suspicion was high. Demographics, dd-cfDNA, antibody status, and biopsies were collected prospectively. Data were analyzed to determine predictive value of dd-cfDNA for identifying grafts at risk for rejection; 19 of 67 patients had dd-cfDNA testing as part of routine monitoring with a median dd-cfDNA score of 0.37 (IQR: 0.19 to 1.10); 48 of 67 patients who had clinical suspicion of rejection had median dd-cfDNA score of 0.47 (0.24 to 2.15). DSA-positive recipients had higher dd-cfDNA scores than those who were negative or had AT1R positivity alone (p = 0.003). There was no association between dd-cfDNA score and strength of DSA positivity; 7 of 48 recipients had a biopsy with a dd-cfDNA score of less than 1 %; 2 showed evidence of rejection. Neither DSA nor AT1R positivity was statistically associated with biopsy-proven rejection. However, dd-cfDNA of greater than 1 % was diagnostic of rejection with sensitivity of 86 % and specificity of 100 % (AUC: 0.996, 0.98 to 1.00; p = 0.002). The authors concluded that dd-cfDNA represented a non-invasive method for early detection of rejection in pediatric renal transplants. The findings of this study showed dd-cfDNA to be highly predictive of histological rejection and superior to other indicators such as graft dysfunction or antibody positivity alone. Moreover, these researchers stated that further studies in larger separate cohorts are needed to validate dd-cfDNA as a useful non-invasive tool for the identification of grafts at high risk of rejection to inform and guide patient monitoring and management, and ultimately improve outcomes.

The authors stated that this study had several drawbacks. This was a prospective clinical study; therefore, there was no standardization of time-points when samples and biopsies were taken, and there were no protocol biopsies. Thus, power to assess dd-cfDNA as a diagnostic test was limited. Data relating to infections were unavailable, and so, it was not possible to conclude whether these may confound ddcfDNA results. The study was also limited by sample size and the granularity of data available.

Furthermore, an UpToDate review on “Kidney transplantation in adults: Investigational methods in the diagnosis of acute renal allograft rejection” (Anglicheau et al, 2021) lists “donor-derived cell-free DNA” as one of the investigational methods.

Nissaisorakarn et al (2022) stated that dd-cfDNA surveillance testing has never been studied in comparison with other surveillance tests. These investigators described their center's clinical experience with routine dd-cfDNA monitoring and examined if monitoring dd-cfDNA by protocol would provide additional information that aids in detection of AR. They implemented the dd-cfDNA (Allosure) surveillance protocol in addition to measurements of serum creatinine, proteinuria, and DSA. These investigators retrospectively reviewed all kidney recipients transplanted from July 2018 to April 2020. A total of 366 dd-cfDNA test results were reviewed from 82 patients. There were 13/366 positive dd-cfDNA tests in 8/82 patients; 5 of the 8 patients had kidney biopsy that showed: 3 rejections (1 antibody-mediated rejection, 1 T-cell-mediated rejection, and 1 mixed), 1 acute tubular necrosis, and 1 transplant glomerulopathy. The remaining 3 patients did not undergo a biopsy and repeat dd-cfDNA testing improved without intervention. In the 353/366 negative dd-cfDNA tests in 74 patients: 7 patients underwent a biopsy: 1 patient with increased creatinine showed borderline cellular rejection, 3 had recurrent disease (membrano-proliferative glomerulonephritis, diabetes mellitus, immunoglobulin A nephropathy), and 3 showed interstitial fibrosis and tubular atrophy. dd-cfDNA levels were not elevated in recipients with infection (BK viruria/viremia, CMV viremia, or urinary tract infection (UTI). The authors concluded that the addition of surveillance dd-cfDNA testing resulted in marginal added benefit. Whether this would offset the cost of testing needs to be further examined. These researchers stated that in this cohort of low-risk patients, the cost of protocol dd-cfDNA testing may not be justified by its limited benefits.

Halloran et al (2022) stated that dd-cfDNA fraction and quantity have both been shown to be associated with allograft rejection. In a prospective, biopsy-matched, multi-center study, these researchers compared the relative predictive power of each of these variables to the combination of the 2; and developed an algorithm incorporating both variables to detect AR in renal allograft biopsies. The first 426 sequential indication

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biopsy samples collected from the Trifecta Trial with microarray-derived gene expression and dd-cfDNA results were included. After exclusions to simulate intended clinical use, 367 samples were analyzed. Biopsies were evaluated using the molecular microscope diagnostic system and histology. Logistic regression analysis examined if combining dd-cfDNA fraction and quantity would enhance predictive value to either alone. The first 149 sequential samples were used to develop a 2-threshold algorithm and the next 218 to validate the algorithm. In regression, the combination of dd-cfDNA fraction and quantity was found to be significantly more predictive than either variable alone ($p = 0.009$ and $p < 0.0001$). In the test set, the AUC of the 2-variable system was 0.88, and performance of the 2-threshold algorithm showed a sensitivity of 83.1 % and specificity of 81.0 % for molecular diagnoses and a sensitivity of 73.5 % and specificity of 80.8 % for histology diagnoses. The authors concluded that this study in KTx patients found that the combination of dd-cfDNA fraction and quantity was more powerful than either dd-cfDNA fraction or quantity alone and validated a novel 2-threshold algorithm incorporating both variables.

The authors stated that this study had several drawbacks. First, the algorithm training only allowed variation of 1 of the 2 metrics (dd-cfDNA quantity), whereas the other was held constant (dd-cfDNA fraction, at 1 %). Second, all biopsies in this study were indication biopsies; therefore, the performance of the 2-threshold algorithm in protocol surveillance biopsies was not assessed. A previous study showed the performance of this 2-threshold algorithm in a clinical cohort that included the use of the assay in a surveillance setting and another study showed that dd-cfDNA fraction performs similarly in surveillance and indication biopsies. Third, given the high time-dependency of the rejection states, the use of these findings to protocol surveillance biopsies could not be evaluated without knowing the details of the protocol and patient selection, which differed from center to center. These researchers suggested that the present results anticipate the findings that will be obtained in protocol biopsies in terms of the relationship between quantity and fraction of dd-cfDNA in the rejection states, once the prior probabilities of rejection in the population being tested are taken into account.

The American Society of Transplant Surgeons (ASTS, 2023) has issued a position statement on donor-derived cell-free DNA that concludes that “We strongly encourage commercial payers to align reimbursement practices for molecular diagnostic testing with clinical guidelines and emerging practice patterns, as well as with Medicare beneficiary coverage patterns”. This is a document created for purposes of payor advocacy, and not an evidence-based guideline supported by a systematic evidence review critically appraising the evidence, and with explicit ratings of the strength of evidence and strength of recommendations (IOM, 2011).

Bromberg et al (2024) noted that standard-of-care (SOC) biomarkers for renal allograft rejection are lagging indicators, signaling existing organ injury. This precludes early intervention when immunological cascades leading to rejection are most susceptible. Donor-derived cell-free DNA (dd-cfDNA) shows promise as an early indicator of rejection, allowing earlier and possibly more effective treatment. This analysis was designed to evaluate this promise approach using real-world dd-cfDNA testing evidence. This retrospective analysis of the prospective, observational ProActive registry study examined dd-cfDNA and serum creatinine levels before biopsy in 424 patients with 1 or more dd-cfDNA test ($n = 1,013$) in the 6 months before biopsy. Of 4,667 enrolled patients, 1,631 patients had 18 months or longer of follow-up data, of which 424 had a biopsy and were included in this analysis; 26 biopsies showed ABMR, 62 showed TCMR, and 336 showed non-rejection; each from a unique patient. dd-cfDNA fractions were significantly elevated 5 months before ABMR biopsies, and 2 months before TCMR biopsies, compared with non-rejection biopsies. In contrast, serum creatinine did not discriminate between rejection and non-rejection in advance, or concurrent with biopsy. Among patients with non-rejection biopsies, eGFR rate was significantly lower in cases with 2 or more increased dd-cfDNA results (1 % or higher), compared with those with 0 or 1 increased dd-cfDNA result. The authors concluded that these findings indicated that dd-cfDNA is an early indicator of biopsy-proven rejection, especially ABMR, suggesting a greater role for dd-cfDNA in surveillance to identify patients at high risk of ongoing or future rejection; therefore, requiring closer monitoring, biopsy, or other management changes. Moreover, these researchers stated that future analysis from the ProActive study further examining the value of dd-cfDNA surveillance are planned.

The authors stated that this trial had several drawbacks. First, the study was observational, meaning the study did not mandate pre-determined protocols in the management of these patients, including standardized

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intervals or frequencies of diagnostic testing or treatment, or requiring a biopsy after an increased dd-cfDNA result. However, given the heterogeneous clinical practice patterns in post-kidney transplant care, capturing the use of dd-cfDNA testing in a real-world clinical setting was best accomplished with an observational registry design. For example, in a controlled trial with a biopsy mandated after increased dd-cfDNA results, it would not have been possible to show that dd-cfDNA is an early indicator of biopsy-proven rejection. Interestingly, if the study had involved blinded dd-cfDNA results, a number of ABMR biopsies likely would have been performed later, suggesting that the lead time of dd-cfDNA elevation in relation to biopsy-proven ABMR reported in this study may in fact be an under-estimate. The fact that dd-cfDNA was shown to be prognostic of biopsy-proven ABMR across a heterogeneous set of management protocols indicated that the test should perform well in a variety of clinical settings. Furthermore, the fact that only a subset of patients had a biopsy when the dd-cfDNA first rose above 1 % meant that it was not possible to conclude if dd-cfDNA was elevated in advance of the histological onset of rejection, or elevated because of ongoing sub-clinical rejection. Lastly, the study was not designed to be a diagnostic clinical validation, and the performance estimates represented those expected in real-world practice.

VitaGraft Kidney (Oncocyte Corporation, Irvine, CA) is a blood-based transplant monitoring test that uses a droplet-digital polymerase chain reaction (dd-PCR) technology to quantify the concentration of donor-derived cell-free (dd-cfDNA) following kidney transplant. A set of single-nucleotide polymorphisms (SNPs) specific to each transplant recipient is determined using dd-PCR. The selection of a subset of SNPs from commonly-shared SNPs across the global population allows for differentiation between the donor and recipient cfDNA. The dd-cfDNA is measured using absolute and relative quantification from the selected personalized assay. VitaGraft Kidney is intended to manage the risk of kidney transplant rejection where an increase in dd-cfDNA may indicate rejection.

In a prospective, observational, single-center study, Kim et al (2024) examined the clinical validity of donor-derived cell-free DNA (dd-cfDNA) in comparison with that of donor specific anti-HLA antibody (DSA) for predicting biopsy-proven rejection (BPR) and severe microvascular inflammation (severe MVI) in kidney transplant recipients (KTRs). This trial included 64 KTRs who underwent the indicated biopsies. Blood samples collected before biopsy were tested for dd-cfDNA and DSA. Biopsy specimens were classified by a renal pathologist according to the Banff classification. The predictive performance of dd-cfDNA and DSA for histological allograft diagnosis was evaluated. KTRs were categorized into the high and low dd-cfDNA groups based on a level of 0.4 %. A total of 18 patients (28.1 %) had positive DSA at biopsy, exhibiting higher dd-cfDNA levels than the DSA-negative patients. BPR and severe MVI incidences were elevated in the high dd-cfDNA group (BPR: 42.9 % versus 3.4 %, $p < 0.001$; severe MVI: 37.1 % versus 3.4 %, $p = 0.001$). Furthermore, elevated glomerulitis and MVI scores were observed in the high dd-cfDNA group. DSA showed the highest predictive value for BPR (AUC = 0.880), whereas dd-cfDNA alone excelled in predicting severe MVI (AUC = 0.855). Combination of DSA and dd-cfDNA (greater than 0.4 %) yielded sensitivities of 80.0 % and 50.0 % with specificities of 90.7 % and 88.0 % for antibody-mediated rejection (ABMR) and severe MVI detection, respectively. The authors concluded that the dd-cfDNA test was a predictive tool for BPR and severe MVI, and it could improve the performance, especially when combined with DSA for BPR. Moreover, these investigators stated that further research should be carried out to validate these findings in a larger cohort and to expand its clinical validity to an immune monitoring marker and a long-term prognostic marker. Furthermore, serial monitoring of dd-cfDNA may also be of interest for examining the therapeutic effect of novel treatment for ABMR, such as the anti-CD38 monoclonal antibody, felzartamab, which is currently evaluated in clinical trials

The authors stated that this study had several drawbacks. First, it was a single-center study with a relatively small sample size (n = 64). However, considering that the study population consisted of patients who underwent kidney transplantation and indication-based biopsies, the number of participants was acceptable. In addition, the number of subjects was comparable to those in the other previously published studies. Second, these investigators only measured the baseline dd-cfDNA level at the time of allograft biopsy. Analyzing the treatment response based on the trend of dd-cfDNA changes in patients diagnosed with rejection through serial testing could further expand the clinical application of dd-cfDNA. Third, in this trial, these researchers measured dd-cfDNA as a fraction of total circulating cell-free DNA. %dd-cfDNA is influenced not only by allograft damage but also by the recipient's total cell-free DNA. In other words, if the release of recipient DNA increases due to factors like infection or exercise, %dd-cfDNA may decrease. As

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previously reported, absolute quantification of dd-cfDNA could aid in avoiding false negative results and improve the predictive value of dd-cfDNA for allograft injury. Moreover, the authors excluded patients who received kidney transplants more than twice or those who were multiple organ transplant recipients due to methodological limitations. Recently, Pettersson et al (2024) presented data on a new next-generation sequencing (NGS)-based dd-cfDNA assay that could quantify dd-cfDNA from multiple grafts both in-vitro using generated mixed samples and in a cohort of patients who had received 2 kidney transplants. Consequently, additional larger studies including patients with multiple transplants are needed to confirm the findings of this trial.

Akifova et al (2024) stated that AMR is a major cause of graft failure limiting long-term graft survival following KTx. Current diagnostic strategy to detect AMR is sub-optimal and requires further improvement. Previously suggested treatment regimens for AMR could not show effectiveness; however, novel therapeutic agents are currently under investigation. Donor-derived cell-free DNA is a novel non-invasive biomarker for allograft injury that has been mainly examined in the context of rejection. Its short-half-life in circulation and injury-dependent release are its key advantages that contribute to its superior diagnostic accuracy, compared to traditional biomarkers. Moreover, previous studies demonstrated that dd-cfDNA-release is well-linked to histological and molecular features of AMR, and thus able to reflect real-time injury. Further observations suggested that dd-cfDNA could be used as a suitable screening tool for early detection of AMR in patients with DSA, as well as for monitoring AMR activity after anti-rejection treatment. The authors concluded that the current evidence suggests that the integration of dd-cfDNA in the graft surveillance of patients with AMR, or those suspicious of AMR (e.g., due to the presence of DSA) has additional value and may aid in improving outcomes in this specific cohort. These researchers stated that dd-cfDNA could be helpful in personalized post-transplantation therapy with the potential to reduce premature graft loss. Moreover, these investigators stated that there are still some limitations and knowledge gaps that deserve consideration and need additional research, such as the biological background of release, optimal frequency of dd-cfDNA testing, comparability and cost of different assays as well as cost-effectiveness of different approaches. Within the context of DSA-positive patients more data are needed for an optimal clinical implementation; and ideally future prospective, randomized studies with robust endpoints and sufficient follow-up are needed to provide more evidence on the clinical impact of complementary dd-cfDNA monitoring on the clinical course of patients with DSA and AMR.

Botella et al. (2024) evaluated whether dd-cfDNA levels can help identify adverse events, particularly rejection, in kidney transplant recipients during the first two years post-transplant. In this study, 30 kidney transplant patients were monitored. dd-cfDNA levels were measured at 15 days, 3, 6, and 9 months post-transplant. The study compared dd-cfDNA results with traditional markers like serum creatinine and previously published reference values. The investigators found that elevated dd-cfDNA (>1%) was found in 6 of 9 patients with active rejection (either antibody-mediated or T-cell mediated). Moderately elevated levels (>0.5%) were observed in 2 additional patients with rejection. The diagnostic performance of dd-cfDNA was modest, with a sensitivity of 50%, a specificity of 61.11%, and a negative predictive value (NPV) of 64.71% and a positive predictive value (PPV) of 46.13%. dd-cfDNA showed better performance than creatinine in detecting T-cell mediated rejection (TCMR), especially at 15 days post-transplant. It may also help distinguish patients with donor-specific antibodies (dnDSAs) who are at risk of rejection. The study found that, while dd-cfDNA shows promise, especially in early detection and in specific rejection types like TCMR, its overall diagnostic accuracy in this study was limited. The authors suggest it could be a complementary tool rather than a standalone diagnostic method.

Chandar et al. (2025) explored the clinical utility of donor-derived cell-free DNA (dd-cfDNA) testing in pediatric kidney transplant recipients, focusing on its role in identifying biopsy-proven acute rejection (BPAR) and immune activation. Conducted at a single center, the study reviewed 106 pediatric patients (mean age 11 years) who underwent dd-cfDNA testing due to rising creatinine levels, the appearance of de novo anti-HLA antibodies (dnHLAab), or other clinical indications. Results showed that patients with dd-cfDNA levels >1% were significantly more likely to have BPAR (59%) and dnHLAab (88.1%), particularly class II antibodies (DQ and DR) and donor-specific antibodies (DSAs). In contrast, low dd-cfDNA levels were associated with immunoquiescent states. Limitations of this study include its single-center design and the fact that the analysis was based on a retrospective chart review, Biopsies were performed only when clinically indicated, not as part of a standardized protocol. The study did not fully account for other variables

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that might influence dd-cfDNA levels, such as infections, medication adherence, or concurrent illnesses, which could affect the interpretation of elevated values.

Nie et al. (2025) investigated whether combining two dd-cfDNA metrics -- fraction (%) and absolute quantity (copies/mL) -- could improve the accuracy of diagnosing kidney transplant rejection. The investigators conducted a prospective study involving 50 kidney transplant recipients who underwent biopsy for suspected rejection (including T-cell-mediated rejection, antibody-mediated rejection, and borderline cases). The investigators reported that a dd-cfDNA fraction cutoff of 1.08% achieved a sensitivity of 93.33%, a specificity of 91.43%, and an Area Under Curve (AUC) of 0.95. The NPV was 96.97% and the PPV was 82.35%. An absolute dd-cfDNA threshold of 32 copies/mL yielded a sensitivity of 80.00%, specificity of 71.43%. and AUC of 0.78. NPV was 89.29% and PPV was 54.55%. The investigators concluded that the combination of dd-cfDNA fraction and absolute quantification significantly enhances diagnostic accuracy compared to using either metric alone. This dual-parameter approach may offer a more reliable, non-invasive method for early detection of kidney allograft rejection. Limitations of the study include its single center design, small sample size, and lack of longitudinal data.

Furthermore, an UpToDate review on “Overview of care of the adult kidney transplant recipient” (Chandraker and Yeung, 2025) states that “Monitoring Kidney Allograft Function -- The routine evaluation of kidney allograft function typically involves monitoring the serum creatinine level and screening for proteinuria. Some patients with evidence of kidney allograft dysfunction and/or proteinuria may require a kidney biopsy to determine the cause of these abnormalities. A number of newer tests are now commercially available for allograft function monitoring, including donor-derived cell-free DNA (dd-cfDNA) testing as well as gene expression profiling. The exact role for how these tests should be used has yet to be determined, but registry studies are underway to better understand their utility. In addition to the individual assays of allograft function, composite scoring systems, such as the Integrative Box (iBox) Scoring System, are being developed, which may be able to predict long-term allograft function using data available at earlier time points post-transplantation”.

Measurement of Angiotensin II Type 1 (AT1) Receptors or AT1 Antibodies for Evaluation of Renal Transplantation Candidates / Recipients

Michielsen and associates (2016) stated that HLA antibodies play a major role in the recipient's immune response against the renal allograft and are an established risk factor for antibody-mediated rejection (AMR) and subsequent impaired graft survival. Evidence originating from HLA-identical donor-recipient pairs indicated that non-HLA antibodies may play a role as well. Numerous non-HLA antibodies have been identified in renal organ transplantation, directed against a heterogeneous subset of both allo- and auto-antigens including MHC Class-I-related chain A (MICA) and Angiotensin II type 1 receptor (AT1R). These researchers discussed the mechanisms predisposing to non-HLA antibody formation, the possible synergy with HLA-antibodies in their pathologic potential and the mechanisms involved in allograft damage. Furthermore, an overview of the identified non-HLA antibodies and antigens and their relation with rejection and graft survival was provided. The authors stated that “ … these studies all indicate that AT1R-antibodies are associated with an increased incidence of graft failure. Remarkably, despite the fact that all studies used the same ELISA-based assay, the prevalence of AT1R-antibodies pre-transplantation ranged from 17 % to 47 %”. Moreover, they noted that the determination of the exact clinical relevance of non-HLA antibodies in renal transplantation is impaired by highly heterogenic study designs including differences in testing methods, immunosuppressive regimens and outcome measures. Considering the technical difficulties of current non-HLA antibodies assays and the large variation in reported incidences of antibodies even with the same assays, continuous efforts to develop reliable and sensitive diagnostic tests are essential.

Pinelli and colleagues (2017) noted that endothelial cell antigens have been reported as potential targets for antibodies in the context of organ transplantation, leading to increased risk for graft failure. Serum samples from 142 consecutive living donor kidney recipients were tested for the presence of antibodies to AT1R, donor endothelial cells, and donor HLA. Graft survival was monitored for 5 years post-transplant, and secondary outcomes, including biopsy-proven rejection, proteinuria, biopsy-proven vasculopathy, and renal function based on serum creatinine were also assessed for the first 2 to 3 years. AT1R antibody levels were

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positive (greater than 17U/ml) in 11.3 %, 18.8 % and 8.1 % of patients pre-transplant, post-transplant and at time of indication biopsy, respectively. XM-ONE assay was positive in 17.6 % of patients pre-transplant (7 IgG+; 15 IgM+; 3 IgG+/IgM+). Overall, 4 patients experienced AMR, 31 borderline cellular rejection (BCR), 19 cellular rejection (CR) and 3 mixed AMR and CR within the first 24months. The authors concluded that while pre-existing and de-novo donor-specific HLA antibodies were associated with graft failure and many secondary outcomes, no statistical association was found for either anti-endothelial or anti-AT1R antibodies, indicating that these tests may not be the best predictors of graft outcome in living donor renal transplantation.

Furthermore, UpToDate reviews on "Evaluation of the potential renal transplant recipient" (Rossi and Klein, 2018), and "Overview of care of the adult kidney transplant recipient" (Chandraker and Yeung, 2018) do not mention measurement of AT1 receptors or AT1 antibodies as a management tool.

Kang and colleagues (2022) noted that anti-AT1R antibodies (AT1R-Abs) have been recognized as non-HLA antibodies associated with allograft rejection and poor allograft outcomes following KTx. In a systematic review and meta-analysis, these researchers examined the risk anti-AT1R-Abs pose for rejection and graft loss among KTx recipients. They systematically searched PubMed, Embase, and the Cochrane Library databases for relevant articles published from inception until June 2021 to identify all studies concerning the role AT1R-Abs play in the clinical outcome following KTx. Two reviewers independently identified studies, abstracted outcome data, and assessed the quality of the studies. The meta-analysis was summarized using the fixed-effects models or random-effects models, according to heterogeneity. The major outcomes included delayed graft function, acute rejection, graft loss, or patient death after transplantation. A total of 21 eligible studies involving a total of 4,023 KTx recipients were included in this review. Meta-analysis results showed that the AT1R-Ab positive kidney transplant (KT) group had a greater incidence of AMR (RR = 1.94, 95 % CI: 1.61 to 2.33, $p < 0.00001$) and graft loss (RR = 2.37, 95 % CI: 1.50 to 3.75, $p = 0.0002$) than did the AT1R-Abs negative KT group. There was no significant statistical difference in delayed graft function rate, T-cell mediated rejection, mixed rejection, acute cellular rejection (ACR), acute rejection, and patient death rate between AT1R-Ab positive KT and AT1R-Ab negative KT groups. The authors concluded that the findings of this study showed that the presence of anti-AT1R-Abs was associated with a significantly higher risk of AMR and graft loss in KTx. Moreover, these researchers stated that future studies are still needed to examine the importance of routine anti-AT1R monitoring and therapeutic targeting.

Complement Inhibitors (e.g., Eculizumab) for the Treatment of Antibody-Mediated Rejection

Wan and colleagues (2018) stated that current treatments for AMR in kidney transplantation are based on low-quality data from a small number of controlled trials. Novel agents targeting B cells, plasma cells, and the complement system have featured in recent studies of AMR. These investigators conducted a systematic review and meta-analysis of controlled trials in kidney transplant recipients using Medline, Embase, and CENTRAL from inception to February 2017. Of 14,380 citations, these researchers identified 21 studies, including 10 RCTs, involving 751 participants. Since the last systematic review conducted in 2011, these researchers found 9 additional studies evaluating plasmapheresis + intravenous immunoglobulin (IVIG) (n = 2), rituximab (n = 2), bortezomib (n = 2), C1 inhibitor (n = 2), and eculizumab (n = 1). Risk of bias was serious or unclear overall and evidence quality was low for the majority of treatment strategies. Sufficient RCTs for pooled analysis were available only for antibody removal, and here there was no significant difference between groups for graft survival (hazard ratio [HR] 0.76; 95 % CI: 0.35 to 1.63; $p = 0.475$). Studies showed important heterogeneity in treatments, definition of AMR, quality, and follow-up. Plasmapheresis and IVIG were used as standard-of-care in recent studies, and to this combination, rituximab appeared to add little or no benefit. Insufficient data were available to assess the efficacy of bortezomib and complement inhibitors. The authors concluded that newer studies evaluating rituximab showed little or no difference to early graft survival, and the efficacy of bortezomib and complement inhibitors for the treatment of AMR remains unclear. Despite the evidence uncertainty, plasmapheresis and IVIG have become standard-of-care for the treatment of acute AMR.

Belimumab for the Treatment of Antibody-Mediated Rejection

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Banham and colleagues (2018) stated that B cells produce allo-antibodies and activate allo-reactive T cells, negatively affecting kidney transplant survival. By contrast, regulatory B cells are associated with transplant tolerance. Immunotherapies are needed that inhibit B-cell effector function, including antibody secretion, while sparing regulators and minimizing infection risk. B lymphocyte stimulator (BLyS) is a cytokine that promotes B-cell activation and has not previously been targeted in kidney transplant recipients. These researchers examined the safety and activity of an anti-BLyS antibody, belimumab, in addition to standard-of-care immunosuppression in adult kidney transplant recipients. They used an experimental medicine study design with multiple secondary and exploratory end-points to gain further insight into the effect of belimumab on the generation of de-novo IgG and on the regulatory B-cell compartment. In a randomized, double-blind, placebo-controlled, phase-II clinical trial, these researchers employed belimumab, in addition to standard-of-care immunosuppression (basiliximab, mycophenolate mofetil, tacrolimus, and prednisolone) at 2 centers. Subjects were eligible if they were aged 18 to 75 years and receiving a kidney transplant and were planned to receive standard-of-care immunosuppression. They were randomly assigned (1:1) to receive either intravenous belimumab 10 mg/kg body weight or placebo, given at day 0, 14, and 28, and then every 4 weeks for a total of 7 infusions. The co-primary end-points were safety and change in the concentration of naive B cells from baseline to week 24, both of which were analyzed in all patients who received a transplant and at least 1 dose of drug or placebo (the modified intention-to-treat [mITT] population). Between September 13, 2013, and February 8, 2015, of 303 patients assessed for eligibility, 28 kidney transplant recipients were randomly assigned to receive belimumab (n = 14) or placebo (n = 14); 25 patients (12 [86 %] patients assigned to the belimumab group and 13 [93 %] patients assigned to the placebo group) received a transplant and were included in the mITT population. These investigators observed similar proportions of adverse events (AEs) in the belimumab and placebo groups, including serious infections (1 [8 %] of 12 in the belimumab group and 5 [38 %] of 13 in the placebo group during the 6-month on-treatment phase; and none in the belimumab group and 2 [15 %] in the placebo group during the 6-month follow-up). In the on-treatment phase, 1 patient in the placebo group died because of fatal myocardial infarction (MI) and acute cardiac failure. The co-primary end-point of a reduction in naive B cells from baseline to week 24 was not met. Treatment with belimumab did not significantly reduce the number of naive B cells from baseline to week 24 (adjusted MD between the belimumab and placebo treatment groups -34.4 cells/µL, 95 % CI: -109.5 to 40.7). The authors concluded that belimumab might be a useful adjunct to standard-of-care immunosuppression in renal transplantation, with no major increased risk of infection and potential beneficial effects on humoral allo-immunity. These preliminary findings need to be validated by well-designed studies.

Genotyping Donors and Recipients Before Renal Transplantation

Huart and colleagues (2018) noted that delayed graft function (DGF) is defined as the need for dialysis within 7 days following KTx. DGF is associated with increased costs, higher risk for acute rejection and decreased long-term graft function. Renal ischemia/reperfusion (I/R) injury plays a major role in DGF occurrence; SNPs in certain genes may aggravate kidney susceptibility to I/R injury, thereby worsening post-transplant outcomes. These investigators presented an extensive review of the literature regarding the putative impact of donor or recipient SNPs on DGF occurrence in kidney transplant recipients (KTRs). Among 30 relevant PubMed reports, 16 articles identified an association between 18 SNPs and DGF. These polymorphisms concerned 14 different well-known genes and 1 not-yet-identified gene located on chromosome 18. They have been categorized into 5 groups according to the role of the corresponding proteins in I/R cascade: oxidative stress, telomere shortening, chemokines, T-cell homeostasis, and metabolism of anti-inflammatory molecules. The remaining 14 studies failed to demonstrate any association between the studied SNPs and the occurrence of DGF. The authors concluded that several polymorphisms in either the donor or the recipient or both have been associated with DGF in KTRs. These polymorphisms are involved in oxidative stress, telomere length, cytokine secretion and modulation, immunity and inflammation. These processes are involved in I/R injury, which is regarded as one of the most important causes of DGF. These researchers stated that identifying the polymorphisms linked to renal I/R may lead to better understand pathophysiology of DGF in KTRs and find new therapeutic targets.

The authors stated that the present review highlighted the state of knowledge in the field of genetic susceptibility to renal I/R. Although SNPs may only have minor impacts per se on gene expression and

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protein function, interactions among multiple SNPs may have a major impact on molecular cascades. Furthermore, some SNPs showed very low frequency. Validation studies are lacking or inadequately powered for most SNPs studied thus far, which may explain the controversial observations. These researchers stated that replication studies will need to include multi-variate analyses to isolate the putative effects of SNPs among other well-established risk factors of DGF. They stated that most importantly, one must clearly distinguish the impact of SNPs in donors versus in recipients versus in both. Polymorphisms involved in I/R severity may be especially relevant in donors, whereas polymorphisms implicated in AR and inflammation may rather concern recipients. These investigators stated that prospective, multi-center studies including patients of various genetic backgrounds are needed to clinically determine the benefits (and harms) of genotyping donors and recipients before KTx.

Routine Stenting of Extravesical Ureteroneocystostomy in Renal Transplantation

Abrol and colleagues (2018) stated that although rare, major urologic complications (MUC) in kidney transplantation can cause significant morbidity, increased cost, and may even lead to graft loss. Ureteric stents are routinely used to prevent MUC, although complications related to their use have been reported. These investigators reviewed the role of routine stenting in preventing MUC in kidney transplantation with extravesical ureteric implantation and performed a meta-analysis of 6 RCTs. A PubMed search was performed for studies on MUC and stents in kidney transplant recipients; RCTs were short-listed for the review following Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines. RevMan 5 was used for statistical analysis, and outcome analysis was done with Cochran-Mantel-Haenszel test using random effect model. A total of 6 trials meeting the criteria were identified. Although stent use did not decrease the incidence of urinary leak (OR] 0.39; 95 % CI: 0.14 to 1.11; p = 0.08) or obstruction (OR, 0.41; 95 % CI: 0.13 to 1.24; p = 0.11), it was associated with a higher incidence of urinary tract infection (UTI; OR, 3.59; 95 % CI: 1.33 to 9.75; p = 0.01). The authors concluded that in the present era of extravesical uretero-vesical anastomosis, routine stenting has a limited role in decreasing MUC and may be associated with higher incidence of UTIs.

Furthermore, an UpToDate review on “Overview of care of the adult kidney transplant recipient” (Chandraker and Yeung, 2019) does not mention routine stenting of extravesical ureteroneocystostomy as a management option of urologic complications in kidney transplantation.

Kidney Molecular Microscope Diagnostic System (MMDx-Kidney)

The Kidney Molecular Microscope Diagnostic System (MMDx-Kidney) refers to mRNA gene expression analysis of 1,494 genes utilizing microarray; it measures mRNA transcript levels in transplant kidney biopsy tissue, with allograft rejection and injury algorithm reported as a probability score.

In a prospective study, Halloran and colleagues (2017) examined the feasibility of real time central molecular evaluation of kidney transplant biopsy samples from 10 North American or European centers. Biopsy samples taken 1 day to 34 years post-transplantation were stabilized in RNAlater, sent via courier over-night at ambient temperature to the central laboratory, and processed (29-hour work-flow) using microarrays to assess T cell- and antibody-mediated rejection (TCMR and ABMR, respectively). Of 538 biopsy samples submitted, 519 (96 %) were sufficient for microarray analysis (average length, 3 mm). Automated reports were generated without knowledge of histology and HLA antibody, with diagnoses assigned based on Molecular Microscope Diagnostic System (MMDx) classifier algorithms and signed out by 1 observer. Agreement between MMDx and histology (balanced accuracy) was 77 % for TCMR, 77 % for ABMR, and 76 % for no rejection. A classification tree derived to provide automated sign-outs predicted the observer sign-outs with greater than 90 % accuracy. In 451 biopsy samples where feedback was obtained, clinicians indicated that MMDx more frequently agreed with clinical judgment (87 %) than did histology (80 %) (p = 0.0042). In 81 % of feedback forms, clinicians reported that MMDx increased confidence in management compared with conventional assessment alone. The authors concluded that real time central molecular assessment was feasible and offered a useful new dimension in biopsy interpretation. These researchers stated that the fact that the clinicians indicated that MMDx would add valuable support for clinical decisions

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beyond their current standard-of-care is encouraging for further development of MMDx testing. The findings of this feasibility study need to be validated by further investigation.

Reeve and associates (2019) previously reported a system for evaluating rejection in kidney transplant biopsies using microarray-based gene expression data, the MMDx. The present study was designed to optimize the accuracy and stability of MMDx diagnoses by replacing single machine learning classifiers with ensembles of diverse classifier methods. These researchers also examined the use of automated report sign-outs and the agreement between multiple human interpreters of the molecular results. Ensembles generated diagnoses that were both more accurate than the best individual classifiers, and nearly as stable as the best, consistent with expectations from the machine learning literature. Human experts had approximately 93 % agreement (balanced accuracy) signing out the reports, and random forest-based automated sign-outs showed similar levels of agreement with the human experts (92 % and 94 % for predicting the expert MMDx sign-outs for TCMR and ABMR, respectively). In most cases disagreements, whether between experts or between experts and automated sign-outs, were in biopsies near diagnostic thresholds. Considerable disagreement with histology persisted. The balanced accuracies of MMDx sign-outs for histology diagnoses of TCMR and ABMR were 73 % and 78 %, respectively. Disagreement with histology was largely due to the known noise in histology assessments.

Furthermore, an UpToDate review on "Clinical features and diagnosis of acute renal allograft rejection" (Brennan et al, 2019) states that "The 2017 Banff Kidney Meeting Report modified the diagnostic criteria for ABMR by stating that both C4d staining and validated molecular assays could serve as potential alternatives to DSAs in the diagnosis of ABMR".

Perfusate Biomarkers Produced During Hypothermic Machine Perfusion for Prediction of Graft Outcomes in Kidney Transplantation

Guzzi and colleague (2020) stated that there is good evidence to support the use of hypothermic machine perfusion (HMP) over static cold storage as the favored preservation method for deceased donor kidneys. However, the utility of HMP as a tool to evaluate the viability of kidneys for transplant is unclear. There is a need to examine if perfusate biomarkers produced during HMP could predict post-transplant outcomes and evaluate the suitability of organs for transplantation. Three different databases (Medline, Embase, Transplant Library) were screened to May 31, 2019. Articles were included if a relationship was reported between 1 or more perfusate biomarkers and post-transplant outcomes. Studies were assessed and graded for methodological quality and strength of evidence. Glutathione S-transferase was the most promising biomarker for predicting delayed graft function (DGF), but its predictive ability was at best moderate. Analysis of primary non-function rates was challenging due to low occurrence rates and small sample sizes. The authors concluded that existing studies were limited in quality and have not yielded biomarkers for kidneys undergoing HMP that are able to predict post-transplant outcomes with sufficient accuracy to support routine clinical use. Moreover, these researchers stated that further studies with larger samples and more robust methodology are needed.

Recombinant Human Erythropoietin for Nephron-Protection in Persons Undergoing Kidney Transplantation

Vlachopanos and associates (2015) stated that DGF due to ischemia-reperfusion injury is a major early complication of KTx. Recombinant human erythropoietin (rHuEPO) has been shown to exert nephron-protective action in animal models. In a meta-analysis, these investigators examined the impact of rHuEPO on DGF in KTx. Eligible studies comparing peri-operative high-dose rHuEPO with placebo or no therapy for prevention of DGF were identified through Medline, CENTRAL, and Transplant Library. Their design and data were assessed by 2 independent reviewers. Among 737 examined studies, 4 RCTs involving 356 recipients of kidney allografts from deceased donors, fulfilled inclusion criteria. Statistical heterogeneity across studies was not significant ($p = 0.98$, $I(2) = 0$ %). In a random effects model, no significant difference was found in the occurrence of DGF (OR: 0,74, 95 % CI: 0.47 to 1.18, $p = 0.21$). At 4 weeks after KTx, the rHuEPO group exhibited higher systolic blood pressure (SBP) (MD: 6.47 mmHg, 95 % CI: 1.25 to 11.68, $p = 0.02$). The authors concluded that peri-operative, high-dose rHuEPO administration did not prevent DGF in

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deceased donor KTx. Furthermore, it was associated with higher SBP leading to safety concerns. These researchers stated that non-erythropoietic rHuEPO derivatives, designed for nephron-protective action without increasing cardiovascular risk, might prove an alternative but still are at early stages of development.

Zhou and colleagues (2020) noted that the protective effect of rHuEPO on KTx has not been established. In a systematic review and meta-analysis, these researchers examined the potential influence of rHuEPO on transplanted kidneys. They identified relevant studies, searched electronic databases (PubMed, Medline, Embase, Ovid, the Cochrane Library, and major nephrology journals) from inception until June 15, 2018; 2 independent reviewers assessed study quality. The systematic review and meta-analysis were performed with fixed- or random-effects models according to heterogeneity, and results were expressed as RR or weighted MD (WMD). A total of 6 RCTs with 435 patients met the inclusion criteria. Compared with placebo, rHuEPO had no statistically significant effect on DGF (RR = 0.89, 95 % CI: 0.73 to 1.07; p = 0.22) and slow graft function (RR = 0.93, 95 % CI: 0.60 to 1.43; p = 0.73). The rHuEPO and control groups did not differ in thrombo-embolic events, mortality, AR, and blood transfusion. A significant difference was found in long-term eGFR (RR = 3.65, 95 % CI: -4.45 to 11.75; p = 0.003). The authors concluded that these findings suggested that rHuEPO had a limited nephron-protective effect in patients undergoing KTx and did not increase the susceptibility to AEs.

Renal Auto-Transplantation for the Treatment of Loin Pain Hematuria Syndrome

Almaiman and associates (2013) reviewed retrospectively their experience with laparoscopic approach to renal auto-transplantation (RAT) in 4 patients using a single iliac incision in the management of loin pain hematuria syndrome (LPHS). Patients with LPHS (all women, mean age of 29.5 years, range of 23 to 36 years) underwent 4 technically successful laparoscopic nephrectomies with RAT, using a single iliac incision to both harvest and transplant the kidney. Hand assistance was used in 2 patients immediately before clamping the renal pedicle. All patients required narcotic analgesics pre-operatively. Mean total surgical time was 4.1 hours. For laparoscopic donor nephrectomy phase, mean operative time was 1.9 hours. The warm ischemia time was 5 mins; the cold ischemia time was 58 mins. The hospital stay was 6 days. None of the patients had abnormal renal function post-operatively; 3 of 4 patients had episodes of iliac fossa pain with effort at the level of the transplantation incision; 2 of 4 patients became morphine-free. The other 2 required a significantly reduced dose of oral narcotics. None of these patients required nephrectomy; median follow-up duration was 9 months. The authors concluded that laparoscopic approach to RAT using a single extended iliac incision in the management of LPHS can be considered as a less invasive treatment compared to open RAT in selected patients. This technique may be extended to patients having other conditions requiring auto-transplantation.

Taba Taba Vakili et al (2014) noted that LPHS is a rare disease with a prevalence of approximately 0.012 %. The most prominent clinical features include periods of severe intermittent or persistent unilateral or bilateral loin pain accompanied by either microscopic or gross hematuria. Patients with LPHS initially present with hematuria, flank pain, or most often both hematuria and flank pain. Kidney biopsies from patients with loin pain hematuria typically reveal only minor pathologic abnormalities. Further, loin pain hematuria syndrome is not associated with loss of renal function or urinary tract infections (UTIs). LPHS-associated hematuria and pain are postulated to be linked to vascular disease of the kidney, coagulopathy, renal vasospasm with microinfarction, hyper-sensitivity, complement activation on arterioles, venocalyceal fistula, abnormal ureteral peristalsis, and intra-tubular deposition of calcium or uric acid microcrystals. Many patients with LPHS also meet criteria for a somatoform disorder, and analgesic medications, including narcotics, are often used to treat LPHS-associated pain. Interventional treatments include renal denervation, RAT, and nephrectomy; however, these methods should be used only as a last resort when less invasive measures have been tried unsuccessfully.

Zubair and colleagues (2016) stated that LPHS, first described in 1967, is a rare pain syndrome, which is not well understood. The syndrome is characterized by severe intermittent or persistent flank pain, either unilateral or bilateral, associated with gross or microscopic hematuria. LPHS is a diagnosis of exclusion as there still is not a consensus of validated diagnostic criteria, though several criteria have been proposed. The wide differential diagnosis would suggest a meticulous yet specific diagnostic work-up depending on the individual clinical features and natural history. Several mechanisms regarding the pathophysiology of LPHS

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have been proposed but without pin-pointing the actual causative etiology, the treatment remains symptomatic. Treatment modalities for LPHS are diverse including simple analgesia, opioid analgesic and RAT. The authors summarized the current understanding regarding the pathophysiology of LPHS along with the steps required for proper diagnosis and a discussion of the different therapeutic approaches for LPHS.

Campsen and co-workers (2019) stated that in patients with LPHS, a response to percutaneous renal hilar blockade (RHB) and a multi-disciplinary team (MDT) evaluation predicts patient's potential RAT success. A pain assessment was performed using a 0 to 10 numeric pain rating scale prior to a percutaneous RHB under computed tomography (CT) guidance. If the pain score was reduced greater than 50 % immediately after the RHB, patients were evaluated for RAT by a MDT. Pre-operative and 1-year post-operative quality-of-life (QOL) surveys were administered to each RAT patient. A total of 43 LPHS patients were referred for RHB. Of the 38 patients who received a RHB, 31 had greater than 50 % reduction in pain scores. Pre- and post-RHB mean pain scores were 6/10 and 0.7/10, respectively, in patients who had greater than 50 % reduction in pain. A total of 22 of the patients who responded favorably then proceeded to RAT; 12 patients had at least 1-year follow-up following RAT. All patients had a meaningful decrease in their pain. Mean pain score at 1 year was 0.8/10 for an 85 % overall reduction in pain; 92 % of patients experienced a greater than or equal to 50 % reduction in pain at 1 year. Mean Beck Depression Inventory (BDI) score (0 to 66) 1 year after RAT decreased from 25.2 pre-op (moderate depression) to 12.8 post-op (minimal depression). The authors concluded that a MDT approach utilizing a RHB should be considered as a tool to select appropriate LPHS patients for RAT to achieve long-term success in reducing chronic pain and depression while increasing QOL.

Furthermore, an UpToDate review on “Loin pain-hematuria syndrome (Hebert et al, 2020) states that “Kidney auto-transplantation has been used in LPHS patients with chronic, severe pain that has been unresponsive to nonsurgical therapies. However, because of the risks associated with kidney auto-transplantation and lack of long-term efficacy, this approach is now regarded as a last resort”.

TruGraf Blood Gene Expression Test

Marsh et al (2019) stated that TruGraf v1 is a laboratory-developed DNA microarray-based gene expression blood test to enable proactive non-invasive serial assessment of kidney transplant recipients with stable renal function. It has been previously validated in patients identified as Transplant eXcellence (TX: stable serum creatinine, normal biopsy results, indicative of immune quiescence), and not-TX (renal dysfunction and/or rejection on biopsy results). TruGraf v1 is intended for use in subjects with stable renal function to measure the immune status as an alternative to invasive, expensive, and risky surveillance biopsies. In this study, simultaneous blood tests and clinical assessments were performed in 192 patients from 7 transplant centers to evaluate TruGraf v1. The molecular testing laboratory was blinded to renal function and biopsy results. Overall, TruGraf v1 accuracy (concordance between TruGraf v1 result and clinical and/or histologic assessment) was 74 % (142/192), and a result of TX was accurate in 116 of 125 (93 %). The negative predictive value (NPV) for TruGraf v1 was 90 %, with a sensitivity 74 % and specificity of 73 %. Results did not significantly differ in patients with a biopsy-confirmed diagnosis versus those without a biopsy. The authors concluded that TruGraf v1 could potentially support a clinical decision enabling unnecessary surveillance biopsies with high confidence, making it an invaluable addition to the transplant physician's tool kit for managing patients. TruGraf v1 testing could potentially avoid painful and risky invasive biopsies, reduce health care costs, and enable frequent assessment of patients with stable renal function to confirm the presence of immune quiescence in the peripheral blood.

In a retrospective study, First et al (2019) examined the clinical utility of the TruGraf test in patient management. Simultaneous blood tests and clinical assessments were performed in 192 patients at 7 transplant centers, and in a prospective observational study they were performed in 45 subjects at 5 transplant centers. When queried regarding whether or not the TruGraf test result impacted their decision regarding patient management, in 168 of 192 (87.5 %) cases the investigator responded affirmatively. The prospective study indicated that TruGraf results supported physicians' decisions on patient management 87 % (39/45) of the time, and in 93 % of cases physicians indicated that they would use serial TruGraf testing in future patient management. A total of 21 of 39 (54 %) reported results confirmed their decision that no intervention was needed, and 17 of 39 (44 %) reported that results specifically informed them that a decision not to

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perform a surveillance biopsy was correct. The authors concluded that TruGraf is the 1st and only non-invasive test to be examined for clinical utility in determining rejection status of patients with stable renal function and showed promise of providing support for clinical decisions to avoid unnecessary surveillance biopsies with a high degree of confidence. These researchers stated that based on the comprehensive clinical and analytical validation data published to-date, and their findings from routine clinical diagnostics, they suggested that broad utilization of the TruGraf v1 blood test has the potential to have a significant positive impact on patient care as a tool to support treatment decisions in managing the health of kidney transplant.

The authors stated that 1 main drawback of the present study was that TruGraf v1 is a new test, and the clinicians involved in this study did not have any prior experience with the test at the beginning of the study. Thus, clinicians were not prepared to make patient management decisions based on TruGraf results, which might under-estimate the changes in patient management once the test is fully integrated in the diagnostic work-up. In addition, these studies were observational in nature, such that the impact of decisions physicians claimed they would have made could not be directly measured. These investigators plan to further demonstrate the clinical utility of TruGraf in supporting decisions to avoid indiscriminate surveillance biopsies, which are largely negative and thus often unnecessary and potentially harmful to patients, by performing interventional studies in which physician decisions are made (rather than simply considered) based on TruGraf test results provided in real time, and outcomes are assessed 3 months afterward. Such studies have already been initiated.

Peddi et al (2020) noted that TruGraf is a blood-based biomarker test that measures differential expression of a collection of genes that have been shown to correlate with surveillance biopsy results. However, in the majority of U.S. transplant centers, surveillance biopsies are not performed. These researchers examined the clinical validity of TruGraf in stable kidney transplant recipients and demonstrated the potential clinical utility of serial TruGraf testing in a center not utilizing surveillance biopsies. Serum creatinine levels, TruGraf testing at multiple time-points, and subsequent clinical follow-up (1 year) were obtained for 28 patients. Overall concordance of TruGraf results, when compared with independent clinical assessment of testing, was 77 % (54/70) for all tests; 79 % (22/28) for test 1, 75 % (21/28) for test 2, and 79 % (11/14) for test 3; the NPL was 98.0 %. Analysis of clinical utility indicated that 77 % of TruGraf results would have been useful in patient management. The authors concluded that the findings of this study indicated the value of serial TruGraf testing in those transplant centers that do not perform surveillance biopsies as part of their standard of care. The high NPV indicated the ability of TruGraf to confirm immune quiescence with a high degree of probability in patients with a Transplant eXcellence (TX) result, without the need to perform a surveillance biopsy.

The authors stated that the drawbacks of the this study were the relatively small sample size (n = 28), the short period of follow-up (1 year), and the fact that TruGraf is a new test and the principal investigators (PIs) involved in this study did not have any prior experience with the test at the beginning of the study. Thus, the PIs were not prepared to make patient management decisions based on TruGraf results, which might have under-estimated the changes in patient management once the test is fully integrated into the diagnostic work-up of kidney transplant recipients. These researchers stated that in the present study, the PIs expressed satisfaction with the ability to confidently confirm stability in 77 % of patients, and led them to independently conclude that TruGraf testing is reasonable and is a promising alternative to surveillance biopsy. With sufficient experience, the PIs may be able to eventually use TruGraf results in future studies to provide optimal care for stable patients undergoing a reduction of immunosuppression.

Furthermore, an UpToDate review on “Kidney transplantation in adults: Investigational methods in the diagnosis of acute renal allograft rejection” (Anglicheau et al, 2020) states that “The TruGraf assay is a DNA microarray-based gene expression blood test that was developed as an alternative to surveillance biopsies to rule out subclinical rejection in patients with stable graft function. All aspects of discovery and external validation of the TruGraf test were performed on blood samples paired with protocol biopsies from prevalent cohorts. How the test performs in patients with graft dysfunction has not been assessed and remains to be studied”.

Plasminogen Activator Evaluation Prior to Kidney Transplantation

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Bukovsky et al (1992) studied the expression of alpha-smooth muscle actin (alpha sm-1) by mesangial cells, and the expression of Thy-1 glycoprotein, antithrombin III (ATIII), and urokinase by tubular epithelial cells in normal kidneys and dysfunctional renal allografts. Kidney biopsies were studied immunocytochemically for changes in each of these markers and the findings were classified into 2 groups and compared with creatinine plasma levels at the time the biopsies were taken. In dysfunctional grafts, mesangial alpha sm-1 and tubular epithelial Thy-1 reactivities were greatly diminished, and urokinase and ATIII were missing from proximal renal tubular epithelial cells. Urokinase, which was absent from normal renal glomeruli, appeared in glomeruli of some dysfunctional allografts. The possible usefulness of these markers in patient evaluations was supported by the finding that the distribution of vinculin, fibronectin, myosin, actin B4, desmin, glomerular HLA-DR, and the tubular expression of CD15 remained unchanged. The authors concluded that these findings suggested that the immunocytochemical localization and evaluation of alpha sm-1, Thy-1, ATIII, and urokinase in kidney allografts may be useful adjuncts in the assessment of function in renal allografts.

Furthermore, an UpToDate review on "Kidney transplantation in adults: Evaluation of the potential kidney transplant recipient" (Rossi and Klein, 2021) does not mention evaluation of plasminogen activator as a management option.

DNA Methylation as a Biomarker of Post-Transplant Complications in Kidney Transplantation

Cristoferi and colleagues (2022) noted that although KTx improves patient survival and QOL, long-term results are hampered by both immune- and non-immune-mediated complications. Current biomarkers of post-transplant complications, such as allograft rejection, chronic renal allograft dysfunction, and cutaneous squamous cell carcinoma (SCC), have a sub-optimal predictive value. DNA methylation is an epigenetic modification that directly affects gene expression and plays an important role in processes such as ischemia/reperfusion injury, fibrosis, and alloreactive immune response. Novel techniques can quickly evaluate the DNA methylation status of multiple loci in different cell types, allowing a deep and interesting study of cells' activity and function; thus, DNA methylation has the potential to become an important biomarker for prediction and monitoring in KTx. In a systematic review, these investigators examined the role of DNA methylation as a potential biomarker of graft survival and complications development in KTx. They carried out a systematic review of several databases. The Newcastle-Ottawa scale and the Jadad scale were employed to examine the risk of bias for observational and randomized studies, respectively. A total of 20 studies reporting on DNA methylation as a biomarker for KTx were included, all using DNA methylation for prediction and monitoring. DNA methylation pattern alterations in cells isolated from different tissues, such as kidney biopsies, urine, and blood, have been associated with ischemia-reperfusion injury and chronic renal allograft dysfunction. These alterations occurred in different and specific loci. DNA methylation status has also proved to be important for immune response modulation, having a crucial role in regulatory T cell definition and activity. Research also focused on a better understanding of the role of this epigenetic modification assessment for regulatory T cells isolation and expansion for future tolerance induction-oriented therapies. The authors concluded that studies included in this review were heterogeneous in study design, biological samples, and outcome. These researchers stated that more coordinated investigations are needed to confirm DNA methylation as a clinically relevant biomarker important for prevention, monitoring, and intervention in KTx. Moreover, these investigators stated that an international agreement on study settings is needed to stimulate further research and achieve the first milestones in the quest for clinically useful biomarkers.

Clarava / Tuteva

Clarava is a pre-transplant prognosis test for the risk of early AR in KTx candidates. Tuteva is a post-transplant test focused upon acute cellular rejection (ACR), including sub-clinical rejection as correlates to histopathology findings in KTx recipients. Both of these tests employ next-generation RNA sequencing to create a defined risk profile for KTx patients, allowing the tailoring of immunosuppressive therapy and identification of rejection events. However, there is currently insufficient evidence that Clarava / Tuteva would improve health outcomes of KTx candidate/recipients.

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Liu et al (2015) noted that KTx is the major therapeutic option for ESRD; however, AR could cause allograft loss in some of these patients. Emerging evidence supports that microRNA (miRNA) dysregulation is implicated in acute allograft rejection. These researchers used next-generation sequencing (NGS) to profile miRNA expression in normal and acutely rejected kidney allografts. Among 75 identified dysregulated miRNAs, miR-10b was the most significantly down-regulated miRNAs in rejected allografts. Transfecting miR-10b inhibitor into human renal glomerular endothelial cells recapitulated key features of acute allograft rejection, including endothelial cell apoptosis, release of pro-inflammatory cytokines (IL-6, TNF-α, interferon-gamma [IFN-γ], and chemokine [C-C motif] ligand 2) and chemotaxis of macrophages whereas transfection of miR-10b mimics had opposite effects. Down-regulation of miR-10b directly derepressed the expression of BCL2L11 (an apoptosis inducer) as revealed by luciferase reporter assay. The authors concluded that miR-10b down-regulation mediates many aspects of disease pathogenicity of acute kidney allograft rejection. These researchers stated that further investigation of the role of miRNA in graft rejection may provide new mechanistic insights and guide the development of novel therapeutics.

Jin et al (2017) stated that KTx is an effective treatment for patients with ESRD. Although advanced immunosuppressive therapy is administered following transplantation, certain patients still suffer from acute allograft rejection. miRNAs have a potential diagnostic and therapeutic value for acute renal allograft rejection; however, their underlying mechanism of action is largely unknown. In this study, an increased level of miR-650 was identified to be associated with the down-regulation of B-cell CLL/lymphoma 11B (BCL11B) expression in acute renal allograft rejection. Furthermore, in-vitro study using human renal glomerular endothelial cells (HRGECs) transfected with a miR-650 mimic revealed that key characteristics of acute renal allograft rejection were observed, including apoptosis, the release of cytokines and the chemotaxis of macrophages, while the effects were reduced in HRGECs transfected with a miR-650 inhibitor. The existence of a conserved miR-650 binding site on the 3'-untranslated region of BCL11B mRNA was predicted by computational algorithms and confirmed by a luciferase reporter assay. Knockdown of BCL11B with small interfering RNA (siRNA) significantly increased the apoptotic rate and significantly decreased the proliferation ability of HRGECs compared with the negative control group. HRGECs transfected with a combination of BCL11B siRNA and the miR-650 mimic demonstrated a significant increase in the rate of apoptosis compared with the control. The authors concluded that these findings suggested that the up-regulation of miR-650 contributed to the development of acute renal allograft rejection by suppression of BCL11B, which led to apoptosis and inflammatory responses. Therefore, miR-650 and BCL11B may represent potential therapeutic targets for the prevention of acute renal allograft rejection.

Hamdorf et al (2017) noted that the control of gene expression by microRNAs (miRNAs, miR) influences many cellular functions, including cellular differentiation, cell proliferation, cell development, and functional regulation of the immune system. Recently, miRNAs have been detected in serum, plasma, and urine and circulating miR profiles have been associated with a variety of diseases. Rejection is one of the major causes of allograft failure and preventing and treating AR are the primary task for clinicians working with transplant patients. Invasive biopsies used in monitoring rejection are burdensome and risky to transplant recipients. Novel and easily accessible biomarkers of AR could make it possible to detect rejection earlier and make more fine-tuned calibration of immunosuppressive or new target treatment possible. The authors concluded that miRNAs are emerging as important regulatory molecules of gene expression. Research has focused on different expression profiles in health and disease. In the field of transplantation, several miRNAs have been described and it has been shown that miRs have the potential to be a novel diagnostic marker; thus, they represent a group of promising candidates for early detection of organ rejection with the potential to affect clinical decision making. Moreover, these researchers stated that further investigation and standardization in the profiling of miRNAs in serum, plasma, and urine samples are needed to find a robust diagnostic marker and to develop insights into pathways responsible for the rejection process as well as novel targets for therapy.

Iwasaki et al (2017) stated that de-novo DSA would not necessarily contribute to chronic antibody-mediated rejection (CAMR) in KTx. These investigators examined if PBMC miRNAs could be predictable biomarkers for CAMR. They carried out microarray profiling of 435 mature miRNAs in pooled samples. Individual analysis revealed that miR-142-5p was significantly ($p < 0.01$) under-expressed in patients with DSA. After DSA production, miR-486-5p and its target PTEN/foxO3 mRNA were significantly over-

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expressed ($p < 0.01$) and under-expressed ($p < 0.01$), respectively, in patients with biopsy-proven CAMR, compared with non-CAMR. The authors concluded that these findings suggested that miRNA expression patterns may serve as non-invasive diagnostic biomarkers to assess immune response and kidney allograft status. Moreover, these researchers stated that further studies should focus on elucidation of the functional importance of each miRNA; and correlation with the patient immune cells might provide new insight into alloreactivity and novel therapeutics for transplant recipients.

Yamamoto et al (2018) stated that non-invasive methods for the early diagnosis of chronic antibody-mediated rejection (cAMR) are desired for patients with de-novo (dn) DSA. These investigators examined the clinical relevance of immune-related gene expression in peripheral blood of KTx recipients. The expression levels of 14 key molecules (Foxp3, CTLA-4, CCR7, TGF-β, IGLL-1, IL-10, ITCH, CBLB, Bcl-6, CXCR5, granzyme B, CIITA, Baff, TOAG-1/TCAIM) related to regulatory/cytotoxic function of immune cells were compared in 93 patients, which were divided into Groups A (clinical cAMR with dn DSA, n = 16), B (sub-clinical cAMR with dn DSA, n = 17), C (negative cAMR with dn DSA, n = 21) and D (stable function without dn DSA, n = 39). CIITA mRNA expression levels in groups B and C were significantly lower than those in group D ($p < 0.01$). Moreover, the CTLA-4 mRNA expression in group A was significantly higher than that in groups B and C ($p < 0.01$). ROC curve analysis suggested that CIITA (AUC = 0.902) and CTLA-4 (AUC = 0.785) may serve as valuable biomarkers of the stage of dn DSA production and clinical cAMR, respectively. The authors concluded that in addition to dn DSA screening, monitoring of CIITA and CTLA-4 in peripheral blood could offer useful information on the time course of the development of cAMR. Moreover, these researchers stated that further prospective, multi-center studies are needed to draw more definitive conclusions.

Lin et al (2021) noted that KTx is an optimal method for treatment of ESRD; however, KTx rejection (KTR) is commonly observed to have negative effects on allograft function. miRNAs are small non-coding RNAs with regulatory role in KTR genesis, the identification of miRNA biomarkers for accurate diagnosis; thus, subtyping of KTR is of clinical significance for active intervention and personalized therapy. These researchers developed an integrative bioinformatics model based on multi-omics network characterization for miRNA biomarker discovery in KTR. Compared with existed methods, the topological importance of miRNA targets was prioritized based on cross-level miRNA-mRNA and protein-protein interaction network analyses. The biomarker potential of identified miRNAs was computationally validated and examined by receiver-operating characteristic (ROC) evaluation and integrated "miRNA-gene-pathway" pathogenic survey. A total of 3 miRNAs (i.e., miR-145-5p, miR-155-5p, and miR-23b-3p) were screened as putative biomarkers for KTR monitoring. Among them, miR-155-5p was a previously reported signature in KTR, whereas the remaining 2 were novel candidates both for KTR diagnosis and subtyping. The ROC analysis convinced the power of identified miRNAs as single and combined biomarkers for KTR prediction in kidney tissue and blood samples. Functional analyses, including the latent crosstalk among HLA-related genes, immune signaling pathways and identified miRNAs, provided new insights of these miRNAs in KTR pathogenesis. The authors proposed a network-based bioinformatics approach and applied to identify candidate miRNA biomarkers for KTR study. Moreover, these researchers stated that biological and clinical validations are needed for translational applications of the findings.

The authors stated that this study had several drawbacks. First, the importance and contribution of miRNAs and genes in the network were prioritized based on their structural characteristics, the functional significance of genes in immune processes related to rejection responses needs to be examined and quantified to improve the sensitivity of miRNAs in KTR management. Second, in addition to KTR and the normal controls, samples from other conditional phenotypes such as infection-mediated or drug toxicity-induced delated graft function (DGF) should be included and compared to test the biomarker specificity of identified miRNAs in rejection since similar clinical symptoms (e.g., decreased amount of urine and increased level of serum creatinine) often occurred in the early stage both of KTR and DGF. Third, it is potentially important that the biomarkers originally identified from tissue biopsies were able to be validated in blood samples as blood could be easily obtained for clinical translation. However, the ROC result from tissue and blood group tended to be heterogeneous in this study. Although many studies reported that the concordance rate for detection of gene variant and expression may be affected by the type and source of sample data, it should be admitted that the blood sample set used for comparison was relatively small. Therefore, the biomarker significance of identified miRNAs in blood needs to be further investigated with the accumulation of enough

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public data to decrease the risk of model over-fitting. On the other hand, large-sample-based multi-center experimental validation and pathogenetic survey are expected to be carried out for future translational applications.

Lai et al (2021) stated that despite advances in post-transplant management, the long-term survival rate of kidney grafts and patients has not improved as approximately 40 % of transplants fails within 10 years following transplantation. Both immunologic and non-immunologic factors contribute to late allograft loss. Chronic kidney transplant rejection (CKTR) is often clinically silent; yet progressive allogeneic immune process leads to cumulative graft injury, deterioration of graft function. Chronic active TCMR and ABMR are classified as 2 principal subtypes of CKTR. While significant improvements have been made towards a better understanding of cellular and molecular mechanisms and diagnostic classifications of CKTR, lack of early detection, differential diagnosis and effective therapies continue to pose major challenges for long-term management. Recent development of high throughput cellular and molecular biotechnologies has allowed rapid development of new biomarkers associated with chronic renal injury, which not only provide insight into pathogenesis of chronic rejection but also allow for early detection. These investigators stated that recent studies have revealed that mi-R21 and miR200b expression in urine are associated with interstitial fibrosis and tubular atrophy (IFTA) and chronic allograft dysfunction (CAD), while circulating miR-150, miR192, miR-200b, and miR-423-3p in plasma are related to IFTA. Meanwhile, expression of miR21, miR-155, and miR-142-3p was up-regulated in the plasma of patients with IFTA, while miR-145-5p and miR-148a were down-regulated. These investigators stated that up-regulation of miR142-5p, and down-regulation of miR-486-5p may serve as biomarkers for early detection of chronic ABMR. These markers could, therefore, be considered as potential markers for CAD.

Furthermore, an UpToDate review on “Kidney transplantation in adults: Evaluation and diagnosis of acute kidney allograft dysfunction” (Kadambi et al, 2022) does not mention next-generation RNA sequencing as a management tool.

Pleximark for Evaluation of Acute Cellular Rejection Following Kidney Transplantation

Pleximark is a functional cell-based blood test for evaluation of the likelihood of ACR following kidney transplantation. It measures the immune response of recipient lymphocytes to donor lymphocytes in cell culture. Recipient T-cytotoxic memory cells, which express the inflammatory marker, CD40 ligand or CD154 are measured with flow cytometry. Results are expressed as an index of rejection, which is a measure of the likelihood of rejection.

Ashokkumar et al (2011) noted that the novel, recently described allo (antigen)-specific CD154+T cells were examined for their association with ACR in 43 adult renal transplant recipients receiving steroid-free tacrolimus following alemtuzumab induction. Single blood samples corresponding to "for cause" allograft biopsies were assayed for CD154+naive or memory T-helper or T-cytotoxic cells in 16-hour mixed leukocyte reaction. Intra- and inter-assay variation was less than 10 % for a variety of conditions. In logistic regression, leave-one-out cross-validation, and ROC analyses, the rejection-risk threshold of allo-specific CD154+T-cytotoxic memory cells (TcMs) associated best with biopsy-proven ACR with a sensitivity/specificity of 88 % in 32 of 43 subjects. Sensitivity/specificity of 100 %/88 % was replicated in blinded prediction in the remaining 11 subjects. Allo-specific CD154+TcM correlated inversely with CTLA4+TcM (Spearman $r = -0.358$, $p = 0.029$) and increased significantly with increasing histological severity of ACR ($p = 2.99E-05$, Kruskall-Wallis). The authors concluded that the strong association between ACR and allo-specific CD154+TcM may be useful in minimizing protocol biopsies among recipients at reduced rejection risk. Moreover, these researchers stated that as a follow-up to this preliminary study; and before any potential future clinical implementation, they plan a longitudinal study in living donor renal transplant recipients to validate these findings.

Ashokkumar et al (2015) stated that belatacept blocks CD28-mediated T-cell co-stimulation and prevents kidney transplant rejection. Understanding T-cell subset sensitivity to belatacept may identify cellular markers for immunosuppression failure to better guide treatment selection. These researchers examined the

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belatacept sensitivity of allo-antigen-specific CD154-expressing-T-cells, whose T-cytotoxic memory (TcM) subset predicts rejection with high sensitivity following non-renal transplantation. The belatacept concentration associated with half-maximal reduction (EC50) of CD154 expression was calculated for 36 T-cell subsets defined by combinations of T-helper (Th), Tc, T-memory and CD28 receptors, following allo-stimulation of peripheral blood leukocytes from 20 normal healthy subjects. Subsets were ranked by median EC50, and by whether subset EC50 was correlated with and thus could be represented by the frequency of other subsets. No single subset frequency emerged as the significant correlate of EC50 for a given subset. Most (n = 25) T-cell subsets were sensitive to belatacept. Less sensitive subsets demonstrated a memory phenotype and absence of CD28 receptor. Potential drug-resistance markers for future validation included the low frequency highly differentiated, Th-memory-CD28-negative T-cells with the highest median EC50, and the least differentiated, high-frequency Tc subset, with the most CD28-negative T-cells, the 3rd highest median EC50, and significant correlations with frequencies of the highest number of CD28-negative and memory subsets. The authors concluded that alloreactive T-cell subsets, which express the inflammatory co-stimulator, CD154, showed wide-ranging susceptibility to CD28-costimulation blockade with belatacept. The less sensitive subsets in this trial were characterized by the presence of the memory marker, and the absence of CD28 and appeared to be distributed in both the Th and Tc compartment. These subsets were suited for clinical validation as markers of relative belatacept resistance, because immunosuppression failure, which could cause transplant rejection has been associated with resistant alloreactive T-cell subsets previously6. These researchers stated that during the clinical validation phase, the relative merits of resistant subsets in the T-helper and T-cytotoxic cell compartments must also include assessment of available cell counts and the dynamic range of a candidate subset to allow reliable measurements for clinical decision-making.

These investigators stated that these findings were at best preliminary; but proved that donor-specific allo-responses of many T-cell subsets could be used for clinical rejection-risk assessment. Furthermore, none of the 11 recipients was treated with belatacept; thus, performance testing of the ThM+CD28- cell, which demonstrated the greatest resistance to belatacept must await additional clinical trials.

Rohan et al (2020) stated that allo-antigen-specific TcM that express CD40 ligand (CD154) in over-night lymphocyte co-culture are strongly associated with ACR observed in "for cause" biopsies for renal allograft dysfunction. Specifically, when the likelihood of rejection is increased, donor-specific allo-specific TcM exceeded those induced by HLA-non-identical third-party cell by 1.15-fold or greater. In a retrospective, single-center study, these researchers examined the performance of allo-specific TcM in primary renal transplant recipients (RTR) at routine clinical visits, crosssectionally at presentation for biopsies, and serially. Performance metrics were sensitivity, specificity, PPV and NPV. A total of 22 primary RTR (median age of 45 years; range of 19 to 72) were tested with allo-specific CD154 + TcM. Samples were obtained at the mean ± SD time interval of 806 ± 239 days following kidney transplantation; 6 of 22 patients experienced biopsy proven TCMR, and a 7th recipient showed ABMR. Of these 7 participants, 6 demonstrated increased likelihood of rejection with allo-specific TcM (sensitivity 83 %); 10 of these 15 participants with no rejection had a negative test (specificity 67 %). False-positive tests were observed in 5 participants; and 6 out of 11 patients with positive tests had ACR/ABMR with a PPV of 54 %, while 10 out of 11 participants with negative tests were non-rejecters with a NPV of 91 %. The authors concluded that allo-specific T-cytotoxic memory cells distinguished primary RTR with quiescent allografts from those with dysfunction. With serial surveillance measures, this test system may facilitate decisions to manage immunosuppression in RTR.

The authors stated that this trial has several drawbacks. First, it was a retrospective, single-center study with a limited number of patients (n = 22). Most of the patients in the study were monitored with a single cross-sectional test. The number of patients with BK virus nephropathy were small. Serial testing of more patients with BK nephropathy would aid in better understanding the false positivity. A larger longitudinal experience with allo-specific CD154 + TcM may aid in understanding the long-term implications of monitoring donor-specific T-cell alloreactivity.

Lai et al (2021) noted that despite advances in post-transplant management, the long-term survival rate of renal grafts and patients has not improved as approximately 40 % of transplants fail within 10 years following transplantation. Both immunologic and non-immunologic factors contribute to late allograft loss. CKTR is often a clinically silent yet progressive allogeneic immune process that leads to cumulative graft

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injury, deterioration of graft function. Chronic active TCMR and chronic active ABMR are classified as 2 principal subtypes of CKTR. While significant improvements have been made towards a better understanding of cellular and molecular mechanisms and diagnostic classifications of CKTR, lack of early detection, differential diagnosis and effective therapies continue to pose major challenges for long-term management. Recent development of high throughput cellular and molecular biotechnologies has allowed rapid development of new biomarkers associated with chronic renal injury, which not only provide insight into pathogenesis of chronic rejection but also allow for early detection. In parallel, several novel therapeutic strategies have emerged that may hold great promise for improvement of long-term graft and patient survival. The authors concluded that this mini review provided updates and insights into the latest development of promising novel biomarkers for diagnosis and novel therapeutic interventions to prevent and treat CKTR. Moreover, these investigators stated that there have been significant attentions drawn to quantify allo-reactive CD8+ T cells as potential cellular biomarkers of rejection, or tolerance. Ashokkumar et al (2009) found that allo-specific CD154+ T-cytotoxic memory cells were associated with rejection risk in liver transplant recipients. Limited data showed that an increase in CD154+ subset is implicated in acute kidney transplant rejection. They stated that with improved understanding of cellular mechanisms underlying CKTR and advances in the multi-color flow cytometry analyses combining with more recent development of single-cell genomics studies, it is conceivable that more precise cellular biomarkers will be identified for CKTR.

These researchers stated that several considerations need to be addressed before these biomarkers could be employed in the clinical practice for kidney transplants. First, sensitivity, specificity, PPV and NPV must be considered, and ROC curves need to be thoroughly evaluated for their clinical utility. Second, integration of different biomarkers is needed for accurate diagnosis. Third, robust validation studies and standardization of measurements are needed to identify new biomarkers. Fourth, timing required for generating results and cost of assessment should be reasonable.

Furthermore, an UpToDate review on “Kidney transplantation in adults: Clinical features and diagnosis of acute renal allograft rejection” (Brennan et al, 2022) does not mention the use of Pleximark / measurement of donor and third party-induced CD154+T-cytotoxic memory cells as a management tool.

Artificial Intelligence/Machine Learning Method for Predicting Graft Survival in Kidney Transplantation

Naqvi et al (2021) noted that KTx is the optimal treatment for patients with ESRD. Short- and long-term kidney graft survival is influenced by a number of donor and recipient factors. Predicting the success of KTx is important for optimizing kidney allocation. These researchers predicted the risk of kidney graft failure across 3 temporal cohorts (within 1 year, within 5 years, and after 5 years following a transplant) based on donor and recipient characteristics. They analyzed a large data set comprising over 50,000 kidney transplants covering an approximate 20-year period. These investigators used machine learning (ML)-based classification algorithms to develop prediction models for the risk of graft failure for 3 different temporal cohorts. Deep learning-based autoencoders were applied for data dimensionality reduction, which improved the prediction performance. The influence of features on graft survival for each cohort was studied by examining a new non-overlapping patient stratification approach. The models predicted graft survival with area under the curve scores of 82 % within 1 year, 69 % within 5 years, and 81 % within 17 years. The feature importance analysis elucidated the varying influence of clinical features on graft survival across the 3 different temporal cohorts. The authors applied ML to develop risk prediction models for graft failure that demonstrated a high level of prediction performance. Acknowledging that these models performed better than those reported in the literature for existing risk prediction tools, future studies will focus on how best to incorporate these prediction models into clinical care algorithms to optimize the long-term health of kidney recipients. These researchers stated that as a next step, they plan to incorporate the prediction models into clinical care at the time of allocation; models that best predict short- and long-term kidney graft survival may be used as a pragmatic prognostic tool to aid clinicians in maximizing the best possible matching of donors and recipients while preserving existing allocation rules that are used to promote equity.

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The authors stated that a drawback of this study was the removal of censored instances. These researchers removed all successful cases that were censored before 8 years following transplant. Although this type of approach has previously been used, including censored cases is a potential consideration for future analyses.

Ravindhran et al (2023) stated that the variations in outcome and frequent occurrence of kidney allograft failure continue to pose important clinical and research challenges despite recent advances in KTx. In a systematic review, these researchers examined the current use of artificial intelligence (AI)/ML models in KTx and conducted a meta-analysis of these models in the prediction of graft survival. This review was registered with the PROSPERO database and all peer-reviewed original studies that reported ML model-based prediction of graft survival were included. Quality assessment was carried out by the criteria defined by Qiao and risk-of-bias assessment was carried out using the PROBAST tool. The diagnostic performance of the meta-analysis was assessed by a meta-analysis of the area under the receiver operating characteristic curve and a hierarchical summary receiver operating characteristic plot. A total of 31 studies met the inclusion criteria for the review and 27 studies were included in the meta-analysis. A total of 29 different ML models were used to predict graft survival in the included studies; 9 studies compared the predictive performance of ML models with traditional regression methods; 5 studies had a high-risk of bias and 3 studies had an unclear risk of bias. The area under the hierarchical summary receiver operating characteristic curve was 0.82 and the summary sensitivity and specificity of ML-based models were 0.81 (95 % CI: 0.76 to 0.86) and 0.81 (95 % CI: 0.74 to 0.86), respectively, for the overall model. The diagnostic odds ratio (DOR) for the overall model was 18.24 (95 % CI: 11.00 to 30.16) and 29.27 (95 % CI: 13.22 to 44.46) based on the sensitivity analyses. The authors concluded that prediction models using ML methods may improve the prediction of outcomes following KTx by the integration of the vast amounts of non-linear data.

The authors stated that drawbacks of this review included the methodological shortcomings arising out of the substantial heterogeneity within the included studies and the difficulty in arriving at a single summary estimate of overall model performance. The evidence presented in this review has also suggested that most model predictions were based on a very large amount of retrospective data with limited external validation. In an ideal ML setting, a specific prospective data entry based on clinical experience should be combined with graft outcome data and then analyzed over an interval of time in conjunction with actual outcomes. The most recent study comparing the predictive abilities of ML models versus conventional statistical models performed in a large database has suggested that AI/ML models are not significantly superior to conventional regression-based models. Although the authors agreed that no substitute could replace human intelligence or clinical experience, the results of this review have shown that the best prediction of difficult outcomes such as graft survival, which took into account numerous pre-operative, operative, and post-operative outcomes, was difficult with human intelligence alone. An informed and well-guided decision is best taken by combining clinical experience and a well-designed prediction model. These well-designed prediction models can only be developed by ML tools given the vast amounts of data needed to build a reliable model. These investigators summarized the current available evidence, identified the best ML models suited for these outcomes, and the key challenges that need to be addressed to accurately guide future research. They stated that while the use of AI/ML in KTx is still in its infancy, such models have a significant future role, not only in the prediction of graft survival, but also in organ matching, diagnostics, and management pathways.

Panniculectomy in Kidney Transplant Candidates

Promny et al (2021) stated that obesity is often present in patients suffering from ESRD; however, over-weight KTx candidates are a challenge for the transplant surgeon. Obese patients tend to develop a large abdominal panniculus after weight loss creating an area predisposed to wound-healing disorders. Due to concerns regarding graft survival and post-operative complications following KTx, obese patients are frequently refused in this selective patient cohort. In a retrospective study, these investigators examined the effect of panniculectomies on post-operative complications and transplant candidacy in an inter-disciplinary setting. They reviewed 10 cases of abdominal panniculectomies carried out in patients with ESRD before KTx was performed. The median body mass index (BMI) was 35.2 kg/m2 (range of 28.5 to 53.0 kg/m2) at 1st transplant-assessment versus 31.0 kg/m2 (range of 28.0 to 34.4 kg/m2) at panniculectomy, and 31.6 kg/m2 (range of 30.3 to 32.4 kg/m2) at KTx. These researchers observed no major post-operative complications following panniculectomy and minor wound-healing complications in 2 patients. All aside

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from 1 patient became active transplant candidates 6 weeks after panniculectomy. No post-transplant wound complications occurred in the transplanted patients. The authors concluded that abdominal panniculectomy was feasible in patients suffering ESRD with no major post-operative complications; thus, converting previously ineligible patients into KTx candidates.

The authors stated that this study had several drawbacks. First, there was an inherent selection bias, with the possibility that this study population did not reflect the ESRD patient population at large. Second, the low number of patients, as these researchers examined a highly selective group of patients. In a prospective study design a comparison of panniculectomy patients with a similar patient population in BMI who did not undergo a panniculectomy before KTx would be of high interest. Third, these investigators could not present any information regarding long-term graft survival. Nevertheless, they intended to show the feasibility of abdominal panniculectomy in obese patients suffering from ESRD and enable the patients' addition to the transplant waiting list.

Laspro et al (2024) noted that as the obesity crisis in the U.S. continues, some renal transplantation centers have liberalized their BMI criteria necessary for transplant eligibility. More individuals with larger body-habitus related co-morbidities with ESRD now qualify for KTx. Surgical modalities from other fields also interact with this patient population. In order to evaluate surgical outcomes of panniculectomy in the context of KTx and ESRD, these investigators carried out a systematic review following the PRISMA 2020 guidelines. Due to a paucity of existing primary studies, these researchers retrospectively collected data on patients with ESRD undergoing panniculectomy from the American College of Surgeons (ACS)' National Surgical Quality Improvement Program (NSQIP) to examine outcomes of body contouring in this patient population. From the systematic review, a total of 783 ESRD patients underwent panniculectomy among the studies identified. Of these, 91 patients underwent panniculectomy simultaneously to KTx while 692 had their pannus resected before KTx. The most common complication was hematoma followed by wound dehiscence. From the NSQIP database, 24,868 patients met the inclusion criteria for analysis. In the setting of KTx status, patients with diabetes, hypertension requiring medication, and requiring dialysis were more likely to suffer post-operative complications (OR 1.31, 1.15, and 2.2, respectively). However, upon sub-analysis of specific types of complications, the only retained association was between diabetes and wound complication. The authors concluded that these preliminary findings demonstrated that panniculectomy in ESRD patients appeared to be safe, although with a nominal increased risk for complications. These investigators stated that pannus resection did not appear to impact post-transplantation outcomes, including long-term allograft survival. These researchers stated that larger, higher-powered, randomized studies are needed to confirm the safety, utility, and medical benefit of panniculectomy in the context of KTx.

CXCL10 Urine Test

The CXCL10 Urine Test is a non-invasive urine test that detects the CXCL10 chemokine in human urine samples to monitor kidney transplants. The test is performed by Thermo Fisher Scientific One Lambda Laboratories (Fishers, IN). It can detect elevated levels of CXCL10, which are associated with inflammation and early graft dysfunction.

Rabant et al (2016) monitored the urinary C-X-C motif chemokine (CXCL)9 and CXCL10 levels in 1,722 urine samples from 300 consecutive KTx recipients collected during the 1rst post-transplantation year and examined their predictive value for subsequent AR. The trajectories of urinary CXCL10 revealed an early increase at 1 month ($p = 0.0005$) and 3 months ($p = 0.0009$) in patients who subsequently developed AR. At 1 year, the AR-free allograft survival rates were 90 % and 54 % in patients with CXCL10:creatinine (CXCL10:Cr) levels of less than 2.79 ng/mmoL and greater than 2.79 ng/mmoL at 1 month, respectively ($p < 0.0001$), and 88 % and 56 % in patients with CXCL10:Cr levels of less than 5.32 ng/mmoL and greater than 5.32 ng/mmoL at 3 months ($p < 0.0001$), respectively. CXCL9:Cr levels also were associated, albeit less robustly, with AR-free allograft survival. Early CXCL10:Cr levels predicted clinical and sub-clinical rejection and both T cell- and antibody-mediated rejection. In 222 stable patients, CXCL10:Cr at 3 months predicted AR independent of concomitant protocol biopsy results ($p = 0.009$). The authors concluded that although its PPV was low, a high NPV suggested that early CXCL10:Cr might predict immunological quiescence on a triple-drug calcineurin inhibitor-based immunosuppressive regimen in the 1st post-

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transplantation year, even in clinically and histologically stable patients. Moreover, these researchers stated that prospective, clinical studies are needed to ascertain the clinical utility of this test.

Ho et al (2018) noted that urinary CXCL10 is a promising biomarker for sub-clinical tubule-interstitial inflammation; however, limited data were available regarding its correlation with micro-vascular inflammation (MVI). In addition, no study has examined if concomitant serum CXCL10 would improve the discrimination for (MVI. These investigators examined if serum/urinary CXCL10 would reflect sub-clinical inflammation within different renal compartments. Patients (n = 107) with 107 surveillance biopsies were classified as: normal histology (n = 47), normal histology with polyomavirus BK (BKV) or cytomegalovirus (CMV) viremia (n = 17), moderate-severe tubule-interstitial inflammation (tubulitis ≥ 2, n = 18), pure MVI (n = 15), and isolated v lesions (n = 10). Serum and urinary CXCL10 ELISA was carried out. An independent validation set was evaluated for urine CXCL10: normal histology (n = 14), normal histology with BKV or CMV viremia (n = 19), tubulitis ≥ 2 (n = 15), pure MVI (n = 41), and isolated v lesions (n = 14). Elevated urinary CXCL10 reflected inflammation within the tubule-interstitial (urinary CXCL10/creatinine, 1.23 ng/mmol versus 0.46 ng/mmol; p = 0.02; AUC, 0.69; p = 0.001) and microvascular compartments (urinary CXCL10/creatinine, 1.72 ng/mmol versus 0.46 ng/mmol; p = 0.03; AUC, 0.69; p = 0.02) compared to normal histology. Interestingly, urinary CXCL10 was predominantly elevated with peritubular capillaritis, but not glomerulitis (p = 0.04). Additionally, urinary CXCL10 corresponded with BKV, but not CMV viremia (p = 0.02). These urine CXCL10 findings were confirmed in the independent validation set. Lastly, serum CXCL10 was elevated with BKV and CMV viremia but was not associated with microvascular or vascular inflammation (p ≥ 0.19). The authors concluded that urinary CXCL10 was associated with alloimmune and virus compartment-specific inflammation. These researchers independently confirmed that urinary CXCL10 was associated with MVI, and extended these findings to sub-clinical MVI. They noted that this was a highly selected cohort, suggesting that its diagnostic performance may be modest. The elevation of urinary CXCL10 in MVI was predominantly related to peritubular capillaritis, and this may be due to the close proximity of the peritubular capillaries to the urine.

Basak Oktay et al (2022) examined post-transplantation graft functions non-invasively by using urinary CXCL10 and metabolome analysis. The 65 living-donor KTx recipients in this cohort underwent renal biopsy to examine possible graft dysfunction. Subjects were divided into 2 groups, according to pathology reports: chronic allograft dysfunction (CAD; n = 18) and antibody-mediated/humoral allograft rejection (AMR; n = 16). The control group was composed of KTx recipients with stable health (n = 33). These investigators carried out serum creatinine, blood urea nitrogen (BUN), cystatin C, urine protein, CXCL10, and metabolome analyses on specimens from the patients. BUN, creatinine, cystatin C, urine protein, leucine + isoleucine, citrulline, and free/acetyl/propionyl carnitine levels were significantly higher in patients with CAD and AMR, compared with the control individuals. CXCL10 levels were significantly elevated in patients with AMR, compared with patients with CAD and controls. CXCL10 (AUC = 0.771) and cystatin C (AUC = 0.746) were significantly higher in the AMR group, compared with the CAD group (p < 0.02). The authors concluded that CXCL10 and metabolome analyzes were useful for evaluation of graft functions. In addition, CXCL10 might be useful as a supplementary non-invasive screening test for diagnosis of allograft rejection. These researchers stated that the findings of this study suggested that CXCL10 had potential as a non-invasive differentiator of the causes of graft dysfunction.

The authors stated that this trial had several drawbacks. First, the relatively small number of patients (n = 65) was the principal limitation in this study. Second, as a consequence of having few patients in the cohort, the pathological subgroups were limited. For instance, cellular rejection, viral infections (such as BK virus), and systemic inflammatory diseases were not included due to insufficient number of patients having those conditions. The inclusion of these pathologic manifestations, in which increased inflammation is observed, may affect the sensitivity and specificity values of CXCL10 in pathological differentiation. Third, the subjects in this trial could not be evaluated with long-term dynamic follow-up, and concurrent serum levels of parameters measured in urine were not analyzed. These researchers stated that further investigations are needed to confirm these preliminary findings by cross-validation in different centers with larger patient numbers and pathological subgroups. CXCL10 analysis in spot urine testing with ELISA method and metabolome analysis using the MS/MS method in dried spot urine yield same-day results. Fourth, for CXCL10 analysis, although the ELISA method with assay procedures could provide same-day results, separate analysis for each patient is impractical and not cost-effective in routine practice. In the future, it will

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be important to develop immunoassay methods that yield results within a shorter turn-around time and that can be applied in a more practical manner.

Janfeshan et al (2024) stated that the challenges of long-term graft survival and the side effects of current immunosuppressive therapies in KTx highlight the need for improved drugs with fewer adverse effects. Biomarkers play an important role in quickly detecting post-transplant complications, with new biomarkers showing promise for ongoing monitoring of disease and potentially reducing the need for unnecessary invasive biopsies. The chemokines such as CXCL10, are promising protein biomarkers for AR, with urine samples being a desirable source for biomarkers. These investigators analyzed the available evidence on the potential role of urinary CXCL10 protein in predicting kidney graft injuries. The authors concluded that the identification of specific and sensitive biomarkers could potentially reduce unnecessary biopsies, resulting in more individualized treatment plans and improved health outcomes. These investigators stated that while urinary CXCL10 levels have been examined as an important inflammatory chemokine in post-KTx outcomes, relying solely on CXCL10 is insufficient in determining graft complications; thus, considering other critical clinical parameters alongside CXCL10 may facilitate early detection and intervention in graft-related complications.

Van Loon et al (2024) noted previous studies have shown the diagnostic potential of urinary CXCL9 and CXCL10 for KTx rejection; however, their benefit in addition to clinical information has not been reported. In a prospective, single-center study, these investigators examined the diagnostic performance for detecting AR of urinary CXCL9 and CXCL10 when integrated with clinical information. They analyzed 1,559 biopsy-paired urinary samples from 622 KTx carried out between April 2013 and July 2019 at a single transplant center in Belgium. External validation was performed in 986 biopsy-paired urinary samples. These researchers quantified urinary CXCL9 (uCXCL9) and CXCL10 (uCXCL10) using an automated immunoassay platform and normalized the values to urinary creatinine. Urinary chemokines were incorporated into a multi-variable model with routine clinical markers (eGFR, DSA, and polyoma viremia) (integrated model). This model was then compared with the tissue diagnosis according to the Banff classification for AR. Chemokines integrated with routine clinical markers had high diagnostic value for detection of AR (n = 150) (ROC area under the curve 81.3 % [95 % CI: 77.6 to 85.0]). The integrated model would aid in avoiding 59 protocol biopsies per 100 patients when the risk for rejection was predicted to be below 10 %. The performance of the integrated model was similar in the external validation cohort. The authors concluded that the use of an integrated model of urinary chemokines and clinical markers for non-invasive monitoring of rejection could enable a reduction in the number of biopsies. Urinary chemokines may be useful non-invasive biomarkers whose use should be further studied in prospective, randomized trials to clarify their role in guiding clinical care and the use of biopsies to detect rejection following KTx. Moreover, these researchers noted that the cross-sectional nature of this trial obviated investigating the evolution over time and prediction of future rejection.

Barrett-Chan et al (2024) stated that urinary CXCL10/creatinine (uCXCL10/Cr) is proposed as an effective biomarker of sub-clinical rejection in pediatric KTx recipients. These researchers modeled urinary CXCL10/creatinine testing implementation in the clinical setting. Banked urine samples at a single center were tested for uCXCL10/Cr to validate published thresholds for rejection diagnosis (greater than 80 % specificity). The PPV for rejection diagnosis for uCXCL10/Cr-indicated biopsy was modeled with 1st-positive versus 2-test-positive approaches, with accounting for changes associated with UTI, BK and CMV viremia, and subsequent recovery. A total of 70 patients aged 10.5 ± 5.6 years at transplant (60 % men) had n = 726 urine samples with n = 236 associated biopsies (no rejection = 167, borderline = 51, and Banff 1A = 18). A threshold of 12 ng/mmol was validated for Banff 1A versus no-rejection diagnosis (AUC = 0.74, 95 % CI: 0.57 to 0.92). The 1st-positive test approach (n = 69) did not resolve a clinical diagnosis in 38 cases (55 %), whereas the 2-test approach resolved a clinical diagnosis in the majority as BK (n = 17/60, 28 %), CMV (n = 4/60, 7 %), UTI (n = 8/60, 13 %), clinical rejection (n = 5/60, 8 %), and transient elevation (n = 18, 30 %). In those without a resolved clinical diagnosis, PPV from biopsy for sub-clinical rejection was 24 % and 71 % (p = 0.017), for 1st-test versus 2-test models, respectively. After rejection treatment, uCXCL10/Cr level changes were all concordant with change in it-score. Sustained uCXCL10/Cr after CMV and BK viremia resolution was associated with later AR. The authors concluded that urinary CXCL10/Cr reliably identified kidney allograft inflammation. These investigators stated that the findings of this trial revealed the potential clinical utility of serial uCXCL10/Cr for non-invasive monitoring of sub-clinical

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allograft inflammation. Persistent uCXCL10/Cr elevation should prompt additional investigations for CMV, BKV, or UTI and in the absence of a clinically apparent cause should indicate a kidney biopsy to rule out subclinical rejection. Moreover, these researchers stated that future implementation trials are needed to examine the clinical and cost benefit of uCXCL10/Cr monitoring.

The authors stated that drawbacks of this trial included the lack of uniformity in the follow-up of patients involved. As data were compiled from both retrospective patients and patients undergoing ongoing monitoring, the schedule and interval between uCXCL10/Cr testing and biopsies differed significantly between participants. While the majority of patients with ≥ Banff1A rejection had uCXCL10/Cr of greater than 12 ng/mmol, there were notably cases that fell below this diagnostic threshold. Further investigations are needed to understand whether this represents mechanistic differences in rejection pathophysiology in the type of inflammation that is present or differences in rejection severity. There were also incomplete data to track outcome after infection and rejection in cases where repeat uCXCL10/Cr did not coincide with follow-up. The authors were unable to characterize all UTI cases and only focused on those who had elevated uCXCL10/Cr paired with the infection. It has recently been reported that urinary detection of BV virus infection was associated with uCXCL10 in the absence of viremia, and inclusion of urinary PCR or decoy cell testing may provide more complete detection of clinical BKV infection when uCXCL10 was elevated.

Furthermore, an UpToDate review on "Investigational methods in the diagnosis of acute kidney allograft rejection" (Anglicheau, 2024) states that "A number of urinary immune-related proteins have been identified as biomarkers of acute rejection of the kidney allograft. As examples, elevated urinary levels of chemokine (C-X-C motif) ligands 9 and 10 (CXCL9 and CXCL10) have been associated with acute TCMR …. Urinary levels of CXCL10 are also elevated in patients with ABMR. In a study of 244 kidney transplant patients, elevated urinary CXCL10 levels had a PPV and NPV of 55 and 86 % for the diagnosis of ABMR. However, increased urinary CXCL9 and CXCL10 levels have also been detected in patients with BK polyomavirus (BKPyV) infection, suggesting that these chemokines may represent markers of allograft inflammation rather than specific diagnostic markers of acute rejection".

Donor-Derived Cell-Free DNA for Consideration of Tapering of Immunosuppression Therapy in Kidney Transplant Recipients

Osuchukwu et al (2024) stated that immune-suppression therapy (IST) is required for allograft survival but can cause significant adverse effects. Donor-derived cell-free DNA is an non-invasive biomarker for active rejection in KTx; however, evidence supporting the use of dd-cfDNA testing in IST management is limited. In an observational, single-center, proof-of-concept (POC) study, these researchers examined the use of dd-cfDNA to inform tapering of IST in KTx recipients. dd-cfDNA testing was carried out in 21 KTx recipients who were considered good candidates for mycophenolic acid (MPA) reduction. Subjects with dd-cfDNA of less than 1 % at the 1st visit (enrollment) had their MPA dosage reduced; those with dd-cfDNA of 1 % or higher had their MPA dosage maintained. Subjects were monitored with dd-cfDNA for 6 additional visits. Of 21 patients enrolled in the study, 17 were considered low risk for rejection by dd-cfDNA and underwent MPA reduction; 4 patients were considered high risk for rejection by dd-cfDNA and had their initial MPA dosage maintained. Of the 4 patients considered high risk for rejection by dd-cfDNA, 1 experienced chronic allograft nephropathy and graft loss, and another received an indication biopsy that showed no evidence of rejection. Of the 17 patients considered low risk for rejection by dd-cfDNA, none experienced allograft rejection; dd-cfDNA was used for surveillance in a 6-month period following MPA reduction; no untoward results were noted. The authors concluded that this POC study reported the use of dd-cfDNA to directly inform IST management in a cohort of KTx recipients who were candidates for IST reduction. Moreover, these researchers stated that as the patients in this POC study were at a median of 5 years post-KTx, further investigations are needed to evaluate this approach in the 1st year following transplantation, when physicians routinely make decisions regarding IST reduction.

The authors stated that this study had several drawbacks. First, the small sample size (n = 21) necessitated more studies to confirm the observed effects. Second, the study was not controlled, and no protocol biopsies were conducted, with the designation of clinical stability relying on traditional non-invasive markers. Third, the starting MPA dosages varied across the cohort. Although this was reflective of the situation usually

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experienced in clinical care, it was possible that the impact of MPA dose reduction on dd-cfDNA differed depending on the starting dose. Furthermore, dd-cfDNA was only used to inform IST decisions at the 1st visit, and no subsequent management changes were made based on dd-cfDNA results. Fourth, the follow-up period of this trial was short, which may not be sufficient to capture long-term outcomes, such as chronic rejection and graft survival. Furthermore, these investigators stated that future studies are needed to consider the value of serial dd-cfDNA testing for IST optimization. In addition, absolute quantification of dd-cfDNA could be considered, because some reports have indicated that dd-cfDNA quantity may be more representative of organ status as it is not affected by host cfDNA factors (e.g., infection) and has superior accuracy in KTx rejection detection compared with dd-cfDNA fraction alone.

myOLARIS-KTdx for Diagnosis and Prognosis of Kidney Allograft Rejection

myOLARIS – KTdx, an urinary metabolite-based assay, is a non-invasive approach to differentiate rejection from polyomavirus-associated nephropathy (PVAN) infection (a complication associated with over-immunosuppression) in renal transplant recipients.

Heidari et al (2021) stated that although AMR accounts for 20 % to 30% of all acute renal allograft rejections, introducing biomarkers for a timely detection of allograft rejection has remained challenging. These investigators examined novel diagnostic biomarkers of AMR by assessing urine proteome in renal transplant patients. A total of 36 patients with renal transplantation including 22 AMR patients and 14 patients with stable renal function (control group) were enrolled in this trial. Urinary samples were collected and Label free quantification (LFQ) proteomics technique was employed on urinary samples and data were subjected to Random Forest (RF) algorithm to predict main candidate proteins contributing in AMR. Lastly, the use of candidate diagnostic biomarkers was examined in new sets of AMR subjects, stable patients and healthy volunteers. A total of 1,020 proteins were detected in urine proteome. RF algorithm predicted 20 differentially expressed proteins with the highest sensitivity and specificity and combination of EGF, COL6A, and NID-1 was identified as possible panel for early diagnosis of AMR. Applicability of EGF as diagnostic biomarker was validated in urinary samples of independent set of AMR subjects. This was the 1st urinary proteomics study in AMR patients showing that urinary EGF might be used as early diagnostic biomarker for AMR. The authors concluded that although several possible biomarkers have been proposed to predict AMR, ineffectiveness of current urinary biomarkers in early diagnosing of AMR patients and in distinguishing AMR subjects from patients with stable kidney function casts doubts on their applicability in the clinical setting.

Ramalhete et al (2022) noted that renal transplantation is currently the treatment of choice for end-stage renal disease (ESRD), enabling a QOL superior to dialysis. Despite this, all transplanted patients are at risk of allograft rejection processes. The gold-standard diagnosis of graft rejection, based on histological analysis of kidney biopsy, is prone to sampling errors and carries high costs and risks associated with such invasive procedures. In addition, the routine clinical monitoring, based on urine volume, proteinuria, and serum creatinine, usually only detects alterations after graft histologic damage and does not differentiate between the diverse etiologies. Thus, there is a need for new biomarkers enabling to predict, with high sensitivity and specificity, the rejection processes and the underlying mechanisms obtained from minimally invasive procedures to be implemented in routine clinical surveillance. These new biomarkers should also detect the rejection processes as early as possible, ideally before the 78 clinical outputs, while enabling balanced immunotherapy in order to minimize rejections and reducing the high toxicities associated with these drugs. Proteomics of biofluids, collected via non-invasive or minimally invasive analysis (e.g., blood or urine) present inherent characteristics that may provide biomarker candidates. The authors reviewed biofluids proteomics toward biomarkers discovery that specifically identify sub-clinical, acute, and chronic immune rejection processes while allowing for the discrimination between cell-mediated or antibody-mediated processes. These researchers stated that in time, these biomarkers will result in patient risk stratification, monitoring, and personalized and more efficient immunotherapies toward higher graft survival and patient QOL.

Rainey et al (2023) stated that renal transplantation is the optimal therapeutic option for most patients with ESRD given the significantly lower morbidity and mortality rates compared to remaining on dialysis. Rejection and graft failure remain common in transplant recipients with limited improvement in long-term

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transplant outcomes despite therapeutic advances. There is a need in the development of non-invasive biomarkers that specifically monitor graft function and predict transplant pathologies that affect outcomes. Despite the potential of proteomic investigatory approaches, no candidate biomarkers of sufficient sensitivity or specificity have translated into clinical use. In a scoping review, these investigators examined protein findings and protein pathways implicated in the literature, and flagged putative biomarkers worth validating in independent patient cohorts. This review followed the Joanna Briggs' Institute Methodology for a scoping review. MedlineALL, Embase, Web of Science Core Collection, Scopus and Google Scholar databases were searched from inception until December 2022. Abstract and full text review were undertaken independently by 2 reviewers; and data were collated using a pre-designed data extraction tool. A total of 101 studies met the inclusion criteria. The majority were retrospective, single-center studies of small sample size. Mass spectrometry was the most used technique to examine differentially expressed proteins between diagnostic groups and studies identified various candidate biomarkers such as immune or structural proteins. The authors concluded that putative immune or structural protein candidate biomarkers have been identified by means of proteomic techniques in multiple sample types including urine, serum and fluid used to perfuse donor kidneys. The most consistent findings implicated proteins associated with tubular dysfunction and immunological regulatory pathways such as leukocyte trafficking. Moreover, these researchers stated that, clinical translation and adoption of candidate biomarkers is limited, and these will require comprehensive evaluation in larger prospective, multi-center studies.

Sirolli et al (2023) stated that KT is the optimal therapeutic strategy for patients with ESRD. The key to post-transplantation management is careful surveillance of allograft function. Kidney injury may occur from several different causes that require different patient management approaches; however, routine clinical monitoring has several limitations and detects alterations only at a later stage of graft damage. Accurate new non-invasive biomarker molecules are needed for continuous monitoring following KT in the hope that early diagnosis of allograft dysfunction will result in an improvement in the clinical outcome. The advent of "omics sciences", and in particular of proteomic technologies, has revolutionized medical research. Proteomic technologies allow researchers to attain the identification, quantification, as well as functional characterization of proteins/peptides in biological samples such as urine or blood via supervised or targeted analysis. Many studies have examined proteomic techniques as potential molecular markers discriminating among or predicting allograft outcomes. Proteomic studies in KT have examined the whole transplant process: donor, organ procurement, preservation, as well as post-transplant surgery. These investigators reviewed the most recent findings on proteomic studies in the setting of KT to better understand the potential of this new diagnostic approach.

The authors noted that to translate proteomic biomarkers from bench to bedside is a challenge. One major obstacle is that only a few transplant studies ranged beyond the discovery phase of biomarker development to include fully independent validation studies showing the clinical utility of the biomarkers proposed. Most studies were also of limited size. These researchers stated that adequately-powered clinical studies including larger patient cohorts are needed to identify robust clinical biomarkers. The increased availability of high-definition equipment such as mass spectrometry in hospital biochemistry services is also of crucial importance in favoring the introduction of proteomic-based tools in clinical practice. Lastly, collaboration between researchers entailing data sharing as well as the realization of a central proteome data-base is also a critical objective. A comprehensive urine proteome data-base was recently generated from a variety of urine samples, including KT recipients with acute rejection or stable graft. The data-base may serve as a reference resource for facilitating the discovery of potential urinary proteomic biomarkers.

Rroji et al (2024) stated that KT significantly improves the lives of patients with ESRD; however, transplant success is threatened by the acute and chronic rejection mechanisms due to complex immune responses against the new organ. These investigators noted that the ongoing research into biomarkers holds promise for revolutionizing the early detection and monitoring of the graft health. Liquid biopsy techniques offer a new avenue, with several diagnostic, predictive, and prognostic biomarkers showing promise in detecting and monitoring kidney diseases and an early and chronic allograft rejection. The authors concluded that evaluating the protein composition related to KT results could result in identification of biomarkers that provide insights into the graft functionality. Non-invasive proteomic biomarkers can improve clinical outcomes and change the way how kidney transplants are evaluated for patients and physicians if they succeed in this transition. The advancement in proteomic technologies, leads toward a significant

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improvement in understanding of the protein markers and molecular mechanisms linked to the outcomes of KT. Moreover, these researchers stated that the road from discovery to the use of such proteins in clinical practice is long, with a need for continuous validation and beyond the singular research team with comprehensive infra-structure and across research groups collaboration.

Lazarou et al (2025) noted that renal transplantation stands as the preferred treatment for ESRD, significantly improving both the QOL and longevity of life compared to dialysis. In recent years, the survival rates for patients and grafts have markedly increased as a consequence of innovative strategies in desensitization protocols for incompatible transplants and advancements in immunosuppressive therapies. For renal transplant recipients, preventing allograft rejection is of critical importance, necessitating the use of immunosuppressive medications. Regular follow-up appointments are essential, as monitoring the function of the renal allograft is critical. Presently, established biomarkers such as serum creatinine, eGFR, proteinuria, as well as albuminuria are often used to evaluate allograft function. However, these biomarkers have limitations, as elevated levels often indicate significant allograft damage only after it has occurred; thus, limiting therapeutic options and the potential for restoring graft function. Furthermore, kidney biopsies, while considered the gold standard for diagnosing rejection, are invasive and carry associated risks. Consequently, the identification and development of new, sensitive, and specific biomarkers like dd-cfDNA, microRNAs (e.g., miR-21, miR-155), and sCD30 for allograft rejection are crucial. To tackle this challenge, intensive ongoing research using cutting-edge technologies, including "omics" approaches, like genomic techniques, proteomics, or metabolomics, is uncovering a variety of promising new biomarkers.

These researchers stated that progress in biomarker research presents promising alternatives that can enable prompt interventions and enhance patient outcomes. A variety of biomarkers, including donor-derived cell-free DNA, specific antibodies, and gene expression profiles, offer valuable insights into both acute and chronic rejection; thus, aiding in the customization of immunosuppressive treatments and improving graft survival rates. Donor-specific antibodies (DSAs), especially those that bind complement, are considered important in diagnosing renal allograft rejection. Studies have shown that the presence of complement-binding DSAs is associated with an increased risk of graft failure, making them relevant biomarkers in the context of ABMR. Furthermore, dd-cfDNA is emerging as a promising biomarker for the diagnosis of allograft rejection. It provides a non-invasive means to detect injury to the graft, and its levels can correlate with the severity of rejection. These investigators noted that as research advances, incorporating these biomarkers into clinical settings will be important for optimizing renal transplant care and tackling the challenges posed by graft rejection and dysfunction. Ultimately, the strategic use of biomarkers has the potential to provide better long-term results for renal transplant recipients and to enhance the understanding of the immunological mechanisms underlying rejection.

Furthermore, UpToDate reviews on “Kidney transplantation in adults: Evaluation and diagnosis of acute kidney allograft dysfunction” (Kadambi et al, 2025), “Kidney transplantation in adults: Clinical features and diagnosis of acute kidney allograft rejection” (Brennan et al, 2025), and “Kidney transplantation in adults: BK polyomavirus-associated nephropathy” (Lamaye and Brennan, 2025) do not mention urinary biomarkers as a management tool.

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